Hochschule
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Design and Implementation of a Real-Time Impedance Controller for a 7-DOF Collaborative Robot

Entwurf und Implementierung eines Echtzeit-Impedanzreglers für einen 7-achsigen kollaborativen Roboter

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Declaration of Authorship

I hereby declare that this master's thesis entitled

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ORANGE The argument and presentation are sufficient for the present thesis. Minor polishing may still be possible.

YELLOW The content is relevant, but the explanation is dense, repetitive, weakly supported, or harder to follow than necessary.

RED The passage retains a generic, templated, or textbook-restatement pattern that can read as machine-produced. This is a stylistic assessment, not a claim about authorship or an AI-detector result.

GREY A reasoning step, evidence link, validation, or required measurement is absent, uncertain, or inconsistent elsewhere.

Orange means sufficient, not perfect. A grey assessment does not make the reported data invalid; it identifies where the current evidence does not fully support the surrounding interpretation. Red is reserved for prose structure rather than technical correctness. Where a section contains both sound work and one important weakness, the more serious colour is used and the box names the exact concern.

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GREEN The passage has been read, checked and settled. Its wording is final and is not edited again.

BLUE Either new thesis text in a passage not yet settled, or a comment raised against a settled passage.

Green is not a judgement that the passage is strong. It records that the passage has been controlled and found correct, and that it already satisfies what the thesis requires of it. Nothing inside it is reworded, tightened, extended, or restructured afterwards. Rewording, sharpening, and stylistic improvement are not grounds for changing a settled passage, and a revised section that carries no blue at all is the normal case rather than an oversight.

Where a settled passage appears to need something, the text is left as it stands and the concern is raised beside it instead, as

The clearance gate holds the pose the tool reached and freezes the descend clock, until a bare Enter releases it. **comment:** “the frozen clock also governs the grind gate, which may be worth stating here”

A comment is a remark about the text, not a part of it. It appears only in this annotated copy and never in the submitted thesis, so a settled passage reads identically in both. It is raised only where the context or the reasoning is genuinely incomplete without the addition, not for anything that would merely read better.

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Abstract

ORANGE — sufficient. The abstract states the problem, implemented mechanism, principal measured ordering, null-space result, and scope limits without overloading the reader with the full parameter matrix.

Angular tool–surface mismatch causes edge-first contact. Under rigid position control, the robot resists the aligning rotation and develops contact forces and moments.

A real-time, phase-dependent Cartesian impedance controller was implemented for a seven-degree-of-freedom (DOF) Franka Emika Panda. During contact set-up, stiffness and damping defined about a selected virtual centre of compliance p_c were transformed to the tool centre point. The resulting coupling allowed the commanded pressing motion along the surface normal n_s to generate a corrective moment.

The experiments varied directional stiffness and the direction and magnitude of the compliance-centre lever r_c. Within the tested range, the in-plane compliance-centre position produced the largest change. The assisting lever was perpendicular to the commanded rotation axis. Reversing the commanded tool orientation offset also reversed the assisting lever direction. This selection gave about 6° improvement in both tangent-axis tests. The best tested lever magnitude depended on the commanded rotation axis and offset angle. Higher rotational stiffness reduced alignment, with axis-dependent behaviour.

For four commanded offsets in the tangent plane, a direction-selected lever improved alignment in every case, while one fixed lever reduced it.

Without any lever, the contact alone corrected an offset about one surface tangent and increased one about the other. Defining the lever in the tool frame rather than in surface coordinates changed the response once the tool turned.

Complementary Cartesian pose-hold trials showed that projected null-space damping reduced redundant motion. Both tested conditioning torque commands returned the mini-

mum singular value $\sigma_{\min}$ to approximately its initial level after the disturbance.

Kurzfassung

ORANGE — sufficient. The German summary remains aligned with the English abstract in structure, certainty, findings, and limitations.

Eine kleine Winkelabweichung führt dazu, dass eine Werkzeugkante die Oberfläche zuerst berührt. Bei starrer Positionsregelung behindert der Roboter die ausrichtende Drehung, sodass Kontaktkräfte und -momente entstehen.

Für einen 7-achsigen Franka Emika Panda wurde ein phasenabhängiger kartesischer Echtzeit-Impedanzregler implementiert. Während des Kontaktaufbaus wurden Steifigkeit und Dämpfung vom ausgewählten virtuellen Nachgiebigkeitszentrum p_c zum Werkzeugmittelpunkt transformiert. Die Kopplung ließ die Anpressbewegung entlang der Oberflächennormalen n_s ein korrigierendes Moment erzeugen.

Untersucht wurden richtungsabhängige Steifigkeit sowie Richtung und Betrag des Hebels r_c am Nachgiebigkeitszentrum p_c. Dessen Lage in der Ebene hatte im untersuchten Bereich den größten Einfluss. Der unterstützende Hebel stand senkrecht zu der vorgegebenen Drehachse. Bei Umkehr des vorgegebenen Orientierungsversatzes des Werkzeugs kehrte sich auch die Hebelrichtung um. Diese Auswahl verbesserte die Ausrichtung in beiden Tangentenachsen-Versuchen um etwa 6°. Die beste geprüfte Hebellänge hängte von der vorgegebenen Drehachse und dem Winkel des Orientierungsversatzes ab. Eine höhere Rotationssteifigkeit verringerte die Ausrichtungsreaktion achsenabhängig.

Bei vier vorgegebenen Versätzen in der Tangentialebene verbesserte ein richtungsabhängig gewählter Hebel stets die Ausrichtung, ein fester Hebel verringerte sie.

Ohne Hebel korrigierte der Kontakt allein einen Versatz um die eine Flächentangente und vergrößerte einen um die andere. Wurde der Hebel im Werkzeugframe statt in Flächenkoordinaten definiert, änderte sich die Reaktion, sobald sich das Werkzeug drehte.

Ergänzende kartesische Halteversuche zeigten, dass projizierte Nullraumdämpfung die

redundante Gelenkbewegung verringerte. Beide geprüften Konditionierungsdrehmomente stellten den kleinsten Singulärwert $\sigma_{\min}$ nach der Störung annähernd wieder her.

Contents

List of Figures

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List of Abbreviations

ORANGE — sufficient. The abbreviations are separated from the symbol list and their printed forms are controlled centrally.

CSV	Comma-Separated Values
DH	Denavit–Hartenberg
DOF	Degree of Freedom
EE	End-Effector
E-stop	Emergency Stop
FCI	Franka Control Interface
FCU	Franka Control Unit
F/T	Force/Torque
IP	Internet Protocol
LDLT	Lower–Diagonal–Lower-Transpose Factorisation
OS	Operating System
PREEMPT_RT	Real-Time Preemption Patch
ROS	Robot Operating System
SE(3)	Special Euclidean Group in three dimensions
SO(3)	Special Orthogonal Group in three dimensions
SVD	Singular Value Decomposition
TCP	Tool Centre Point

List of Symbols

ORANGE — sufficient. The notation is extensive but well organised, and the unit column makes the mixed translational and rotational quantities traceable.

Frames and Kinematics

$\{0\}$	Robot base frame	[-]
$\{i\}$	Joint frame i	[-]
$\{\mathrm{EE}\}$	End-effector (tool) frame	[-]
n	Number of joints ($n = 7$)	[-]
$q,\ \dot{q}$	Joint position / velocity vector	[rad], [rad/s]
q_0	Home or reference joint configuration	[rad]
x	Stacked Cartesian end-effector state	[m], [rad]
$x,\ y,\ z$	Scalar Cartesian coordinate directions or components; not the stacked state x	[m]
p	Position / translation vector	[m]
R	Rotation matrix	[-]
$R_x,\ R_y,\ R_z$	Elementary rotations about x, y, z	[-]
$\alpha,\ \beta,\ \gamma$	Generic rotations about the x, y, z axes	[rad]
T	Homogeneous transformation matrix	[-]

$J(q)$	Geometric Jacobian; linear rows above angular rows	[m/rad], [-]
z_{i-1}	Rotation axis of joint i, expressed in the base frame	[-]
p_{i-1}	Point on the axis of joint i, expressed in the base frame	[m]
$x_{\mathrm{EE}}, y_{\mathrm{EE}}, z_{\mathrm{EE}}$	End-effector frame axes (base frame)	[-]

Pose and Error

$p_{\mathrm{EE}},\ p_{\mathrm{TCP}}$	End-effector / tool-centre-point position; $p_{\mathrm{TCP}} = p_{\mathrm{EE}}$ in the implemented model	[m]
$R_{\mathrm{EE}},\ T_{\mathrm{EE}}$	End-effector orientation / transformation	[-]
$p_d,\ R_d,\ T_d$	Desired position / orientation / transformation	[m], [-]
T_{err}	Relative (error) transformation	[-]
ΔR	Current-to-desired relative rotation, $R_{\mathrm{EE}}^{\top} R_d$	[-]
ϕ	Orientation-error angle	[rad]
$u,\ [u]_{\times}$	Current-EE-frame orientation-error axis / its skew matrix	[-]
e_p	Translational (position) error	[m]
e_R	Base-frame rotational error used by the command	[rad]
$e_{R,\mathrm{EE}}$	Rotational error expressed in the current EE frame	[rad]
$e_{p,c},\ e_c$	Compliance-centre position error / stacked error	[m], [rad]

e	Stacked Cartesian error	[m], [rad]
$\dot{x}$	Cartesian velocity (twist)	[m/s], [rad/s]
$\dot{p}_{\mathrm{EE}},\ \dot{p}_d,\ \omega_{\mathrm{EE}}$	Measured linear end-effector velocity, desired linear velocity, and angular end-effector velocity	[m/s], [m/s], [rad/s]

Cartesian Impedance Gains

$K_p,\ D_p$	Generic translational stiffness / damping block; a frame subscript is added when needed	[N/m], [N s/m]
$K_R,\ D_R$	Generic rotational stiffness / damping block; a frame subscript is added when needed	[N m/rad], [N m s/rad]
$R_{\mathrm{task}},\ T_R$	Generic task-frame orientation and corresponding block rotation; $R_{\mathrm{task}} = R_{\mathrm{surface}}$ for the implemented task	[-]
$K_{p,\mathrm{task}},\ D_{p,\mathrm{task}}$	Local translational stiffness / damping in the chosen task frame	[N/m], [N s/m]
$K_{R,\mathrm{task}},\ D_{R,\mathrm{task}}$	Local rotational stiffness / damping in the chosen task frame	[N m/rad], [N m s/rad]
$K_{p,0},\ D_{p,0}$	Translational stiffness / damping used in the base-frame command	[N/m], [N s/m]
$K_{R,0},\ D_{R,0}$	Rotational stiffness / damping used in the base-frame command	[N m/rad], [N m s/rad]
$K_0,\ D_0$	Full 6×6 stiffness / damping used in the base-frame command	K: [N/m], [N m/rad]; D: [N s/m], [N m s/rad]

Dynamics and Torques

F	Cartesian wrench	[N], [N m]
$f,\ m$	Commanded Cartesian force / moment	[N], [N m]
f_{press}	Elastic press against the surface, $f_{\text{press}} = K_p e_p$; the force entering the compliance-centre moment m_c	[N]
$K,\ D$	Full Cartesian stiffness / damping matrices	K: [N/m], [N m/rad]; D: [N s/m], [N m s/rad]
Λ	Operational-space (Cartesian) inertia matrix	[kg], [kg m], [kg m^2]
$M(q)$	Joint-space mass matrix	[kg m^2]
ζ	Directional coefficient in the inertia-scaled damping calculation	[-]
τ	Joint torque vector	[N m]
$\tau_c,\ \tau_g$	Coriolis / gravity compensation torque	[N m]
τ_{cart}	Cartesian impedance torque, $J^\top(q)F$	[N m]
τ_{cmd}	Complete commanded joint torque	[N m]

Surface and Contact Geometry

n_s	Calibrated physical-plane normal used by the controller and evaluation	[-]
$t_1,\ t_2$	Surface tangent directions (unit)	[-]
$a_s,\ \tilde{t}_1$	Surface-frame reference direction / projected tangent before normalisation	[-]
R_{surface}	Surface task frame $[\,t_1\ \ t_2\ \ n_s\,]$ in the base frame	[-]
$p_i,\ r_{\text{probe,EE}}$	Captured marked-corner point in the base frame / fixed corner offset in the EE frame	[m], [m]
$R_i,\ \bar{R}$	Captured tool-calibration rotation / mean of the three estimation rotations	[-]
$\bar{n}_B$	Invariant base-frame direction in the tool-normal calibration	[-]
$U,\ \Sigma,\ V$	Singular-value decomposition factors of $\bar{R}$	[-]
$n_{\text{EE}},\ n_T,\ n_d$	Configured EE-frame tool normal / current base-frame tool normal / signed target direction	[-]
s_a	Tool-axis target sign, with $n_d = s_a n_s$	[-]
K_{t_1}, K_{t_2}, K_n	Translational stiffness along $t_1,\ t_2$, and n_s	[N/m]
$K_{r,t_1}, K_{r,t_2}, K_{r,n}$	Rotational stiffness entries in surface-frame column order	[N m/rad]
p_s	Point on the configured surface	[m]

p_c	Centre of compliance, held constant from the geometric clearance transition	[m]
r_c	Compliance-centre-to-TCP lever, $p_{\mathrm{TCP}} - p_c$	[m]
h	Selected normal offset of the centre of compliance	[m]
p_{contact}	Physical contact point	[m]
r_{contact}	Contact-to-TCP lever, $p_{\mathrm{TCP}} - p_{\mathrm{contact}}$; transfers a measured moment and is distinct from r_c	[m]
$M_{\mathrm{TCP}},\ M_{\mathrm{contact}}$	Estimated external moment change about the TCP / about the contact point	[N m]
f_C	Environment-on-tool contact force used in the geometric sketch	[N]

Compliance and Alignment

$K_{\text{task}},\ D_{\text{task}}$	Full local Cartesian stiffness / damping in the chosen task frame	K: [N/m], [N m/rad]; D: [N s/m], [N m s/rad]
$\text{Ad}(r_c)$	Compliance-centre-to-TCP point-shift adjoint	[-]
K_c, D_c	Block-diagonal stiffness / damping at the centre of compliance	K: [N/m], [N m/rad]; D: [N s/m], [N m s/rad]
$F_c,\ F_{\text{TCP}}$	Cartesian wrench at the centre of compliance / corresponding wrench at the TCP	[N], [N m]
$K_{\text{TCP}}, D_{\text{TCP}}$	Centre-of-compliance gains mapped by congruence to the TCP	K: [N/m], [N m/rad]; D: [N s/m], [N m s/rad]
$\theta_{\text{align}},\ \Delta\theta_{\text{align}}$	EE-inferred tool-axis alignment angle / before–after improvement	[rad]
$\Delta\theta_{\text{set}},\ \Delta\theta^S_{\text{set}}$	Finite set-up rotation in the base frame / resolved in the surface frame	[rad]
$\phi_{\text{set}},\ u_{\text{set}}$	Set-up rotation angle / unit rotation axis	[rad], [-]
$r_{c,t},\ \rho_c$	Selected tangential compliance-centre lever / its magnitude	[m]
$\Delta p_g,\ s_g$	Grinding contact-point displacement / its magnitude	[m]
$\Delta p_{\text{TCP}},\ s_{\text{TCP}}$	TCP displacement / its magnitude	[m]
F_n	Normal magnitude of the model-estimated external-force change from the clearance transition	[N]
$\hat{f}_{\text{ext}},\ \hat{f}_{\text{ref}}$	Model-estimated external force / clearance-transition reference	[N]

Quasi-Static Stiffness and Damping Design

$k_{p,i},\ k_{R,i}$	Principal translational / rotational stiffness	[N/m], [N m/rad]
k_i	Directional translational or rotational stiffness	[N/m] or [N m/rad]
λ_i	Directional Cartesian inertia used for damping selection	[kg] or [kg m^2]
$d_{\mathrm{dir},i}$	Directional damping coefficient	[N s/m] or [N m s/rad]

Null-Space and Singular Value Decomposition

J^+	Thresholded-SVD Moore–Penrose pseudoinverse	[rad/m], [-]
$N_q,\ N_\tau$	Joint-velocity / torque null-space projector	[-]
$\rho,\ \sigma_{\mathrm{thr}}$	Relative SVD tolerance / resulting singular-value threshold	[-], [m/rad]
$\sigma_i,\ \sigma_{\min}$	Jacobian singular value / σ_6 conditioning indicator; its value follows the length unit of the linear rows	[m/rad]
$U_J,\ \Sigma_J,\ V_J$	Singular value decomposition factors of J	[-], [m/rad], [-]
$c,\ c_i$	Joint-velocity coefficients in the right-singular-vector basis / coefficient for direction i	[rad/s]
$u_i,\ v_i$	Left / right singular vectors	[-]
$v_6,\ v_7$	σ_6-associated direction / structural null direction	[-]
$\dot{q}_{\mathrm{null}}$	Projected joint velocity in the numerical null space	[rad/s]

F_d	Commanded point-force-equivalent disturbance magnitude	[N]
τ_{dist}	Joint-torque-equivalent commanded disturbance	[N m]
d_{null}	Null-space damping	[N m s/rad]
k_σ	Sigma-conditioning torque magnitude	[N m]
α_{probe}	Null-direction sampling angle	[rad]
$\delta_\sigma,\ \Delta_\sigma$	Sigma-comparison deadband / sampled difference	[m/rad]
s	Deadbanded conditioning-direction selector	[-]
$\tau_d,\ \tau_\sigma,\ \tau_{\text{null}}$	Projected damping / sigma / total null-space torque	[N m]

Miscellaneous

T_s	Control sampling time (1 ms)	[s]
$I,\ I_3,\ I_4,\ I_6,\ I_7$	Identity matrices	[-]

Chapter 1

Introduction

GREEN — revised and confirmed. The chapter was revised in full and every paragraph is settled. No further editing is to be applied to it, and any remaining concern is raised as a comment rather than as a change.

1.1 Motivation

Industrial manipulators commonly apply position control to track commanded trajectories. In surface-finishing applications, the actual surface pose may differ from the programmed pose because of placement tolerances. The surface is positioned manually, the tool is held in a gripper, and the residual angle between them is unknown before contact. Under rigid position control, this angular deviation cannot be absorbed by the commanded motion and generates high contact forces.

Torque-controlled collaborative robots permit contact compliance to be specified directly, and they account for a growing share of industrial deployments [14, 26]. Joint torque sensing and a torque-level command interface let the controller specify how the robot yields under contact, not only where it goes. This capability is particularly useful in surface-finishing tasks, where a small angular mismatch between the tool and the surface can generate undesired contact forces and moments. The objective is therefore not only to reduce these loads, but also to enable passive angular alignment of the grinding face during contact. The controller should allow the tool to rotate toward the surface and reduce the angular mismatch while maintaining the commanded normal contact and the subsequent tangential grinding motion. The resulting alignment is evaluated relative to the physical surface.

Impedance control specifies this yielding behaviour as a dynamic relation between motion and interaction force [13]. Cartesian impedance control defines this relation in task space, meaning in terms of the Cartesian position and orientation of the end-effector rather than the individual joint coordinates. The compliance can therefore be adjusted separately for different translational and rotational directions. This direction-dependent behaviour is important during contact. A tool moving across a surface should maintain its normal position, while small lateral deviations should not generate large forces. The same principle applies to orientation. When the edge of a tilted tool contacts the surface, the contact wrench contains both a force and a moment. Stiffness values chosen according to the contact geometry allow the tool to rotate slightly toward the surface instead of fully resisting this motion. The resulting angular response is evaluated from the end-effector pose relative to the physical surface. The stiffness and damping matrices therefore determine both the contact response and the free-space tracking behaviour.

1.2 Related Work

The relevant literature comprises three areas: Cartesian impedance control, real-time implementation on torque-controlled robots, and the placement of the centre of compliance p_c. The first area is Cartesian impedance control itself. The controller uses standard serial-manipulator kinematics [23, 4, 19]: homogeneous transformations for representing the end-effector pose and the Jacobian for relating joint motion to Cartesian motion. Khatib established the operational-space formulation, in which a Cartesian wrench F is mapped to joint torques τ through $J^\top$ [15]; this mapping is adopted in the controller. Hogan defined the impedance relation between end-effector motion and interaction force [13]. Albu-Schäffer et al. implemented Cartesian impedance control using joint-torque sensing [1], which corresponds to the torque-controlled interface used in this thesis.

The second area concerns the real-time implementation of Cartesian impedance control on torque-controlled robots. Mayr et al. implemented a modular Cartesian impedance controller for the Franka Emika Panda that combines real-time model access, null-space control, signal treatment, and command limiting [17]. The controller developed in this thesis uses the libfranka model interface to obtain the required robot state and model quantities during the real-time control loop. It also includes a selectable null-space torque τ_{null} for influencing the redundant joint motion without changing the commanded Cartesian task.

Recent work has applied compliant and force-controlled strategies directly to robotic surface finishing. Alt et al. combined perception, planning, and force-controlled execution for robotic surface treatment [2]. Min et al. integrated demonstrated motion and force skills with impedance control on a 7-DOF collaborative robot [18]. Wu et al. used adaptive variable impedance control to regulate the contact force during robotic grinding [29]. These studies address force-controlled surface treatment. The present work instead examines contact-induced angular alignment through directional Cartesian stiffness and a virtual centre of compliance p_c.

The third area concerns the location of the centre of compliance p_c. Fasse and Broenink formulate spatial impedance about a selected point, so that translation and rotation become coupled through the position of this point [7]. This formulation provides the theoretical basis for the compliance-centre shift used in this thesis. The same principle appears mechanically in remote-centre-compliance devices, in which elastic elements are arranged so that the tool behaves as if it rotates about a virtual point located away from the mechanism itself [20, 28]. More recent mechanically compliant and variable-compliance-centre strategies apply this principle to peg-in-hole insertion [27, 12]. In the mechanically compliant mechanisms, elastic bending produces the alignment motion rather than software control alone. In these approaches, the compliance is arranged so that contact forces reduce misalignment instead of increasing it. Most of these studies focus on insertion tasks and define the compliance centre p_c through either the mechanical design or a strategy adapted to the hole geometry. In this thesis, an equivalent effect is created in software by shifting the virtual centre of compliance p_c away from the TCP. The independent effects of this position and the axis-specific stiffness during planar tool alignment are then investigated on a torque-controlled robot.

Suomalainen et al. experimentally investigated how compliance-centre placement and compliant-axis selection influence alignment in an assembly task [25]. Friedrich et al. subsequently proposed an active remote centre of compliance p_c that combines passive compliance with active force control [11]. These approaches further illustrate the role of compliance location in contact alignment. The present work examines a software-defined virtual compliance centre p_c for planar tool–surface alignment.

1.3 Problem Statement

The angular relation between the tool and the surface plane is uncertain before contact because both the surface placement and the tool mounting carry tolerances. Under rigid position control, this residual angle produces an undesired contact moment rather than a corrective motion. A tool tilted with respect to the surface cannot be brought flat by translational compliance alone, because the wrench that closes the gap must also contain a moment. Translational compliance therefore does not by itself produce the required corrective rotation. Placing the centre of compliance p_c away from the tool centre point (TCP) couples translation and rotation, so that pressing the tool into the plane also causes it to rotate toward the plane. This placement defines a task-to-impedance mapping rather than a physical pivot. The resulting response was evaluated from the motion measured during contact.

The alignment response cannot be predicted from the gain values alone because it also depends on the contact geometry, the lever direction, the robot configuration, and the compliance of the tool mounting. The experiments therefore compare the effects of a commanded tilt representing the initial angular mismatch between the tool and the surface, the directional stiffness, and the compliance-centre placement under otherwise matched conditions. Chapters 2 and 3 introduce the frame conventions and controller formulation required for these comparisons.

comment: “Use “commanded tool orientation offset” in the problem statement and throughout the thesis wherever the commanded angular offset is meant. Use “commanded rotation axis” or “commanded rotation direction” for its components. Retain “tilt” only for a physical inclination or a calibrated surface angle.”

The overall research question is: how can directional Cartesian stiffness and a virtual centre of compliance p_c be selected so that surface contact aligns a tilted tool with a plane? The experimental investigation is organised around three linked subquestions:

RQ1: How do rotational stiffness about the tilted tangent and translational stiffness perpendicular to it affect contact-induced alignment?

RQ2: How do the direction, sign, and magnitude of the tangential compliance-centre lever r_c affect alignment for different tilt axes, signs, and magnitudes?

RQ3: Can the signed tilt direction be used to derive a geometric, tilt-dependent compliance-centre selection rule that extends beyond the two principal surface tangents?

1.4 Scope and Contributions

Within this scope, the experiments evaluate how a commanded tilt representing the initial angular mismatch between the tool and the surface, the rotational and translational stiffness in the surface directions, and the position of a virtual centre of compliance p_c influence contact-induced alignment. The three research questions treat these variables as one progression from the alignment baseline, through stiffness and compliance-centre effects, to a tilt-dependent selection rule.

The controller is a phase-scheduled Cartesian impedance controller written in C++ using libfranka and Eigen. It runs in the 1 kHz torque-control loop of the Franka Emika Panda. The commanded Cartesian wrench F is mapped to the Cartesian impedance torque τ_{cart} through $J^\top$, and the Coriolis torque τ_c is obtained from the robot model and added to the command. Each control phase uses its own Cartesian stiffness values, while the damping is calculated from an estimate of the Cartesian inertia Λ.

A surface-fixed coordinate frame is constructed from the surface geometry. Its axes consist of the unit surface normal n_s and two mutually orthogonal unit vectors t_1 and t_2 lying in the surface plane. The directions t_1 and t_2 therefore represent the two independent tangential directions along the surface. This frame is used to define the translational and rotational stiffness values according to the contact geometry.

Each run follows a fixed sequence of control phases. First, the tool is brought to the commanded orientation and approaches the surface up to a specified geometric clearance. Based on the rectangular grinding-face geometry and the current tool orientation, the controller predicts the point on the tool expected to reach the surface first. For this purpose, the four tool corners are compared along the approach direction. A single leading corner is selected for a general tilt. If two neighbouring corners are equally close to the surface, their midpoint is used as the predicted contact point at the centre of the leading edge. If all four corners are equally close to the surface, the centre of the grinding face is used. This predicted contact point is fixed when the clearance is reached and remains unchanged during set-up and grinding.

During contact set-up, the desired position of the predicted contact point is moved gradually from the clearance position toward the surface and slightly beyond the surface plane. The physical surface prevents the tool from reaching this virtual target, so the resulting position error e_p generates the normal pressing force. Because the robot command is defined at the TCP, the controller calculates the corresponding TCP target using the fixed geometric offset between the TCP and the predicted contact point.

The desired tool orientation remains equal to the orientation reached at the clearance point, but it is not enforced rigidly. The finite rotational stiffness allows the actual tool orientation to change when the contact forces produce a moment, so the tool can rotate toward the surface during set-up.

During this phase, the 6×6 stiffness and damping matrices defined at the virtual centre of compliance p_c are transformed to the TCP through an adjoint congruence transformation. The resulting off-diagonal blocks couple translation and rotation, allowing the commanded pressing motion to generate a corrective moment that helps the tool align with the plane surface during set-up. The effects of this translation–rotation coupling are investigated in this thesis. After contact set-up, the controller proceeds to the tangential grinding motion.

The software implements this subsequent grinding phase, including an optional tangential sweep. During the reported runs, the grinding phase maintained the achieved press reference at the set-up tangential coordinates. The quantitative evaluation measures the contact-induced alignment established during set-up.

The point-shift adjoint and congruence transformation are established theory. The implementation contribution lies in their phase-dependent, real-time integration with the contact sequence and controller described in Chapter 3.

Accordingly, the contributions of this thesis are divided into implemented controller components and experimentally established findings. The implemented work includes a frame-consistent real-time Cartesian impedance controller for the 7-DOF Franka Emika Panda.

It also includes a spatial compliance-centre formulation in which the 6×6 stiffness and damping matrices are defined at a selected virtual centre of compliance p_c and transformed to the TCP. In addition, a contact and evaluation methodology was developed in which the physical surface orientation and the grinding-face axis are defined independently, allowing the alignment of the tool to be evaluated relative to the surface.

The experimental findings address the three research questions as one progression. First, the measurements quantify how rotational stiffness about the tilted tangent and translational stiffness perpendicular to it affect contact-induced alignment. The response was axis-dependent, and the tested combination of low translational and high rotational stiffness produced a non-additive response. Within the investigated range, however, these stiffness changes were smaller than the effect of moving the virtual centre of compliance p_c.

Second, the compliance-centre study establishes distinct effects of lever direction, sign, and magnitude. The assisting tangential lever was perpendicular to the tilt axis. Reversing the commanded tilt reversed the assisting lever sign on both principal surface tangents. A systematic magnitude sweep and the extended reversed-tilt settings further showed that the best tested magnitude depends on the tilt axis, tilt sign, and commanded tilt magnitude. In three conditions, the largest measured improvement occurred at the upper boundary of the tested range.

Third, these observations lead to a tilt-dependent geometric selection rule: the tangential lever is chosen perpendicular to the signed tilt axis so that the press-induced moment opposes the angular error. Case C tested this rule at four tangent-plane directions using one common 60 mm magnitude. The selected lever assisted in every direction, whereas holding one lever fixed reduced the matched improvement by 32–48 %. This extends the contribution beyond separate t_1 and t_2 comparisons. The matched reductions support tilt-dependent lever selection across the tested directions.

Projected joint damping and two-point minimum-singular-value $\sigma_{\min}$ conditioning were also implemented for the redundant seventh joint. Complementary Cartesian pose-hold trials evaluated both terms using an internally generated joint-torque disturbance equivalent to a point force at an offset point on link 3. The point was located 0.200 m along the link-frame z-axis. The results describe the null-space response to this commanded disturbance.

1.5 Thesis Structure

The thesis is structured as follows. Chapter 2 introduces the impedance law, the frame conventions, and the compliance-centre shift. Chapter 3 describes their real-time implementation on the Franka Emika Panda. Chapter 4 presents the experimental methodology, while Chapter 5 reports and discusses the corresponding results. Chapter 6 summarises the main conclusions and outlines future work.

Chapter 2

Theoretical Background

GREEN — revised and confirmed. The chapter was revised in full and every paragraph is settled. No further editing is to be applied to it, and any remaining concern is raised as a comment rather than as a change.

comment: “In the settled prose, use “commanded tool orientation offset” for the deliberate angular command and “commanded rotation axis” or “commanded rotation direction” for its components. Retain “tilt” only for a physical inclination or a calibrated surface angle.”

This chapter presents the mathematical basis of the Cartesian impedance controller. The required frame conventions and pose errors are introduced first. The subsequent sections derive the kinematic mappings, the impedance wrench F, the compliance-centre transformation, and the null-space torque τ_{null}.

2.1 Reference Frames

The following right-handed coordinate frames are used in the theoretical formulation and the controller [23, 4].

- **Base frame $\{0\}$:** The base frame is fixed to the robot mounting surface. Its origin is the robot base reference point used by the kinematic model. For the upright Panda configuration used in this thesis, the z_0-axis is normal to the mounting surface, while the x_0- and y_0-axes lie in the mounting plane. Unless stated otherwise, Cartesian positions, directions, velocities, forces, and moments are expressed in this frame.

- **Joint frames** $\{1\}, \{2\}, \ldots, \{n\}$**:** These frames are attached to the successive links of the kinematic chain according to the Denavit–Hartenberg convention [5, 4]. Joint i rotates about z_{i-1}, and p_{i-1} denotes a point on this axis. The joint frames define the transformations between consecutive links and the indexing used in the forward kinematics and geometric Jacobian $J(q)$.
- **End-effector frame** $\{\text{EE}\}$**:** The end-effector frame is attached to the TCP and moves with the tool. Its position p_{EE} and orientation R_{EE}, both expressed relative to the base frame, define the current Cartesian pose of the robot. The axes x_{EE}, y_{EE}, and z_{EE} describe the current tool orientation. The configured tool approach axis is defined in this frame and is transformed to the base frame using R_{EE}.
- **Desired end-effector frame** $\{d\}$**:** The desired frame represents the commanded TCP pose. Its origin is p_d, and its orientation is R_d, both expressed relative to the base frame. The difference between the desired and current end-effector frames provides the translational and rotational errors used by the impedance controller.
- **Surface task frame** $\{S\}$**:** The surface task frame is tied to the configured surface rather than to the moving tool. Its origin is placed at the configured surface point p_s. Its orientation R_{surface} is defined by the two surface tangents t_1 and t_2 and the surface normal n_s:

$$R_{\text{surface}} = \begin{bmatrix} t_1 & t_2 & n_s \end{bmatrix}. \tag{2.1}$$

  The directions t_1 and t_2 lie in the surface plane and are mutually orthogonal, while n_s is normal to the surface. This frame is used to describe the commanded tilt and to assign stiffness and damping values according to the two tangential directions and the normal direction. For the contact task, the task-frame orientation R_{task} equals the surface-frame orientation:

$$R_{\text{task}} = R_{\text{surface}}. \tag{2.2}$$

Figure 2.1 shows these frames together with the pose error between the current and the desired end-effector frame.

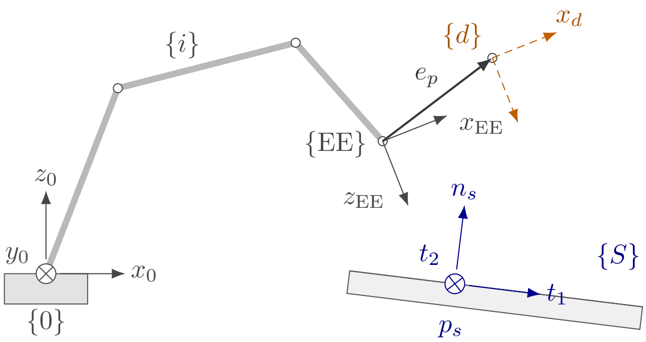


Figure 2.1: Frames used in the controller and the end-effector pose error.

2.2 Rigid-Body Transformations

The positions and orientations of the frames introduced in Section 2.1 are related through rotation matrices and homogeneous transformations.

2.2.1 Rotation Matrices

An orientation is represented by a rotation matrix $R \in SO(3)$, satisfying

$$R^\top R = I, \qquad \det R = 1, \qquad R^{-1} = R^\top. \tag{2.3}$$

Here, I is the 3×3 identity matrix and $\det(\cdot)$ denotes the determinant.

The notation ${}^bR_a$ denotes the orientation of frame $\{a\}$ expressed in frame $\{b\}$. Its columns contain the axes of frame $\{a\}$ expressed in frame $\{b\}$. A vector v is transformed from frame $\{a\}$ to frame $\{b\}$ according to

$${}^bv = {}^bR_a\, {}^av. \tag{2.4}$$

The elementary right-handed rotations use the rotation angles α, β, and γ about the x-, y-, and z-axes, respectively:

$$R_x(\alpha) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\alpha & -\sin\alpha \\ 0 & \sin\alpha & \cos\alpha \end{bmatrix}, \; R_y(\beta) = \begin{bmatrix} \cos\beta & 0 & \sin\beta \\ 0 & 1 & 0 \\ -\sin\beta & 0 & \cos\beta \end{bmatrix}, \; R_z(\gamma) = \begin{bmatrix} \cos\gamma & -\sin\gamma & 0 \\ \sin\gamma & \cos\gamma & 0 \\ 0 & 0 & 1 \end{bmatrix}. \tag{2.5}$$

2.2.2 Homogeneous Transformations

A position is represented by a vector $p \in \mathbb{R}^3$. The homogeneous transformation ${}^bT_a$ combines the rotation and translation from frame $\{a\}$ to frame $\{b\}$ [4, 23]:

$$ {}^bT_a = \begin{bmatrix} {}^bR_a & {}^bp_a \\ 0^\top & 1 \end{bmatrix}, \tag{2.6} $$

where ${}^bp_a$ is the position of the origin of frame $\{a\}$ expressed in frame $\{b\}$. A point expressed in frame $\{a\}$ is transformed to frame $\{b\}$ according to

$$ {}^bp = {}^bR_a \, {}^ap + {}^bp_a. \tag{2.7} $$

Successive transformations are composed by matrix multiplication:

$$ {}^cT_a = {}^cT_b \, {}^bT_a. \tag{2.8} $$

2.2.3 End-Effector Pose

The controlled pose consists of the end-effector position

$$ p_{\mathrm{EE}} = \begin{bmatrix} p_{x,\mathrm{EE}} & p_{y,\mathrm{EE}} & p_{z,\mathrm{EE}} \end{bmatrix}^\top \in \mathbb{R}^3 \tag{2.9} $$

and the orientation $R_{\mathrm{EE}} \in SO(3)$. Both are expressed in the base frame and combined in the end-effector transformation T_{EE}:

$$ T_{\mathrm{EE}} = \begin{bmatrix} R_{\mathrm{EE}} & p_{\mathrm{EE}} \\ 0^\top & 1 \end{bmatrix}. \tag{2.10} $$

In the implementation, T_{EE} is obtained directly from the reported robot state, and the orientation is not converted to an Euler-angle representation.

2.2.4 Cartesian Pose Error

The desired end-effector transformation T_d is

$$T_d = \begin{bmatrix} R_d & p_d \\ 0^\top & 1 \end{bmatrix}, \tag{2.11}$$

where p_d and R_d are the desired TCP position and orientation, respectively. The current end-effector pose is denoted by T_{EE}.

The relative pose-error transformation T_{err} maps the current end-effector frame to the desired frame:

$$T_{\mathrm{err}} = T_{\mathrm{EE}}^{-1} T_d = \begin{bmatrix} R_{\mathrm{EE}}^\top R_d & R_{\mathrm{EE}}^\top (p_d - p_{\mathrm{EE}}) \\ 0^\top & 1 \end{bmatrix}. \tag{2.12}$$

When the current and desired poses coincide, $T_{\mathrm{err}} = I_4$. The rotational and translational blocks of eq. (2.12) therefore both represent zero error in this case.

Position error. The translational block of eq. (2.12) expresses the position error e_p in the current end-effector frame. The position error e_p used by the Cartesian impedance law is expressed in the base frame:

$$e_p = p_d - p_{\mathrm{EE}} \in \mathbb{R}^3. \tag{2.13}$$

A positive position error e_p therefore points from the current TCP position toward the desired TCP position.

Orientation error. The relative rotation ΔR is the rotational block of eq. (2.12):

$$\Delta R = R_{\mathrm{EE}}^\top R_d. \tag{2.14}$$

It represents the rotation required to carry the current end-effector orientation to the desired orientation. If both orientations coincide, then $\Delta R = I$.

For use in the impedance controller, ΔR is represented by a rotation angle ϕ and a unit rotation axis $u \in \mathbb{R}^3$ [23, 19]. The angle follows from

$$\mathrm{tr}(\Delta R) = 1 + 2\cos\phi, \tag{2.15}$$

and is therefore

$$\phi = \arccos\left(\frac{\mathrm{tr}(\Delta R) - 1}{2}\right), \qquad \phi \in [0, \pi]. \tag{2.16}$$

Rodrigues' formula expresses the relative rotation as

$$\Delta R = I + \sin\phi\, [u]_\times + (1 - \cos\phi)[u]_\times^2, \tag{2.17}$$

where

$$[u]_\times = \begin{bmatrix} 0 & -u_z & u_y \\ u_z & 0 & -u_x \\ -u_y & u_x & 0 \end{bmatrix}, \qquad [u]_\times^\top = -[u]_\times. \tag{2.18}$$

Subtracting the transpose of ΔR isolates its skew-symmetric part:

$$\frac{1}{2}\left(\Delta R - \Delta R^\top\right) = \sin\phi\, [u]_\times. \tag{2.19}$$

For $\sin\phi \neq 0$, the rotation axis is obtained from

$$u = \frac{1}{2\sin\phi} \begin{bmatrix} \Delta R_{32} - \Delta R_{23} \\ \Delta R_{13} - \Delta R_{31} \\ \Delta R_{21} - \Delta R_{12} \end{bmatrix}. \tag{2.20}$$

Multiplying the unit axis by the angle gives the current-frame orientation error $e_{R,\mathrm{EE}}$. Rotating it into the base frame gives the controller orientation error e_R:

$$e_{R,\mathrm{EE}} = \phi u, \qquad e_R = R_{\mathrm{EE}} e_{R,\mathrm{EE}} \in \mathbb{R}^3. \tag{2.21}$$

The direction of e_R defines the axis of the required correction, while its magnitude is the angular error in radians. When the current and desired orientations coincide, $\phi = 0$ and $e_R = 0$.

The complete Cartesian error e stacks the position error e_p and orientation error e_R used by the impedance controller:

$$e = \begin{bmatrix} e_p \\ e_R \end{bmatrix} \in \mathbb{R}^6. \tag{2.22}$$

2.3 Kinematics of a 7-DOF Manipulator

Having defined the end-effector pose as the controlled variable, its relation to the joint configuration is introduced next.

2.3.1 Forward Kinematics

Forward kinematics maps the joint-configuration vector q

$$q = \begin{bmatrix} q_1 & q_2 & \cdots & q_n \end{bmatrix}^\top \in \mathbb{R}^n \tag{2.23}$$

to the end-effector pose. Using the joint frames introduced in Section 2.1, the transformation from the base frame $\{0\}$ to the final link frame $\{n\}$ is

$${}^0T_n(q) = {}^0T_1(q_1)\, {}^1T_2(q_2) \cdots {}^{n-1}T_n(q_n). \tag{2.24}$$

In the standard Denavit–Hartenberg convention [5, 4, 23], joint i rotates by q_i about the axis z_{i-1}. The transformation between two consecutive joint frames is

$${}^{i-1}T_i(q_i) = R_z(q_i)\, T_z(d_i)\, T_x(a_i)\, R_x(\alpha_i), \tag{2.25}$$

where a_i, d_i, and α_i describe the fixed geometry between the consecutive joint frames, while q_i is the revolute-joint coordinate.

The complete transformation gives the end-effector pose in the base frame:

$${}^0T_n(q) = T_{\mathrm{EE}}(q) = \begin{bmatrix} R_{\mathrm{EE}}(q) & p_{\mathrm{EE}}(q) \\ 0^\top & 1 \end{bmatrix}. \tag{2.26}$$

This convention establishes the joint and frame indexing used in the subsequent Jacobian $J(q)$ formulation. In the implemented controller, the forward-kinematic chain is not evaluated separately; the current end-effector pose is obtained from the libfranka model interface.

2.3.2 Differential Kinematics

Differential kinematics maps the joint-velocity vector $\dot{q}$ to the end-effector twist $\dot{x}$:

$$\dot{x} = \begin{bmatrix} \dot{p}_{\mathrm{EE}} \\ \omega_{\mathrm{EE}} \end{bmatrix} = J(q)\dot{q}, \tag{2.27}$$

where $\dot{p}_{\mathrm{EE}} \in \mathbb{R}^3$ and $\omega_{\mathrm{EE}} \in \mathbb{R}^3$ are the linear and angular velocities of the end-effector, respectively. The joint-velocity vector is

$$\dot{q} = \begin{bmatrix} \dot{q}_1 & \dot{q}_2 & \cdots & \dot{q}_n \end{bmatrix}^\top \in \mathbb{R}^n, \tag{2.28}$$

and $J(q) \in \mathbb{R}^{6\times n}$ is the geometric Jacobian. For the 7-DOF Panda, $n = 7$. The end-effector twist and the Jacobian used in the controller are expressed in the base frame. This relation provides the Cartesian velocity $\dot{x}$ required by the damping term of the impedance controller.

2.3.3 Geometric Jacobian

The geometric Jacobian $J(q)$ is written column-wise as

$$J(q) = \begin{bmatrix} J_1 & J_2 & \cdots & J_n \end{bmatrix} \in \mathbb{R}^{6\times n}, \tag{2.29}$$

where J_i describes the contribution of joint i to the end-effector twist. Since all joints of the Panda are revolute, each column is given by

$$J_i = \begin{bmatrix} z_{i-1} \times (p_{\mathrm{EE}} - p_{i-1}) \\ z_{i-1} \end{bmatrix} \in \mathbb{R}^6, \tag{2.30}$$

where z_{i-1} is the axis of joint i, and p_{i-1} is a point on that axis. Both are expressed in the base frame. The upper block describes the linear velocity generated at the end-effector, while the lower block describes the corresponding angular velocity [23, 4].

Because the joint axes and their positions depend on the current configuration, the Jacobian also depends on q. In the implemented controller, $J(q)$ is obtained directly from the libfranka model interface.

The Jacobian describes the local relation between joint velocities and the end-effector twist. For a robot with n joints it is written column-wise,

$$\begin{bmatrix} \dot{p}_{\mathrm{EE}} \\ \omega_{\mathrm{EE}} \end{bmatrix} = J(q)\dot{q}, \qquad J(q) = \begin{bmatrix} J_1 & J_2 & \cdots & J_n \end{bmatrix} \in \mathbb{R}^{6\times n}, \tag{2.31}$$

where each column J_i is the contribution of joint i. The Panda has only revolute joints, so with z_{i-1} the unit vector along the axis of joint i and p_{i-1} a point on that axis, both expressed in the base frame, the column is

$$J_i = \begin{bmatrix} z_{i-1} \times (p_{\mathrm{EE}} - p_{i-1}) \\ z_{i-1} \end{bmatrix} \in \mathbb{R}^6. \tag{2.32}$$

This standard form follows from the instantaneous velocity of a point under a revolute motion [23, 4]. The vector $p_{\mathrm{EE}} - p_{i-1}$ is the lever arm from the joint axis to the end-effector, so the cross product $z_{i-1} \times (p_{\mathrm{EE}} - p_{i-1})$ is the linear velocity that rotation of joint i produces at the end-effector, while the lower block z_{i-1} is the corresponding angular velocity. Each column therefore stacks a linear contribution above an angular one.

Because z_{i-1}, p_{i-1} and p_{EE} all depend on the current joint configuration, the Jacobian is configuration dependent, and the influence of each joint on the end-effector motion changes as the robot moves.

2.4 Cartesian Impedance Control

The geometric Jacobian $J(q)$ is also used through its transpose to map a Cartesian wrench F to joint torques τ [15]. Before introducing this mapping, the desired Cartesian interaction behaviour is defined.

In this thesis, Cartesian impedance control is formulated as a spring–damper relation between the end-effector motion and the commanded wrench [13, 21, 17]. The Cartesian pose error e, defined in Section 2.2.4, produces a restoring wrench through the stiffness matrix K, while the Cartesian velocity $\dot{x}$ produces an opposing wrench through the damping matrix D. Their entries determine the translational and rotational compliance in the selected task directions [22].

2.4.1 Impedance Wrench

The impedance wrench is evaluated in the robot base frame. The corresponding translational stiffness and damping matrices are $K_{p,0}, D_{p,0} \in \mathbb{R}^{3\times3}$, while the rotational matrices are $K_{R,0}, D_{R,0} \in \mathbb{R}^{3\times3}$. These matrices are not generally diagonal in the base frame because the direction-dependent gains are defined relative to the task frame. Their base-frame representation therefore changes with the task-frame orientation, even when the local gain values remain unchanged.

The commanded Cartesian force f is given by the translational spring–damper law [13, 17]

$$f = K_{p,0}e_p + D_{p,0}\left(\dot{p}_d - \dot{p}_{\mathrm{EE}}\right) \in \mathbb{R}^3, \tag{2.33}$$

where $\dot{p}_d$ and $\dot{p}_{\mathrm{EE}}$ are the desired and measured linear velocities expressed in the base frame. For a fixed position setpoint, $\dot{p}_d = \mathbf{0}$, and the damping term becomes $-D_{p,0}\dot{p}_{\mathrm{EE}}$.

The commanded Cartesian moment m is

$$m = K_{R,0}e_R - D_{R,0}\omega_{\mathrm{EE}} \in \mathbb{R}^3, \tag{2.34}$$

where ω_{EE} is the measured end-effector angular velocity expressed in the base frame. Since no angular reference velocity is commanded, the rotational damping acts directly against ω_{EE}.

The Cartesian wrench F stacks the force and moment:

$$F = \begin{bmatrix} f \\ m \end{bmatrix} = \begin{bmatrix} K_{p,0}e_p + D_{p,0}\left(\dot{p}_d - \dot{p}_{\mathrm{EE}}\right) \\ K_{R,0}e_R - D_{R,0}\omega_{\mathrm{EE}} \end{bmatrix} \in \mathbb{R}^6. \tag{2.35}$$

The wrench is ordered as force followed by moment, and all its components are expressed in the base frame. It is mapped to joint torques τ_{cart} through the Jacobian transpose in the following subsection.

2.4.2 Joint-Torque Control Law

By the principle of virtual work, the commanded Cartesian wrench F is mapped to joint torques τ_{cart} through the transpose of the geometric Jacobian $J(q)$ [15]:

$$\tau_{\text{cart}} = J^{\top}(q)F, \tag{2.36}$$

where $\tau_{\text{cart}} \in \mathbb{R}^n$ is the Cartesian impedance torque contribution and $J^{\top}(q) \in \mathbb{R}^{n\times 6}$. Substituting eq. (2.35) gives

$$\tau_{\text{cart}} = J^{\top}(q) \begin{bmatrix} K_{p,0}e_p + D_{p,0}\left(\dot{p}_d - \dot{p}_{\text{EE}}\right) \\ K_{R,0}e_R - D_{R,0}\omega_{\text{EE}} \end{bmatrix}. \tag{2.37}$$

The gain matrices in this expression are the base-frame matrices obtained from the selected task-frame gains. A change in the surface-frame orientation therefore changes the Cartesian wrench F and the resulting joint-torque contribution τ_{cart}, even when the local gain values remain unchanged.

2.4.3 Coriolis and Gravity Compensation in libfranka

In a general model-compensated torque controller, the commanded joint torque τ_{cmd} combines Cartesian impedance, gravity, and Coriolis/centrifugal compensation:

$$\tau_{\text{cmd}} = \tau_{\text{cart}} + \tau_g(q) + \tau_c(q, \dot{q}), \tag{2.38}$$

where $\tau_g(q)$ is the gravity torque and $\tau_c(q, \dot{q})$ is the Coriolis and centrifugal torque.

These terms are handled differently by the Franka Control Interface. The Coriolis vector provided by the libfranka model is added explicitly to the command because it is not compensated internally. A separate gravity term is not added. In external joint-torque mode, the robot compensates gravity and motor friction internally, and the commanded torque is interpreted as excluding gravity [9]. Adding $\tau_g(q)$ would therefore compensate gravity twice.

The implemented commanded torque $\tau_{\text{cmd,impl}}$ is consequently

$$\tau_{\text{cmd,impl}} = \tau_{\text{cart}} + \tau_c(q, \dot{q}) = J^\top(q)F + \tau_c(q, \dot{q}), \tag{2.39}$$

which follows the official libfranka Cartesian impedance example [10]. The null-space torque τ_{null} introduced later is added to this expression to obtain the complete commanded torque in eq. (2.97).

2.4.4 Commanded and Estimated Wrench

Two wrenches appear in this work, and they are obtained in opposite directions. The commanded wrench F is formed from eq. (2.35) and reaches the joints through $J^\top$. The robot reports a second one. Its translational part is the model-estimated external force $\hat{f}_{\text{ext}}$, inferred from the measured joint torques after the dynamic model has been subtracted.

The commanded wrench needs no such inference. The stiffness and the pose error are both held by the controller. $K_{p,0}e_p$ is therefore known while the motion is commanded. Against a rigid plane the tool barely moves, and the reference sets the error.

At rest the two are related by the contact. The torques from $J^\top F$ are balanced by the contact reaction through the same Jacobian. $\hat{f}_{\text{ext}}$ then follows $-F$, up to model error, friction, and motion absorbed by the tool. While the reference still moves, the damping term $D_{p,0}\dot{p}_d$ adds to the commanded force. It leaves when the reference stops.

The estimate is an independent measurement of a load the controller already knows. It is not the only route to that load. Agreement between the two is used in chapter 5 as a check on the impedance.

2.5 Surface Task Frame

The contact experiment uses a task frame tied to the surface geometry. This frame is defined before the stiffness and damping matrices are introduced, so that the local gain entries can later be interpreted as normal and tangential values instead of as abstract base-frame directions.

Let the unit surface normal $n_s \in \mathbb{R}^3$, with $\|n_s\| = 1$, be expressed in the robot base frame. To complete the frame, choose an auxiliary direction $a_s \in \mathbb{R}^3$ that is not parallel to n_s.

This auxiliary direction is only used to define a repeatable tangent direction on the surface. The first tangent vector is obtained by projecting a_s onto the plane orthogonal to n_s, and the second tangent vector is then obtained by a cross product. With $\tilde{t}_1$ denoting the unnormalised projected tangent, the construction is a Gram–Schmidt orthonormalization step applied to the surface normal n_s and the auxiliary direction [23, 4]:

comment: "Replace "auxiliary direction" with "surface-frame reference direction". Define a_s as the selected vector used only to fix the orientation of t_1 within the surface plane."

$$\begin{aligned} \tilde{t}_1 &= a_s - n_s \left(n_s^\top a_s \right), \\ t_1 &= \frac{\tilde{t}_1}{\|\tilde{t}_1\|}, \\ t_2 &= n_s \times t_1. \end{aligned} \tag{2.40}$$

Here, t_1 and t_2 are unit tangent vectors lying in the surface plane. The construction is valid as long as $\|\tilde{t}_1\| \neq 0$, which means that the auxiliary direction a_s must not be parallel to the normal n_s. To match the implemented controller and its parameter order, the surface task frame collects the two tangents first and the normal last,

$$R_{\text{surface}} = \begin{bmatrix} t_1 & t_2 & n_s \end{bmatrix}. \tag{2.41}$$

The column order is fixed: every surface-frame three-vector in the implementation uses $[\,\text{tangent } 1, \text{tangent } 2, \text{normal}\,]^\top$. In the gain representation below, the contact task uses this frame by setting $R_{\text{task}} = R_{\text{surface}}$. This makes stiffness values such as K_{t_1}, K_{t_2}, and K_n local surface-frame quantities in that same order.

2.6 Stiffness and Damping Representation

Direction-dependent stiffness and damping gains are most conveniently defined in a task frame whose axes coincide with the desired principal compliance directions. The corresponding matrices are diagonal in that frame but are not generally diagonal when expressed in the robot base frame.

Let $R_{\text{task}} \in SO(3)$ denote the orientation of the task frame relative to the base frame. Its columns contain the task-frame axes expressed in the base frame. The local gains $K_{p,\text{task}}$, $D_{p,\text{task}}$, $K_{R,\text{task}}$, and $D_{R,\text{task}}$ are rotated into the corresponding base-frame gain matrices:

$$\begin{aligned} K_{p,0} &= R_{\text{task}} K_{p,\text{task}} R_{\text{task}}^{\top}, \quad & D_{p,0} &= R_{\text{task}} D_{p,\text{task}} R_{\text{task}}^{\top}, \\ K_{R,0} &= R_{\text{task}} K_{R,\text{task}} R_{\text{task}}^{\top}, \quad & D_{R,0} &= R_{\text{task}} D_{R,\text{task}} R_{\text{task}}^{\top}. \end{aligned} \tag{2.42}$$

For an impedance without translation–rotation coupling in the task frame, the full task-frame stiffness K_{task} and damping D_{task} are

$$K_{\text{task}} = \begin{bmatrix} K_{p,\text{task}} & 0 \\ 0 & K_{R,\text{task}} \end{bmatrix}, \qquad D_{\text{task}} = \begin{bmatrix} D_{p,\text{task}} & 0 \\ 0 & D_{R,\text{task}} \end{bmatrix}. \tag{2.43}$$

The block rotation matrix T_R applies the task-frame rotation to the translational and rotational blocks:

$$T_R = \operatorname{diag}\left(R_{\text{task}}, R_{\text{task}}\right), \tag{2.44}$$

The corresponding full base-frame stiffness K_0 and damping D_0 are

$$K_0 = T_R K_{\text{task}} T_R^{\top}, \qquad D_0 = T_R D_{\text{task}} T_R^{\top}. \tag{2.45}$$

For the contact task, the task frame is the surface frame, whose orientation is

$$R_{\text{task}} = R_{\text{surface}} = \begin{bmatrix} t_1 & t_2 & n_s \end{bmatrix}. \tag{2.46}$$

For example, the local translational stiffness is defined as

$$K_{p,\text{task}} = \operatorname{diag}\left(K_{t_1}, K_{t_2}, K_n\right), \tag{2.47}$$

and its base-frame representation is

$$K_{p,0} = R_{\text{surface}} \operatorname{diag}\left(K_{t_1}, K_{t_2}, K_n\right) R_{\text{surface}}^{\top}. \tag{2.48}$$

The local rotational-stiffness entries K_{r,t_1}, K_{r,t_2}, and $K_{r,n}$ follow the same axis order:

$$K_{R,\text{task}} = \text{diag}\left(K_{r,t_1}, K_{r,t_2}, K_{r,n}\right), \tag{2.49}$$

and the damping matrices are constructed analogously. A change in the configured surface orientation therefore changes the corresponding base-frame gain matrices even when the local gain values remain unchanged.

During approach, contact set-up, and grinding, the reported contact runs used this surface-frame construction for both translational and rotational gain blocks. For set-up and grinding, the translational entries $[2000, 2000, 800]\,\mathrm{N/m}$ were assigned to $[t_1, t_2, n_s]$ before transformation to the base frame. The active gain matrices were based on the configured surface frame and were not rotated with the current end-effector orientation R_{EE}.

comment: “Remove the numerical K_p entries from the theory chapter. Introduce their values later with the controller configuration.”

2.7 Centre of Compliance

The task-frame transformation of eq. (2.45) rotates the principal stiffness directions while leaving the impedance reference point at the end-effector origin, identified here with the tool centre point (TCP). Independently of this orientation choice, the virtual spring can be centred at a different point. Shifting the impedance reference from the TCP to a centre of compliance p_c transforms a block-diagonal impedance at p_c into a coupled impedance at the TCP [19, 7, 24, 16]. This construction is applied during contact set-up to generate translation–rotation coupling that supports tool alignment.

Figure 2.2 shows this geometry for the two signs of the tangential lever.

Two levers occur in this thesis and they are separate quantities. The lever r_c of this section is a control parameter: it places the virtual spring and is selected before a run. The contact lever

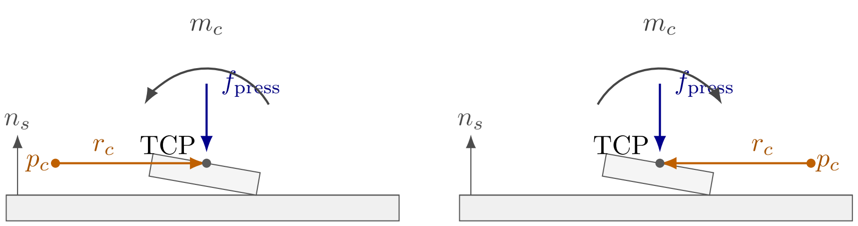
$$r_{\text{contact}} = p_{\text{TCP}} - p_{\text{contact}} \in \mathbb{R}^3 \tag{2.50}$$

runs from the physical contact point to the TCP and is an evaluation quantity: it carries an estimated external moment between those two points, as eq. (3.11) states. It follows from the tool geometry, whereas r_c follows from a setting, and neither appears in the other's expression.

Three forces occur in this thesis and they carry separate symbols, so that the one entering a cross product is never in doubt:

- f is the commanded Cartesian force of the control law, the translational half of the wrench F that eq. (2.36) maps to joint torques;
- f_{press} is the elastic press this section works with, $f_{\text{press}} = K_p e_p$, the part of the commanded force that the compression against the surface produces;
- $\Delta\hat{f}_{\text{ext}}$ is the model-estimated external force change since first contact, a measurement rather than a command, whose normal magnitude is F_n.

The moment m_c of this section is elastic and follows from f_{press} alone. It is the moment that press exerts at the TCP through the lever, namely the cross product of the vector running from the TCP to the centre of compliance with the press, in that order. The vector in this direction is $-r_c$, which is the reversal that produces the sign in eq. (2.57): written with r_c itself the cross product points the other way. The measured contact wrench, and the form it takes during the experiment, is treated in section 5.6.2.



(a) lever selected from the tilt (b) lever with the opposite sign

Figure 2.2: Moment produced by the compliance-centre lever at both lever signs.

For a dual-arm assembly task, Suomalainen et al. experimentally showed that compliance-centre placement and compliant-axis selection affect the alignment response [25]. This task-level result motivates the placement study in this thesis. The point-shift adjoint below remains derived from the spatial-transformation formulation.

The lever r_c from the centre of compliance p_c to the TCP, expressed in the base frame, is

$$r_c = p_{\text{TCP}} - p_c \in \mathbb{R}^3, \qquad p_{\text{TCP}} = p_{\text{EE}}. \tag{2.51}$$

Let $[r_c]_\times$ denote the skew-symmetric matrix satisfying $[r_c]_\times a = r_c \times a$. Under the small-angle approximation, the compliance-centre position error $e_{p,c}$ is

$$e_{p,c} = e_p + [r_c]_\times e_R. \tag{2.52}$$

The positive sign follows from the selected lever convention. The vector from the TCP to the centre of compliance p_c is $-r_c$, and therefore

$$e_R \times (-r_c) = [r_c]_\times e_R.$$

The compliance-centre error e_c stacks the translational error $e_{p,c}$ and the rotational error e_R. The point-shift adjoint $\mathrm{Ad}(r_c)$ maps the corresponding TCP error to this stacked error:

$$e_c = \begin{bmatrix} e_{p,c} \\ e_R \end{bmatrix} = \mathrm{Ad}(r_c) \begin{bmatrix} e_p \\ e_R \end{bmatrix}, \qquad \mathrm{Ad}(r_c) = \begin{bmatrix} \mathbf{I}_3 & [r_c]_\times \\ 0 & \mathbf{I}_3 \end{bmatrix}. \tag{2.53}$$

The point-shift adjoint $\mathrm{Ad}(r_c)$ changes the reference point of the impedance. By contrast, $T_R = \mathrm{diag}(R_{\text{task}}, R_{\text{task}})$ in eq. (2.45) changes its principal directions.

After any required task-frame rotation, the decoupled centre-of-compliance stiffness K_c and damping D_c are

$$K_c = \begin{bmatrix} K_p & 0 \\ 0 & K_R \end{bmatrix}, \qquad D_c = \begin{bmatrix} D_p & 0 \\ 0 & D_R \end{bmatrix}. \tag{2.54}$$

All blocks in eq. (2.54) are expressed in the base-frame orientation. The compliance-centre wrench F_c is

$$F_c = K_c e_c.$$

Transforming this power-conjugate wrench gives the corresponding TCP wrench F_{TCP}:

$$F_{\text{TCP}} = \mathrm{Ad}(r_c)^\top F_c.$$

The resulting TCP stiffness K_{TCP} is therefore

$$K_{\mathrm{TCP}} = \mathrm{Ad}(r_c)^\top K_c \mathrm{Ad}(r_c) = \begin{bmatrix} K_p & K_p[r_c]_\times \\ [r_c]_\times^\top K_p & K_R + [r_c]_\times^\top K_p [r_c]_\times \end{bmatrix}. \tag{2.55}$$

The TCP damping D_{TCP} uses the same point shift:

$$D_{\mathrm{TCP}} = \mathrm{Ad}(r_c)^\top D_c \mathrm{Ad}(r_c) = \begin{bmatrix} D_p & D_p[r_c]_\times \\ [r_c]_\times^\top D_p & D_R + [r_c]_\times^\top D_p [r_c]_\times \end{bmatrix}. \tag{2.56}$$

The off-diagonal blocks introduce translation–rotation coupling. A translational error can therefore generate a restoring moment, while a rotational error can generate a restoring force. This coupling is absent from the block-diagonal impedance defined directly at the TCP.

For a purely translational error, $e_R = 0$, the coupling reduces to a single term. With

$$f_{\mathrm{press}} = K_p e_p,$$

the elastic moment at the TCP is

$$m_c = [r_c]_\times^\top f_{\mathrm{press}} = -r_c \times f_{\mathrm{press}}. \tag{2.57}$$

A commanded tool orientation offset turns the tool about one of the surface tangents. The lever is chosen perpendicular to that tangent within the tangent plane, so that the moment the press generates opposes the commanded rotation. Reversing the commanded offset reverses the lever with it.

The right-handed surface frame satisfies

$$t_1 \times t_2 = n_s, \qquad n_s \times t_1 = t_2, \qquad n_s \times t_2 = -t_1. \tag{2.58}$$

For a selected tangential lever magnitude $\rho_c > 0$, the tangential compliance-centre lever $r_{c,t}$ is therefore

$$r_{c,t} = -\rho_c t_2 \quad \text{about } t_1, \qquad r_{c,t} = +\rho_c t_1 \quad \text{about } t_2. \tag{2.59}$$

Because the set-up motion is commanded primarily along the surface normal, the normal component dominates the Cartesian force, so the force can be approximated as

$$f \approx -F_n n_s, \tag{2.60}$$

where $F_n > 0$ is the normal-force magnitude. The remaining tangential components, such as those arising from tangential pose errors or contact friction, are treated as comparatively small. Substitution into eq. (2.57) gives

$$m_c \approx -F_n \rho_c t_1 \quad \text{about } t_1, \qquad m_c \approx -F_n \rho_c t_2 \quad \text{about } t_2. \tag{2.61}$$

The generated moment therefore acts opposite to the commanded rotation in both cases. Reversing the orientation offset reverses both the selected lever and the generated moment.

The coupling vanishes for $r_c = 0$. Conversely, if K_p is nonsingular, zero coupling implies $r_c = 0$. Since $\mathrm{Ad}(r_c)$ is invertible, the congruence transformation also preserves definiteness:

$$K_c \succeq 0 \implies K_{\mathrm{TCP}} \succeq 0, \qquad K_c \succ 0 \implies K_{\mathrm{TCP}} \succ 0. \tag{2.62}$$

The same statements apply to D_c and D_{TCP}. Negative off-diagonal entries therefore represent coupling terms rather than negative stiffness directions.

The congruence transformation preserves the symmetry and definiteness of the source gain matrices. Closed-loop robot–controller–contact behaviour with phase switching was assessed experimentally under the operating conditions reported in Chapter 4.

This formulation is a local small-motion model in which r_c is held fixed while the wrench is evaluated. In the implemented set-up mode, the selected lever remains constant, and the mapped gain matrices are therefore constant during the corresponding motion segment. If p_c or r_c were updated continuously, the gain matrices would become configuration dependent.

The local linearised centre of compliance p_c is consequently a selected point rather than a measured physical pivot. It need not coincide with the physical contact point p_{contact}. The observed motion also depends on the finite rotational stiffness, the lever direction, the environmental constraint, friction, and the additional active controller terms introduced later.

2.8 Null-Space Torque

The Franka Emika Panda has seven joints, while the controlled end-effector task contains six components: three translational and three rotational. The robot is therefore kinematically redundant. The Cartesian impedance behaviour at the end-effector is the primary objective, and the remaining joint-space freedom can be used for a secondary objective.

In this thesis, the secondary joint-space action consists of a null-space damping contribution and an active singular-value conditioning contribution. Both are projected using the joint-torque null-space projector N_τ. The required null-space direction, pseudoinverse, and projector are introduced before the two torque contributions are defined.

2.8.1 SVD-Based Null-Space Characterisation

The geometric Jacobian $J(q)$ relates the current joint velocity $\dot{q}$ to the end-effector velocity $\dot{x}$:

$$\dot{x} = J(q)\dot{q}, \qquad J(q) \in \mathbb{R}^{6\times 7}, \tag{2.63}$$

where

$$\dot{x} = \begin{bmatrix} \dot{p}_{\text{EE}} \\ \omega_{\text{EE}} \end{bmatrix} \in \mathbb{R}^6 \tag{2.64}$$

contains the three linear and three angular velocity components of the end-effector.

A joint velocity $\dot{q}$ belongs to the kinematic null space when

$$J(q)\dot{q} = 0. \tag{2.65}$$

Such a joint velocity $\dot{q}$ changes the robot configuration while preserving the end-effector position and orientation to first order. The Jacobian $J(q)$ is the first-order term in the expansion of the forward kinematics about the current configuration, so a finite displacement along the null space leaves a pose error of second order in its magnitude. When $J(q)$ has full row rank, the null space of the 6×7 Jacobian $J(q)$ is one-dimensional.

The singular value decomposition (SVD) is applied to the Jacobian $J(q)$ at the current joint configuration q [3, 6]:

$$J(q) = U_J \Sigma_J V_J^\top, \qquad U_J \in \mathbb{R}^{6\times 6}, \quad \Sigma_J \in \mathbb{R}^{6\times 7}, \quad V_J \in \mathbb{R}^{7\times 7}. \tag{2.66}$$

The Jacobian $J(q)$ is the input to the decomposition. The SVD returns the matrices U_J, Σ_J, and V_J, with the singular values contained in Σ_J.

The columns of V_J form an orthonormal basis of the seven-dimensional joint-velocity space. The joint velocity $\dot{q}$ can therefore be expressed as

$$\dot{q} = V_J c = \sum_{i=1}^{7} c_i v_i, \tag{2.67}$$

where v_i is the i-th column of V_J, and $c \in \mathbb{R}^7$ contains the components of $\dot{q}$ along these basis directions.

Similarly, the columns of U_J form an orthonormal basis of the six-dimensional end-effector-velocity space. Substituting $\dot{q} = V_J c$ into eq. (2.63) gives

$$\dot{x} = J(q) V_J c = U_J \Sigma_J c. \tag{2.68}$$

Thus, V_J provides the basis directions in which the joint velocity $\dot{q}$ is expressed, while U_J provides the corresponding basis directions in which the end-effector velocity $\dot{x}$ is expressed. The matrix Σ_J scales the mapping between these directions.

For the 6×7 Jacobian $J(q)$, the ordered Jacobian singular values σ_i occupy the diagonal of the singular-value matrix:

$$\Sigma_J = \begin{bmatrix} \operatorname{diag}(\sigma_1, \sigma_2, \ldots, \sigma_6) & 0_{6\times 1} \end{bmatrix}, \tag{2.69}$$

where

$$\sigma_1 \geq \sigma_2 \geq \cdots \geq \sigma_6 \geq 0. \tag{2.70}$$

For the first six basis directions,

$$Jv_i = \sigma_i u_i, \qquad i = 1, \ldots, 6, \tag{2.71}$$

where u_i is the i-th column of U_J. A joint velocity $\dot{q}$ in the direction v_i,

$$\dot{q} = c_i v_i, \tag{2.72}$$

therefore produces the end-effector velocity

$$\dot{x} = c_i \sigma_i u_i. \tag{2.73}$$

The singular value σ_i scales the mapping from the joint-velocity basis direction v_i to the corresponding end-effector-velocity basis direction u_i.

The seventh column of V_J corresponds to the zero column of Σ_J and therefore satisfies

$$Jv_7 = 0. \tag{2.74}$$

A null-space joint velocity $\dot{q}$ may be scaled by the scalar amplitude c_7:

$$\dot{q} = c_7 v_7 \tag{2.75}$$

therefore changes the joint configuration without producing an end-effector velocity. When the Jacobian $J(q)$ has full row rank, v_7 spans its one-dimensional kinematic null space.

The minimum task-producing singular value $\sigma_{\min}$ is

$$\sigma_{\min} = \sigma_6. \tag{2.76}$$

The quantities v_7 and σ_6 have different purposes. The vector v_7 represents the redundant joint-motion direction, whereas σ_6 describes the weakest of the six mappings from joint velocity $\dot{q}$ to end-effector velocity. As σ_6 approaches zero, one end-effector velocity direction becomes increasingly difficult to produce, indicating an approach to a loss of Jacobian $J(q)$ row rank. In this thesis, $\sigma_{\min}$ is used to compare the conditioning of nearby joint configurations and to select the direction of the conditioning torque.

2.8.2 SVD-Based Pseudoinverse

The Jacobian pseudoinverse J^+ maps end-effector-velocity components back to the corresponding joint-velocity directions. Its calculation uses the singular directions obtained from the SVD.

Small singular values lead to large reciprocal values and can amplify numerical errors. A relative singular-value threshold is therefore applied. Let $\rho \geq 0$ denote the configured relative SVD tolerance. The corresponding absolute singular-value threshold σ_{thr} is calculated from the largest singular value σ_1:

$$\sigma_{\text{thr}} = \rho \sigma_1. \tag{2.77}$$

A singular direction is retained when

$$\sigma_i > \sigma_{\text{thr}}. \tag{2.78}$$

The Jacobian pseudoinverse J^+ is defined as

$$J^+ = \sum_{\substack{i=1 \\ \sigma_i > \sigma_{\text{thr}}}}^{6} \frac{1}{\sigma_i} v_i u_i^\top \in \mathbb{R}^{7\times 6}. \tag{2.79}$$

For each retained singular direction, J^+ maps an end-effector-velocity component along u_i to the corresponding joint-velocity direction v_i and scales it by $1/\sigma_i$. Directions at or below σ_{thr} receive a reciprocal value of zero.

2.8.3 Null-Space Projectors

The pseudoinverse is used to separate a joint-space vector into a task-related component and a null-space component. For an arbitrary joint velocity $\dot{q}$, this decomposition is

$$\dot{q} = \underbrace{J^{+}J\dot{q}}_{\text{task-related component}} + \underbrace{\left(I_7 - J^{+}J\right)\dot{q}}_{\text{null-space component}} . \tag{2.80}$$

The product $J^{+}J$ projects $\dot{q}$ onto the joint-space directions that contribute to the end-effector velocity. Since J includes all three linear and all three angular velocity components, this task-related component accounts for the complete controlled end-effector pose at the velocity level.

The complementary joint-velocity projector N_q is

$$N_q = I_7 - J^{+}J \in \mathbb{R}^{7\times 7}. \tag{2.81}$$

Applying N_q to an arbitrary joint velocity $\dot{q}$ gives the projected null-space velocity $\dot{q}_{\text{null}}$:

$$\dot{q}_{\text{null}} = N_q\dot{q}. \tag{2.82}$$

When J has full row rank and all six task-producing singular directions are retained,

$$J\dot{q}_{\text{null}} = JN_q\dot{q} = 0. \tag{2.83}$$

The projected joint velocity $\dot{q}_{\text{null}}$ can therefore change the robot configuration while preserving the end-effector position and orientation to first order.

A corresponding decomposition is used for a secondary joint torque τ_0. The joint-torque projector N_τ is the transpose of the velocity projector:

$$N_\tau = N_q^{\top} = I_7 - J^{\top}J^{+\top} \in \mathbb{R}^{7\times 7}. \tag{2.84}$$

The secondary torque can then be written as

$$\tau_0 = \underbrace{J^{\top}J^{+\top}\tau_0}_{\text{task-related component}} + \underbrace{N_\tau\tau_0}_{\text{null-space component}} . \tag{2.85}$$

The first component lies in the joint-torque subspace associated with Cartesian wrenches

F through $J^\top$. The second component is retained as the secondary null-space torque τ_{null}:

$$\tau_{\text{null}} = N_\tau \tau_0. \tag{2.86}$$

The transpose relation follows from the mechanical power $\tau^\top \dot{q}$ between joint torques τ and joint velocities $\dot{q}$. Thus, N_q is applied to joint velocities $\dot{q}$, while N_τ is applied to secondary joint torques τ_0.

For the SVD-based pseudoinverse, $J^+ J$ is an orthogonal and symmetric projector. Hence,

$$N_q = N_\tau \tag{2.87}$$

holds analytically. The separate symbols distinguish the quantity on which the projector acts. In the implementation, N_τ is additionally symmetrised to reduce floating-point asymmetry.

When J has full row rank and all six task-producing singular directions are retained,

$$N_q = N_\tau = v_7 v_7^\top. \tag{2.88}$$

In this case, projection retains only the component along the redundant direction v_7. When a singular value lies at or below the selected threshold, its associated right-singular direction is also assigned to the numerical null space.

2.8.4 Null-Space Damping Torque

The current joint velocity $\dot{q}$ is separated into task-related and null-space components according to eq. (2.80). The null-space damping torque τ_d acts on the component $N_q \dot{q}$:

$$\tau_d = -d_{\text{null}} N_\tau \dot{q} = -d_{\text{null}} N_q \dot{q}, \tag{2.89}$$

where $d_{\text{null}} \geq 0$ is the null-space damping coefficient. The negative sign makes the torque oppose motion along the redundant joint-space direction and dissipate the associated motion.

2.8.5 Singular-Value Conditioning Torque

The second contribution applies an active torque along the redundant direction v_7. Its sign is selected by comparing $\sigma_{\min}$ at two nearby joint configurations.

The probe angle α_{probe} specifies the joint-space step from the current configuration q. The sampled configurations in the positive and negative null directions are q_+ and q_-:

$$q_+ = q + \alpha_{\text{probe}} v_7, \qquad q_- = q - \alpha_{\text{probe}} v_7. \tag{2.90}$$

The Jacobian $J(q)$ is recalculated at both configurations. The minimum singular values σ_+ and σ_- correspond to q_+ and q_-, respectively:

$$\sigma_+ = \sigma_{\min}(J(q_+)), \qquad \sigma_- = \sigma_{\min}(J(q_-)). \tag{2.91}$$

The sampled singular-value difference Δ_σ is

$$\Delta_\sigma = \sigma_+ - \sigma_-. \tag{2.92}$$

The direction selector s is defined as

$$s = \begin{cases} +1, & \Delta_\sigma > \delta_\sigma, \\ -1, & \Delta_\sigma < -\delta_\sigma, \\ 0, & |\Delta_\sigma| \leq \delta_\sigma, \end{cases} \tag{2.93}$$

where $\delta_\sigma \geq 0$ is the comparison deadband. Within this deadband, the selector is set to zero and the conditioning contribution is suppressed.

The selected direction is invariant to the sign convention of v_7 returned by the SVD. Reversing v_7 exchanges q_+ and q_- and reverses s, so the product sv_7 remains unchanged.

The conditioning-torque magnitude k_σ is specified independently of the probe angle.

The vector v_7 represents the instantaneous null-space direction at the current configuration. For a finite probe angle, q_+ and q_- provide local approximations of the nonlinear constant-pose self-motion. The corresponding preservation of the end-effector pose is therefore a first-order property of the sampled direction.

The selector uses the sign of Δ_σ to choose the sampled direction with the larger value of $\sigma_{\min}$. This produces a two-point directional conditioning strategy based on local comparison rather than on the magnitude of the sampled variation.

The vector sv_7 defines the selected secondary joint-space direction. Projecting it with N_τ gives the singular-value conditioning torque τ_σ:

$$\tau_\sigma = k_\sigma N_\tau \left(sv_7\right), \tag{2.94}$$

where $k_\sigma \geq 0$ specifies the commanded torque magnitude.

When the Jacobian has full row rank and all six task-producing singular directions are retained, v_7 lies in the one-dimensional null space and

$$N_\tau v_7 = v_7. \tag{2.95}$$

The conditioning contribution therefore acts along the selected redundant joint direction, while the damping contribution opposes motion within the projected subspace.

2.8.6 Complete Null-Space Torque

Combining the damping and conditioning contributions gives

$$\tau_{\text{null}} = \tau_d + \tau_\sigma = -d_{\text{null}} N_\tau \dot{q} + k_\sigma N_\tau \left(sv_7\right) \in \mathbb{R}^7. \tag{2.96}$$

When the Jacobian $J(q)$ has full row rank, its null space is one-dimensional and is spanned by v_7. If the Jacobian loses row rank, the null space becomes multidimensional and the projector includes the additional numerical-null-space directions. The damping contribution acts on motion in all these directions through $N_\tau \dot{q}$. The implemented conditioning term retains its two probes along $+v_7$ and $-v_7$.

The activation of the damping and conditioning contributions is a controller configuration choice. The available combinations and the configuration used in the reported experiments are described in section 3.4.

2.9 Complete Joint-Torque Command

The complete joint-torque command τ_{cmd} combines the Cartesian impedance torque τ_{cart}, the null-space torque τ_{null}, and the Coriolis compensation τ_c:

$$\tau_{\text{cmd}} = \tau_{\text{cart}} + \tau_{\text{null}} + \tau_c(q, \dot{q}) = J^\top(q)F + \tau_{\text{null}} + \tau_c(q, \dot{q}). \tag{2.97}$$

Here, $\tau_{\text{cart}} = J^\top(q)F$ is the Cartesian impedance torque, τ_{null} is the secondary null-space torque, and $\tau_c(q, \dot{q})$ is the Coriolis and centrifugal compensation obtained from the robot model and added in the control callback. A separate gravity term is not added because gravity compensation is handled internally by the Franka Control Interface, as described in Section 2.4.3.

Chapter 3

Controller Design and Implementation

GREEN — revised and confirmed. The chapter was revised in full and every paragraph is settled. No further editing is to be applied to it, and any remaining concern is raised as a comment rather than as a change.

comment: "In the settled prose, use "commanded tool orientation offset" for the deliberate angular command and "commanded rotation axis" or "commanded rotation direction" for its components. Retain "tilt" only for a physical inclination or a calibrated surface angle."

This chapter describes the real-time implementation of the Cartesian impedance controller developed in Chapter 2. The controller is written in C++ using libfranka and Eigen and is executed through the Franka Control Interface (FCI) with a nominal control period of 1 ms.

The software combines the Cartesian impedance law, the SVD-based null-space controller, the configured surface and tool geometry, phase-dependent stiffness and damping, and the virtual centre of compliance p_c. These components are coordinated by a phase state machine that orients the tool, approaches the surface, applies the set-up preload, and executes the grinding motion. Additional operating modes provide Cartesian pose holding, set-up-impedance holding, and manual guidance.

The following sections describe the software architecture, the real-time control loop, the implemented Cartesian torque τ_{cart} and null-space torque τ_{null} contributions, the geometric representation of the surface and tool, the phase-dependent impedance parameters, and the execution of the control sequence.

3.1 Software Architecture and Program Execution

The controller is divided into modules for configuration handling, control calculations, real-time execution, operator interaction, and run evaluation. The main program coordinates these modules and manages the execution of successive controller sessions.

Table 3.1: Software modules of the surface-grinding controller.

Module	Responsibility
`main.cpp`	Connects to the robot, prepares the controller session, starts the selected operating mode, and initiates post-run reporting.
`include/`	Defines the controller configuration, mathematical types, run-time data structures, and interfaces shared between the software modules.
`src/config/`	Reads the parameter files and assigns their values to the typed controller configuration.
`src/control/`	Implements the Cartesian impedance law, null-space torque, phase-dependent gain construction, surface and tool geometry, inertia-scaled damping, and robot support functions.
`src/runtime/`	Executes the phase state machine and the 1 kHz joint-torque control callback.
`src/interface/`	Provides the start-up menu, gripper and tool-handling actions, manual guidance, live parameter commands, and terminal output.
`src/report/`	Stores and writes the recorded controller signals and evaluates the completed run.
`tools/measure_plane.cpp`	Provides a separate calibration utility for estimating one point and the orientation of the physical surface plane.
`tools/measure_tools_axis.cpp`	Estimates the grinding-face normal in the end-effector frame from seated tool orientations.

The parameter directory is resolved relative to the controller executable. The complete configuration is distributed over 15 files in the `params/` directory. These files contain the robot connection and logging settings, collision thresholds, gripper parameters, initial posture, surface and tool geometry, phase settings, Cartesian gains, inertia-scaled-damping settings, virtual-compliance-centre parameters, null-space parameters, disturbance-test settings, and manual-guidance settings.

At program start, the parameter files are read to obtain the robot address and initial controller configuration. The robot connection is then established, error recovery is requested, the selected collision behaviour is assigned, and the libfranka robot model is loaded. A separate keyboard thread is also started to receive operator requests during the controller session. These requests are transferred through atomic variables shared with the control loop.

Before every new session, all parameter files are read again and the resulting values are stored in a new `ControllerConfig` object. This permits controller and experiment parameters to be changed between runs without recompiling the program. The loader requires all 15 parameter files and rejects duplicate parameter names across them.

The start-up interface provides four controller modes:

- the complete tool-orientation, surface-approach, set-up, and grinding sequence;
- Cartesian pose holding with the dedicated hold impedance;
- Cartesian pose holding with the set-up impedance; and
- manual guidance from the current robot pose.

The same interface provides the robot-preparation and gripper actions required before torque control. The robot can be moved to the configured initial joint posture, the gripper can be opened or closed, and the tool can be collected from or returned to its configured holder posture. When manual guidance is selected, the subsequent sequence or hold mode begins from the pose captured after hand-guided positioning.

Once the operating mode and initial state have been selected, the phase-dependent stiffness and damping matrices are constructed from the active configuration. These matrices, together with the robot and model interfaces, are passed to the real-time control loop.

Experimental signals are stored during control in a bounded ring buffer whose memory is allocated before the torque callback begins. After the controller session has ended, the stored samples are arranged in chronological order, written to the configured CSV file, and used to generate the terminal run summary. A new session can then be started from the start-up menu with a freshly loaded configuration.

3.2 Real-Time Control Loop

The function `runControlLoop` prepares one controller session and executes the libfranka joint-torque callback. It receives the active controller configuration, the robot and model interfaces, the phase-dependent impedance gains, and the operator-input signals.

Before torque control begins, the current robot state is read once. The reported end-effector transformation provides the initial TCP position p_{start} and orientation R_{start}, while the measured joint configuration provides the initial posture q_{start}. These values initialise the desired pose of the selected sequence or hold mode.

The initial state also provides an external-wrench reference for subsequent contact evaluation. In addition, the controller initialises the configured surface point, the surface frame, the target tool orientation, the descent direction, and the four tool-face corners. The initial phase, phase clocks, operator gates, set-up references, damping cache, and recording buffer are prepared before entering the real-time callback.

The controller is passed to `robot.control` as a joint-torque callback. For every control cycle, the callback receives the current `RobotState` and the elapsed cycle duration and returns the seven commanded joint torques τ_{cmd}, as shown in listing 3.1.

Listing 3.1: Registration of the joint-torque callback through `libfranka`.

```
robot.control(
    [&](const franka::RobotState& state,
        franka::Duration period) -> franka::Torques {

      // Real-time state and torque calculations

      return franka::Torques(tau_array);
    });
```

Figure 3.1 shows the signals combined into the commanded joint torque τ_{cmd} during one control cycle.

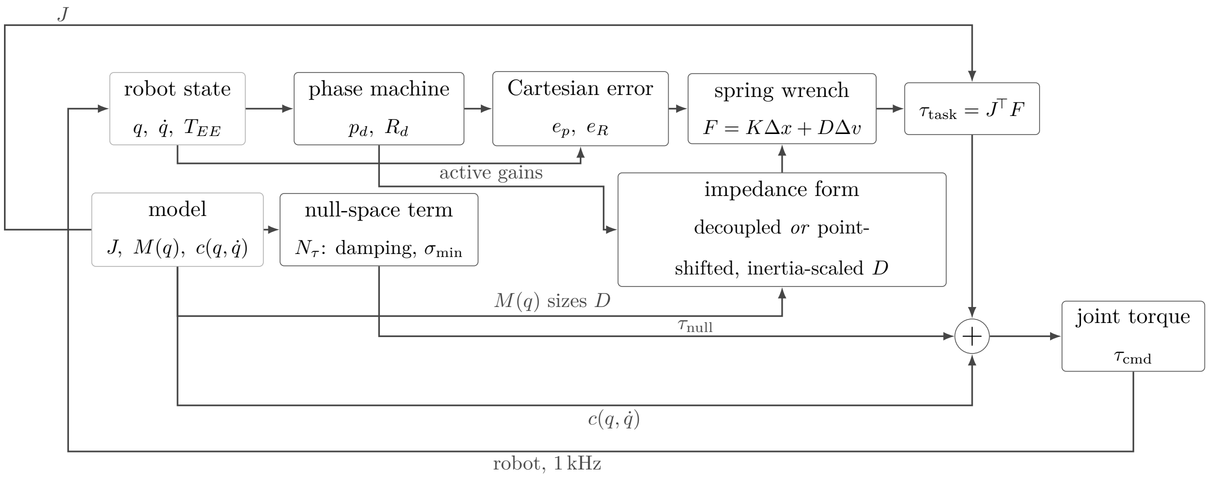


Figure 3.1: Torque contributions combined in one control cycle.

During each callback, the elapsed controller time is updated from the period reported by libfranka. The joint state, end-effector Jacobian $J(q)$, Cartesian velocity $\dot{x}$, end-effector pose, and model-estimated external wrench are then obtained as shown in listing 3.2.

Listing 3.2: Acquisition of the robot-state and model quantities at the beginning of one real-time torque callback.

```
time += period.toSec();

Map<const Vec7> dq(state.dq.data());
Map<const Vec7> q_current(state.q.data());

std::array<double, 42> jacobian_array =
    model.zeroJacobian(Frame::kEndEffector, state);
Map<const Mat6x7> J(jacobian_array.data());

const Vec6 xdot = J * dq;
const Vec3 pdot = xdot.head<3>();
const Vec3 omega = xdot.tail<3>();

Map<const Mat4x4> T_EE(state.O_T_EE.data());
const Vec3 p_EE = T_EE.block<3, 1>(0, 3);
const Mat3 R_EE = T_EE.block<3, 3>(0, 0);

Map<const Vec6> external_wrench(
    state.O_F_ext_hat_K.data());
const Vec3 external_force = external_wrench.head<3>();
const Vec3 external_moment = external_wrench.tail<3>();
```

The vectors `q_current` and `dq` contain the measured joint configuration and velocity. The call to `model.zeroJacobian` returns the 6×7 geometric Jacobian $J(q)$ of the configured libfranka end-effector frame. Multiplication by the measured joint velocity $\dot{q}$ provides the TCP linear velocity `pdot` and angular velocity `omega`, both expressed in the robot base frame.

The field `state.O_T_EE` provides the current end-effector transformation in the base frame. Its translational and rotational blocks are assigned to `p_EE` and `R_EE`, respectively. The controller therefore uses the pose reported by libfranka rather than reconstructing it from a separate forward-kinematic model.

The field `state.O_F_ext_hat_K` provides the model-estimated external force $\hat{f}_{\mathrm{ext}}$ and moment. During surface approach, the transition to set-up is determined geometrically from the tool-face clearance. At this transition, the current estimates are stored as the external-wrench reference. During set-up, the resulting external-moment change is used for the moment-based exit condition. The force and moment estimates are also included in the terminal output, set-up evaluation, and recorded experimental data.

Operator requests are exchanged with the callback through atomic variables. They support transitions between the phase sequence, Cartesian pose hold, set-up-impedance hold, and manual guidance. The same interface receives the gate confirmations and stop request. During the dedicated hold modes, selected null-space and set-up-impedance parameters can also be adjusted. An accepted set-up-impedance change rebuilds the corresponding phase gains and damping cache.

The phase state machine updates the desired TCP position p_d, desired linear velocity $\dot{p}_d$, and desired orientation R_d. Their values depend on the active phase:

- tool orientation maintains the initial TCP position while advancing the orientation command toward the configured target;
- surface approach moves the desired position along the descent direction;
- set-up advances the selected contact point into the surface;
- grinding maintains the achieved preload and can add a tangential trajectory; and
- pose hold regulates the captured Cartesian pose.

After the reference has been updated, the callback calculates the Cartesian position error e_p and orientation error e_R using the conventions introduced in Section 2.2.4. The phase-dependent stiffness and damping matrices are then selected. When a phase gate is active, the dedicated gate holding gains replace the normal phase gains.

The displacement error Δx and velocity error Δv supplied to the impedance calculation are

$$\Delta x = \begin{bmatrix} e_p \\ e_R \end{bmatrix}, \qquad \Delta v = \begin{bmatrix} \dot{p}_d - \dot{p}_{\mathrm{EE}} \\ -\omega_{\mathrm{EE}} \end{bmatrix}. \tag{3.1}$$

The Cartesian impedance calculation returns the commanded wrench F

$$F = \begin{bmatrix} f \\ m \end{bmatrix} \in \mathbb{R}^6. \tag{3.2}$$

Depending on the active mode, this wrench is obtained from either the decoupled impedance law of eq. (2.37) or the point-shifted set-up impedance described in Section 3.7.

The Cartesian torque contribution τ_{cart} is

$$\tau_{\text{cart}} = J^\top(q)F. \tag{3.3}$$

The SVD-based null-space routine calculates the secondary torque τ_{null} according to the selected null-space mode. During the Cartesian pose-hold disturbance experiment described in Section 4.7, an additional joint-torque contribution τ_{dist} is applied. For all other controller operation, the disturbance torque is set to zero, $\tau_{\text{dist}} = 0$.

The Coriolis and centrifugal torque τ_c is obtained from the libfranka model. The complete torque command τ_{cmd} of an impedance-controlled cycle is therefore

$$\tau_{\text{cmd}} = \tau_{\text{cart}} + \tau_{\text{null}} + \tau_{\text{dist}} + \tau_c(q, \dot{q}). \tag{3.4}$$

Manual guidance follows the separate joint-torque command τ_{guide}:

$$\tau_{\text{guide}} = \tau_c(q, \dot{q}) - d_{\text{guide}}\dot{q}, \tag{3.5}$$

where d_{guide} is the configured joint-damping coefficient. After the guided pose is captured, its position, orientation, joint configuration, and the external-wrench reference initialise the resumed sequence or hold mode.

For impedance-controlled cycles, the selected state, reference, error, wrench, phase, null-space, disturbance, and joint-torque quantities are sampled at the configured interval and copied into a preallocated ring buffer. When the buffer capacity is reached, the most recent samples replace the oldest entries.

The callback ends when the configured run duration expires, an operator stop is requested, the maximum approach distance is reached, or libfranka terminates the motion.

After torque control has ended, the ring-buffer entries are placed in chronological order and returned together with the final controller state. File writing and post-run evaluation are performed outside the real-time callback.

3.3 Implemented Cartesian Impedance Controller

The mathematical formulation of the Cartesian impedance controller is given in Section 2.4. This section describes how the pose errors, phase-dependent gains, and Cartesian wrench F are evaluated in the software.

The callback first selects the desired orientation associated with the active phase. During tool orientation, the desired orientation follows the interpolated orientation command. Surface approach uses the final tool-target orientation. At the transition to set-up, the current orientation is captured and held constant throughout set-up and grinding. Cartesian pose hold uses the pose stored for the active hold mode.

The position error e_p and orientation error e_R are evaluated as shown in listing 3.3.

Listing 3.3: Calculation of the phase-dependent Cartesian pose error.

```
const Vec3 e_p = desired.p_d - p_EE;

const Mat3& R_d_used =
    after_contact
        ? R_contact_start
        : (phase == ControlPhase::kPoseHold)
              ? R_d
              : (phase == ControlPhase::kToolOrientation)
                    ? R_orient_command
                    : R_base_tool_target;

const Vec3 e_R = applyRotationalAxisMask(
    params,
    orientationError(R_EE, R_d_used),
    R_base_surface);
```

The function `orientationError` forms the current-to-desired relative rotation and converts it to an axis–angle vector. The resulting local rotation vector is transformed to the robot base frame:

Listing 3.4: Base-frame orientation-error calculation.

```
Mat3 R_error = R_current.transpose() * R_desired;
Eigen::AngleAxisd angle_axis(R_error);

return R_current
       * angle_axis.axis()
       * angle_axis.angle();
```

A rotation angle below the numerical tolerance returns a zero vector. The orientationerror convention therefore corresponds to the corrective base-frame error e_R defined in Section 2.2.4.

The rotational mask is applied after the error has been transformed to the base frame. The function first expresses e_R in the surface frame, selects the enabled components about t_1, t_2, and n_s, and transforms the result back to the base frame. The active configuration enables all three components. The mask acts on the rotational spring error, while rotational damping continues to use the measured angular velocity and the selected damping matrix.

The callback then combines the translational and rotational quantities into the six-dimensional displacement and velocity-error vectors used by the impedance routine. The desired angular velocity is zero in the implemented control sequence.

Listing 3.5: Assembly and evaluation of the Cartesian impedance command.

```
Vec6 dx;
dx.head<3>() = e_p;
dx.tail<3>() = e_R;

Vec6 dv;
dv.head<3>() = desired.pdot_d - pdot;
dv.tail<3>() = -omega;

const Vec6 wrench =
    computeCartesianImpedanceWrench(
        params, phase,
        Kp_used, Dp_used,
        KR_used, DR_used,
        R_base_surface,
        dx, dv, R_EE);

const Vec7 tau_task = J.transpose() * wrench;
```

The variables `Kp_used`, `Dp_used`, `KR_used`, and `DR_used` contain the gains selected for the current phase. Their construction and phase assignment are described in Section 3.6. The implementation variable `wrench` follows the force–moment order used in Chapter 2, while `tau_task` corresponds to the Cartesian impedance torque τ_{cart}.

The function `computeCartesianImpedanceWrench` selects between the decoupled Cartesian impedance and the point-shifted impedance associated with the virtual centre of compliance p_c. The decoupled branch is implemented by applying the translational and rotational gain blocks separately:

Listing 3.6: Decoupled Cartesian impedance branch.

```
Vec6 wrench;
wrench.head<3>() =
    Kp * dx.head<3>() + Dp * dv.head<3>();

wrench.tail<3>() =
```

```
    KR * dx.tail<3>() + DR * dv.tail<3>();

return wrench;
```

This branch is used during tool orientation, surface approach, grinding, and the standard Cartesian pose hold. It is also used during set-up when the virtual centre of compliance p_c is disabled.

The coupled branch is selected when the virtual centre of compliance p_c is enabled during set-up or during the dedicated set-up-impedance hold mode:

Listing 3.7: Selection of compliance-centre coupling.

```
const bool coupled =
    params.use_virtual_compliance_center &&
    (phase == ControlPhase::kSetup ||
     (phase == ControlPhase::kPoseHold &&
      params.use_setup_impedance_hold));
```

In this branch, the translational and rotational gain blocks are assembled into spatial 6×6 matrices and shifted from the configured centre of compliance p_c to the TCP. The construction of these matrices and the two implemented definitions of the compliance-centre lever r_c are described in Section 3.7.

The wrench F returned by either branch is mapped to joint torque τ_{cart} by `J.transpose() * wrench`. The resulting `tau_task` contribution is subsequently combined with the null-space, disturbance, and Coriolis torque contributions in the real-time callback.

3.4 SVD-Based Null-Space Control

The mathematical formulation of the null-space projector, projected damping, and singular-value conditioning torque is presented in Section 2.8. This section describes their implementation in the function `computeNullspaceTorque`, which is evaluated during every impedance-controlled callback.

The function receives the current geometric Jacobian $J(q)$, measured joint velocity $\dot{q}$, robot state, model interface, and active null-space parameters. The current joint configuration is mapped from the robot state, and the full singular value decomposition of the 6×7 Jacobian $J(q)$ is calculated. The pseudoinverse J^+ and joint-torque null-space projector N_τ are then constructed as shown in listing 3.8.

Listing 3.8: Construction of the SVD-based pseudoinverse and joint-torque null-space projector.

```
Map<const Vec7> q_current(state.q.data());

Eigen::JacobiSVD<Mat6x7> svd_current(
    J, Eigen::ComputeFullU | Eigen::ComputeFullV);

const Vec6 singular_values =
    svd_current.singularValues();

const double sigma_max =
    singular_values.maxCoeff();

const double svd_cutoff =
    std::max(
        0.0,
        params.nullspace_svd_relative_tolerance)
    * sigma_max;

Mat7x6 J_pinv = Mat7x6::Zero();

for (int i = 0; i < 6; ++i) {
  if (singular_values(i) > svd_cutoff) {
    J_pinv +=
        (1.0 / singular_values(i))
        * svd_current.matrixV().col(i)
        * svd_current.matrixU().col(i).transpose();
  }
}

Mat7x7 N_tau =
    Mat7x7::Identity()
    - J.transpose() * J_pinv.transpose();

N_tau =
    0.5 * (N_tau + N_tau.transpose());
```

The relative tolerance `nullspace_svd_relative_tolerance` is multiplied by the largest singular value of the current Jacobian $J(q)$. This produces the singular-value threshold used during the current callback. Each singular direction above this threshold contributes to the pseudoinverse with the reciprocal $1/\sigma_i$. The remaining reciprocal entries are set to zero.

The implementation forms the joint-torque projector directly according to the expression derived in Section 2.8.3. Averaging the projector with its transpose removes small numerical asymmetries caused by floating-point calculations.

The measured joint velocity $\dot{q}$ is subsequently projected into the numerical null space. The active `NullspaceMode` determines which secondary torque contributions are evaluated:

Listing 3.9: Projection of the joint velocity and selection of the null-space damping contribution.

```
const Vec7 dq_nullspace = N_tau * dq;

if (params.nullspace_mode ==
    NullspaceMode::kOff) {
  return Vec7::Zero();
}

const bool damping_enabled =
```

```
    params.nullspace_mode ==
        NullspaceMode::kDampingOnly
    ||
    params.nullspace_mode ==
        NullspaceMode::kDampingAndSigma;

Vec7 tau_damping = Vec7::Zero();

if (damping_enabled) {
  tau_damping.noalias() =
      -params.nullspace_damping
      * dq_nullspace;
}

if (params.nullspace_mode ==
    NullspaceMode::kDampingOnly) {
  return tau_damping;
}

const bool sigma_only =
    params.nullspace_mode ==
        NullspaceMode::kSigmaOnly;

const Vec7 sigma_fallback =
    sigma_only
        ? Vec7::Zero()
        : tau_damping;
```

The four selectable modes are defined in the controller as

Listing 3.10: Selectable null-space operating modes.

```
enum class NullspaceMode {
  kOff = 0,
  kDampingOnly = 1,
  kSigmaOnly = 2,
  kDampingAndSigma = 3
};
```

The disabled mode returns zero secondary torque. The damping-only mode returns the negative projected joint-velocity damping. The sigma-only mode evaluates the singular-value conditioning contribution, while the combined mode adds both contributions.

For singular-value conditioning, the seventh column of the current right-singular-vector matrix is selected and normalised. This vector corresponds to the redundant direction v_7 introduced in Section 2.8.1. Two nearby joint configurations are then formed along its positive and negative directions.

Listing 3.11: Construction and evaluation of the two sampled joint configurations.

```
Vec7 n =
    svd_current.matrixV().col(6);

if (n.norm() <= 1e-9) {
  return sigma_fallback;
}

n.normalize();

const double alpha =
    std::abs(
        params.nullspace_probe_step_rad);

if (alpha <= 1e-12) {
  return sigma_fallback;
}
```

```
const Array7 q_plus_array =
    vec7ToArray(
        Vec7(q_current + alpha * n));

const Array7 q_minus_array =
    vec7ToArray(
        Vec7(q_current - alpha * n));

const std::array<double, 42> J_plus_array =
    model.zeroJacobian(
        Frame::kEndEffector,
        q_plus_array,
        state.F_T_EE,
        state.EE_T_K);

const std::array<double, 42> J_minus_array =
    model.zeroJacobian(
        Frame::kEndEffector,
        q_minus_array,
        state.F_T_EE,
        state.EE_T_K);

Map<const Mat6x7> J_plus(
    J_plus_array.data());

Map<const Mat6x7> J_minus(
    J_minus_array.data());

const double sigma_plus =
    smallestSingularValue(J_plus);

const double sigma_minus =
    smallestSingularValue(J_minus);
```

The parameter `nullspace_probe_step_rad` defines the joint-space distance α_{probe} between the current configuration and the two sampled configurations. The Jacobians are evaluated using the current flange-to-end-effector and end-effector-to-stiffness-frame transformations provided in the robot state.

The smallest singular values $\sigma_{\min}(q_{\pm})$ of the sampled Jacobians are compared through the configured deadband. The sign of their difference selects the sampled direction with the larger value of $\sigma_{\min}$.

Listing 3.12: Selection and projection of the singular-value conditioning torque.

```
if (!std::isfinite(sigma_plus) ||
    !std::isfinite(sigma_minus)) {
  return sigma_fallback;
}

const double sigma_difference =
    sigma_plus - sigma_minus;

const double deadband =
    std::max(
        0.0,
        params.nullspace_sigma_deadband);

if (std::abs(sigma_difference)
    <= deadband) {
  return sigma_fallback;
}

const double sigma_direction =
    (sigma_difference > 0.0)
        ? 1.0
        : -1.0;

```

```
const Vec7 best_direction =
    sigma_direction * n;

const Vec7 tau_sigma =
    params.nullspace_sigma_gain
    * N_tau
    * best_direction;

return sigma_only
           ? tau_sigma
           : tau_damping + tau_sigma;
```

The parameter `nullspace_sigma_deadband` specifies the minimum difference required for selecting one sampled direction. Within the deadband, the sigma-only mode returns zero torque and the combined mode preserves the damping contribution.

The vector `best_direction` corresponds to sv_7. Multiplication by the projector preserves its numerical null-space component, while `nullspace_sigma_gain` specifies the conditioning-torque magnitude. The function returns the torque associated with the selected mode.

The active configuration is loaded from `nullspace.conf`. It uses the combined damping and singular-value conditioning mode with the parameters

$$d_{\mathrm{null}} = 2.0\,\mathrm{N\,m\,s/rad}, \qquad k_\sigma = 1.0\,\mathrm{N\,m}, \tag{3.6}$$

$$\alpha_{\mathrm{probe}} = 0.08\,\mathrm{rad}, \qquad \delta_\sigma = 10^{-6}, \qquad \rho = 10^{-4}. \tag{3.7}$$

In eq. (3.6), d_{null} is the null-space damping coefficient and k_σ is the sigma-conditioning torque magnitude. In eq. (3.7), α_{probe} is the null-direction sampling angle and δ_σ is the sigma-comparison deadband. The relative SVD tolerance is denoted by ρ.

During Cartesian pose hold, the operator can adjust the active null-space mode, damping coefficient, conditioning-torque magnitude, and probe step. The requested values are transferred through atomic variables and assigned inside the callback. Live null-space tuning is associated with the standard Cartesian pose hold, while the phase sequence and set-up-impedance hold use the configuration loaded before the controller session.

3.5 Surface and Tool Geometry

The mathematical definition of the surface task frame is given in Section 2.5. In the implementation, this frame is constructed from the calibrated plane orientation and a reference direction for t_1. The resulting frame is then used for the tool-orientation target, the approach direction, the phase-dependent gains, and the surface-relative evaluation quantities.

3.5.1 Surface Task Frame

The plane configuration contains one surface point expressed in the robot base frame, two surface-tilt angles, and a surface-frame reference direction a_s used to define t_1. The parameter loader converts the tilt angles to a unit surface normal n_s. The function `makeSurfaceFrame` uses a_s to fix the in-plane rotation of the surface frame.

Listing 3.13: Construction of the configured surface point, normal, and task frame.

```
params.surface_point =
    getVec3Xyz("surface_point",
               params.surface_point);

const double ax =
    params.surface_tilt_x_deg * M_PI / 180.0;
const double ay =
    params.surface_tilt_y_deg * M_PI / 180.0;

params.surface_normal_base =
    Vec3(std::sin(ay) * std::cos(ax),
         -std::sin(ax),
         std::cos(ay) * std::cos(ax));

params.surface_normal_base.normalize();

Vec3 normal =
    normalizedOrFallback(
        params.surface_normal_base,
        Vec3(1.0, 0.0, 0.0));

Vec3 tangent1 =
    params.surface_tangent1_hint_base
    - normal
      * normal.dot(
          params.surface_tangent1_hint_base);

if (tangent1.norm() <= 1e-9) {
  Vec3 fallback(0.0, 1.0, 0.0);
  if (std::abs(normal.dot(fallback)) > 0.95) {
    fallback = Vec3(0.0, 0.0, 1.0);
  }
  tangent1 =
      fallback - normal * normal.dot(fallback);
}

tangent1.normalize();

Vec3 tangent2 =
    normal.cross(tangent1).normalized();

Mat3 R_base_surface;
R_base_surface.col(0) = tangent1;
R_base_surface.col(1) = tangent2;
R_base_surface.col(2) = normal;
```

The code variable `R_base_surface` corresponds to R_{surface} in Chapter 2. Its columns follow the fixed order

$$[\, t_1,\ t_2,\ n_s \,].$$

The configured surface-frame reference direction a_s is projected onto the surface plane and normalised to obtain t_1. The second tangent is then constructed as $t_2 = n_s \times t_1$, completing the right-handed surface frame in robot base coordinates. A fallback direction is selected when a_s is approximately parallel to the surface normal n_s.

The active configuration uses the calibrated surface point

$$p_s = \begin{bmatrix} 0.5153 & -0.1072 & 0.0031 \end{bmatrix}^\top \text{ m},$$

together with the calibrated surface tilts

$$\alpha_s = -1.585^\circ, \qquad \beta_s = +0.988^\circ.$$

The positive base-frame x-direction is used as the reference direction for constructing t_1. The configuration selector `use_start_as_surface_point` can alternatively assign the captured controller start position to the runtime surface point. The active configuration selects the calibrated point from `surface.conf`:

Listing 3.14: Selection of the surface point used during one controller session.

```
Vec3 surface_point_runtime =
    params.use_start_as_surface_point
        ? p_start
        : params.surface_point;
```

The descent direction is defined once from the surface normal n_s:

Listing 3.15: Surface-normal descent direction.

```
const Vec3 descend_direction =
    -R_base_surface.col(2);
```

It therefore points opposite to n_s, from the free-space starting region toward the configured plane.

3.5.2 Surface-Relative Tool Orientation

The controlled physical tool axis is configured in the end-effector frame. During each callback, its current base-frame direction is obtained from

Listing 3.16: Transformation of the configured tool axis to the robot base frame.

```
Vec3 currentToolAxisInBase(
    const ControllerConfig& params,
    const Mat3& R_EE) {

  return R_EE
         * normalizedOrFallback(
               params.tool_axis_ee,
               Vec3(0.0, 0.0, 1.0));
}
```

The flat alignment direction is selected as either $+n_s$ or $-n_s$ through `tool_axis_target_sign`. The configured angular offsets about the two surface tangents are combined into one rotation vector and applied to this flat direction:

Listing 3.17: Construction of the surface-relative tool-axis target.

```
const double sign =
    (params.tool_axis_target_sign >= 0.0)
        ? 1.0
        : -1.0;

const Vec3 flat_axis =
    sign * R_base_surface.col(2);

const Vec3 rotation_vector =
    (M_PI / 180.0)
    * (params.tool_target_offset_tangent1_deg
          * R_base_surface.col(0)
       + params.tool_target_offset_tangent2_deg
          * R_base_surface.col(1));

const double angle = rotation_vector.norm();

const Vec3 tool_axis_target =
    (angle <= 1e-12)
        ? flat_axis
        : Eigen::AngleAxisd(
              angle,
              rotation_vector / angle)
              * flat_axis;
```

This construction permits the tool axis to be aligned with the plane or assigned a commanded tool orientation offset about t_1 and t_2. The shortest rotation from the tool axis at the captured start orientation to the selected target axis is then applied to the complete start orientation:

Listing 3.18: Initial tool-axis alignment with the captured orientation about the tool axis.

```
const Vec3 tool_axis_start =
    currentToolAxisInBase(
        params, R_start).normalized();

const Mat3 R_axis =
    rotationBetweenUnitVectors(
        tool_axis_start,
        tool_axis_target)
    * R_start;
```

When `command_tool_twist` is enabled, the rectangular-face long axis provides an additional in-plane orientation reference. Its current direction and the selected surface tangent are projected onto the plane perpendicular to the desired tool axis. The configured normal-axis offset then determines the desired twist about that axis.

Listing 3.19: Surface-relative twist command about the desired tool axis.

```
const Vec3 face_long_axis_ee =
    normalizedOrFallback(
        params.tool_contact_half_length_ee,
        Vec3(0.0, 1.0, 0.0));

const auto perpendicular =
    [&tool_axis_target](const Vec3& v) {
      return v
             - v.dot(tool_axis_target)
               * tool_axis_target;
    };

const Vec3 current =
    perpendicular(R_axis * face_long_axis_ee);

const Vec3 zero_reference =
    perpendicular(R_base_surface.col(0));

const Vec3 target =
    Eigen::AngleAxisd(
        (M_PI / 180.0)
            * params.tool_target_offset_normal_deg,
        tool_axis_target)
    * zero_reference.normalized();

const Vec3 current_unit = current.normalized();

double twist =
    std::atan2(
        tool_axis_target.dot(
            current_unit.cross(target)),
        current_unit.dot(target));

const Mat3 R_base_tool_target =
    Eigen::AngleAxisd(
        twist, tool_axis_target)
    * R_axis;
```

For the centred rectangular face, rotations differing by 180° give an equivalent face orientation. The implementation selects the equivalent half-turn nearest to the captured start orientation, thereby reducing the required orientation motion.

The active configuration defines the controlled axis as $+z_{\mathrm{EE}}$ and selects alignment toward the surface through

$$\texttt{tool\_axis\_target\_sign} = -1.$$

The commanded orientation contains an offset of 10° about t_1, 0° about t_2, and an in-plane twist of 90° about the target tool axis. The twist command is enabled in `tool_orientation.conf`.

3.5.3 Rectangular Grinding Face and Clearance

The contact geometry is represented by a rectangular face fixed in the end-effector frame. Its centre, half-width vector, and half-length vector define the four vertices:

Listing 3.20: Construction of the rectangular contact face in the end-effector frame.

```
const Vec3& face_center =
    params.tool_contact_face_center_ee;

const Vec3& half_width =
    params.tool_contact_half_width_ee;

const Vec3& half_length =
    params.tool_contact_half_length_ee;

const std::array<Vec3, 4>
    tool_face_corners_ee = {{
        face_center + half_width + half_length,
        face_center + half_width - half_length,
        face_center - half_width + half_length,
        face_center - half_width - half_length,
    }};
```

The configured face centre is located 20 mm along $+z_{\mathrm{EE}}$. Its half-width and half-length are 20 mm along x_{EE} and 60 mm along y_{EE}, respectively. The represented contact face therefore measures 40 mm $\times$ 120 mm.

Before set-up, the controller transforms all four vertices to the base frame and projects them onto the descent direction. The maximum projection identifies the part of the face that leads during the approach.

Listing 3.21: Selection of the leading face contact point along the descent direction.

```
std::array<double, 4> projection;

double leading_projection =
    (R_EE * tool_face_corners_ee[0])
        .dot(descend_direction);

projection[0] = leading_projection;

for (std::size_t i = 1;
     i < tool_face_corners_ee.size();
     ++i) {
  projection[i] =
      (R_EE * tool_face_corners_ee[i])
          .dot(descend_direction);

  leading_projection =
      std::max(
          leading_projection,
          projection[i]);
}

Vec3 selected = Vec3::Zero();
int selected_count = 0;

for (std::size_t i = 0;
     i < tool_face_corners_ee.size();
     ++i) {
  if (leading_projection - projection[i]
      <= params.tool_contact_feature_tie_tolerance) {
    selected += tool_face_corners_ee[i];
    ++selected_count;
```

```
    }
}

const Vec3 tool_contact_offset_ee =
    selected
    / static_cast<double>(selected_count);
```

The configured tie tolerance groups vertices whose projections differ from the leading value by no more than this tolerance. The grouped vertices are averaged, which permits nearly equal leading projections to represent an edge or the complete face instead of a single corner. The configured tie tolerance is 0.1 mm.

The active configuration enables both contact-point control and leading contact-point selection. The selected offset is transformed to the base frame to obtain the current controlled contact point:

Listing 3.22: Base-frame position of the selected contact point.

```
const Vec3 tool_contact_point =
    p_EE + R_EE * tool_contact_offset_ee;
```

The leading contact point is reevaluated while the tool is being oriented and approached. On entry to set-up, its end-effector-frame offset is stored in `active_tool_contact_offset_ee` and subsequently held constant during set-up and grinding.

During surface approach, the clearance calculation distinguishes the exact leading projection from the position of the averaged controlled contact point:

Listing 3.23: Geometric clearance evaluation during surface approach.

```
const Vec3 surface_normal =
    (-descend_direction).normalized();

const double height_above_surface =
    surface_normal.dot(
        p_EE - surface_point_runtime)
    - leading_contact_projection;

const double controlled_point_height =
    surface_normal.dot(
        tool_contact_point
        - surface_point_runtime);

const Vec3 projected_surface_point =
    tool_contact_point
    - controlled_point_height
      * surface_normal;

const bool clearance_reached =
    height_above_surface
    <= params.descend_surface_clearance;
```

The quantity `height_above_surface` represents the normal clearance of the foremost point of the complete rectangular face. It provides the geometric transition condition. The quantity `controlled_point_height` is the signed height of the averaged controlled contact point. Its projection onto the configured plane provides the surface reference used by the subsequent set-up motion.

When the clearance condition is satisfied, the contact geometry and current pose are captured:

Listing 3.24: Geometry captured at the transition from approach to set-up.

```
active_tool_contact_offset_ee =
    tool_contact_offset_ee;

first_contact_tcp = p_EE;

first_contact_point =
    projected_surface_point;

setup_push_start =
    -controlled_point_height;

R_contact_start = R_EE;

phase = ControlPhase::kSetup;
phase_start_time = time;
```

The internal variable `first_contact_point` stores the projected clearance point on the configured plane. The live signed height determines `setup_push_start`, which produces a continuous desired position when the controller changes from approach to set-up.

During set-up, the desired position of the contact point is advanced from this projected point along the descent direction. The contact target is then converted to the corresponding TCP reference using the orientation captured at the clearance transition:

Listing 3.25: Conversion of the set-up contact target to the desired TCP position.

```
const Vec3 edge_target =
    first_contact_point
    + push * descend_direction;

desired.p_d =
    edge_target
    - R_contact_start
      * tool_contact_offset_ee;

desired.pdot_d =
    setup_push_velocity
    * descend_direction;
```

This construction allows the translational impedance controller to command the selected contact point while the Cartesian control law remains formulated at the TCP.

3.6 Phase-Dependent Gains and Inertia-Scaled Damping

The controller assigns separate Cartesian stiffness and damping matrices to the approach, set-up, grinding, pose-hold, and phase-gate modes. The matrices are constructed before the real-time callback begins, while the inertia-scaled damping matrices are updated on entry to the corresponding phase group.

3.6.1 Phase-Dependent Gain Construction

The function `buildPhaseImpedanceGains` constructs all phase-dependent gain matrices from the active controller configuration. Gains defined along the surface directions are transformed to the robot base frame using

Listing 3.26: Transformation of diagonal surface-frame gains to the robot base frame.

```
PhaseImpedanceGains gains;
gains.R_base_surface = makeSurfaceFrame(params);

auto taskGain = [&](const Vec3& diagonal) -> Mat3 {
  return makeSpatialGainMatrix(
      diagonal,
      gains.R_base_surface);
};

gains.Kp_approach =
    taskGain(params.approach_Kp_diag);
gains.Dp_approach =
    taskGain(params.approach_Dp_diag);
gains.KR_approach =
    taskGain(params.approach_KR_diag);
gains.DR_approach =
    taskGain(params.approach_DR_diag);
```

The function `makeSpatialGainMatrix` maps the diagonal surface-frame gain G_{surface} to the base-frame gain G_0:

$$G_0 = R_{\text{surface}} G_{\text{surface}} R_{\text{surface}}^{\top}, \tag{3.8}$$

where G_{surface} is diagonal in the fixed axis order $[t_1, t_2, n_s]$. The resulting matrix G_0 is expressed in the robot base frame and can be applied directly to the base-frame Cartesian error e.

The gain-frame policy implemented for each controller mode is summarised in table 3.2.

Table 3.2: Coordinate frames used for the phase-dependent Cartesian gains.

Controller mode	Translational gains	Rotational gains
Tool orientation and surface approach	Surface frame $[t_1, t_2, n_s]$	Surface frame $[t_1, t_2, n_s]$
Set-up and grinding	Selectable between base frame $[x, y, z]$ and surface frame $[t_1, t_2, n_s]$	Surface frame $[t_1, t_2, n_s]$
Cartesian pose hold	Base frame $[x, y, z]$	Base frame $[x, y, z]$
Set-up-impedance hold	Same construction as set-up	Same construction as set-up
Phase-gate hold	Base frame $[x, y, z]$	Surface frame $[t_1, t_2, n_s]$

For set-up translation, the parameter `setup_translation_surface_frame` selects the coordinate frame:

Listing 3.27: Selection of the translational set-up gain frame.

```
gains.Kp_setup =
    params.setup_translation_surface_frame
        ? taskGain(params.setup_Kp_surface_diag)
        : params.setup_Kp_diag.asDiagonal();

gains.Dp_setup =
    params.setup_translation_surface_frame
        ? taskGain(params.setup_Dp_surface_diag)
        : params.setup_Dp_diag.asDiagonal();

gains.KR_setup =
    taskGain(params.setup_KR_diag);

gains.DR_setup =
    taskGain(params.setup_DR_diag);
```

The configuration used for the reported contact runs selected the surface-frame branch for set-up and grinding. Its translational entries were $[2000, 2000, 800]$ N/m along $[t_1, t_2, n_s]$. The inactive base-frame keys therefore did not enter the reported commands. The rotational gains were also aligned with the surface tangents and normal.

Cartesian pose-hold gains are assembled as diagonal base-frame matrices:

Listing 3.28: Construction of the pose-hold and phase-gate gains.

```
gains.Kp_hold =
    params.hold_Kp_diag.asDiagonal();
gains.Dp_hold =
    params.hold_Dp_diag.asDiagonal();
gains.KR_hold =
    params.hold_KR_diag.asDiagonal();
gains.DR_hold =
    params.hold_DR_diag.asDiagonal();
```

```

gains.Kp_pause =
    params.pause_hold_Kp_diag.asDiagonal();
gains.Dp_pause =
    params.pause_hold_Dp_diag.asDiagonal();
gains.KR_pause =
    taskGain(params.pause_hold_KR_diag);
gains.DR_pause =
    taskGain(params.pause_hold_DR_diag);
```

The real-time callback selects the active matrices from the current control phase:

Listing 3.29: Selection of the active phase-dependent gain matrices.

```
const Mat3* Kp_phase =
    params.use_setup_impedance_hold
        ? &Kp_setup
        : &Kp_hold;

const Mat3* Dp_phase =
    params.use_setup_impedance_hold
        ? &Dp_setup_eff
        : &Dp_hold_eff;

const Mat3* KR_phase =
    params.use_setup_impedance_hold
        ? &KR_setup
        : &KR_hold;

const Mat3* DR_phase =
    params.use_setup_impedance_hold
        ? &DR_setup_eff
        : &DR_hold_eff;

switch (phase) {
  case ControlPhase::kToolOrientation:
  case ControlPhase::kSurfaceApproach:
    Kp_phase = &Kp_approach;
    Dp_phase = &Dp_approach_eff;
    KR_phase = &KR_approach;
    DR_phase = &DR_approach_eff;
    break;

  case ControlPhase::kSetup:
  case ControlPhase::kGrinding:
    Kp_phase = &Kp_setup;
    Dp_phase = &Dp_setup_eff;
    KR_phase = &KR_setup;
    DR_phase = &DR_setup_eff;
    break;

  case ControlPhase::kPoseHold:
  case ControlPhase::kManualGuidance:
    break;
}
```

Set-up and grinding therefore share the same stiffness and damping matrices. The difference between these phases lies in the reference trajectory and in the selection of the coupled or decoupled impedance law.

The gate-hold gains carry their own frame selectors, one for the translational and one for the rotational block, so a gate can hold in the surface frame where the phase it interrupts does, or in the base frame where the plane plays no part in what is being held. The selection follows the same construction as the set-up gains.

An active phase gate overrides the normal phase matrices with the dedicated gate-hold gains:

Listing 3.30: Gain override during an operator confirmation gate.

```
const Mat3& Kp_used =
    pause_hold_active ? Kp_pause : *Kp_phase;

const Mat3& Dp_used =
    pause_hold_active ? Dp_pause_eff : *Dp_phase;

const Mat3& KR_used =
    pause_hold_active ? KR_pause : *KR_phase;

const Mat3& DR_used =
    pause_hold_active ? DR_pause_eff : *DR_phase;
```

These four matrices are passed to the Cartesian impedance routine during the current callback.

3.6.2 Calculation of Inertia-Scaled Damping

Each phase first receives the damping matrices configured in the parameter files. When inertia-scaled damping is enabled for a phase group, the controller replaces these matrices with values calculated from the current operational-space inertia Λ.

The joint-space inertia matrix M is obtained from the libfranka model and factorised using a lower–diagonal–lower-transpose (LDLT) decomposition,

$$M = LDL^\top,$$

where L is lower triangular and D is diagonal. Eigen's `LDLT` implementation stores this factorisation and uses it to solve

$$MX = J^\top.$$

The solution matrix X is

$$X = M^{-1}J^\top,$$

without explicitly forming M^{-1}. The inverse operational-space inertia Λ^{-1} is then constructed from

$$\Lambda^{-1} = JM^{-1}J^{\top}.$$

Listing 3.31: Calculation of the operational-space inertia estimate.

```
Eigen::LDLT<Mat7x7> mass_ldlt(joint_mass);

const Mat7x6 Minv_Jt =
    mass_ldlt.solve(J.transpose());

Mat6x6 lambda_inv =
    J * Minv_Jt;

lambda_inv =
    0.5 * (lambda_inv
           + lambda_inv.transpose());

Mat6x6 lambda_inv_regularized =
    lambda_inv;

lambda_inv_regularized
    .diagonal().array() += 1e-9;

Eigen::LDLT<Mat6x6> lambda_ldlt(
    lambda_inv_regularized);

Mat6x6 Lambda_base =
    lambda_ldlt.solve(
        Mat6x6::Identity());
```

This calculation gives the regularised base-frame operational-space inertia estimate Λ_0:

$$\Lambda_0 = \left(JM^{-1}J^{\top} + 10^{-9}I_6\right)^{-1}. \tag{3.9}$$

The matrix is symmetrised before and after its factorisation. It is then transformed from the robot base axes to the coordinate frame of the active gain block:

Listing 3.32: Transformation and extraction of directional Cartesian inertia.

```
const Mat6x6 T_task =
    blockDiagonalRotation(R_task);

Mat6x6 Lambda_task =
    T_task.transpose()
    * Lambda_base
    * T_task;

for (int i = 0; i < 3; ++i) {
  result.translational(i) =
      Lambda_task(i, i);

  result.rotational(i) =
      Lambda_task(i + 3, i + 3);
}
```

The first three diagonal entries provide the directional translational inertias, while the last three provide the directional rotational inertias. For each direction i, the inertia-scaled damping coefficient $d_{\mathrm{dir},i}$ is calculated from

$$d_{\mathrm{dir},i} = 2\zeta\sqrt{\lambda_i k_i}, \tag{3.10}$$

where λ_i is the corresponding directional Cartesian inertia, k_i is the directional stiffness, and ζ is the configured damping ratio. For $\zeta = 1$, the expression gives the critical-damping coefficient of the corresponding decoupled direction.

The calculation is implemented independently for the three translational and three rotational directions:

Listing 3.33: Calculation and limiting of the inertia-scaled damping coefficients.

```
for (int i = 0; i < 3; ++i) {
  const double inertia_i =
      std::max(0.0, inertia(i));

  const double stiffness_i =
      std::max(0.0, stiffness(i));

  const double calculated =
      2.0 * zeta
      * std::sqrt(
          inertia_i * stiffness_i);

  damping(i) =
      std::min(
          max_damping,
          std::max(
              min_damping(i),
              calculated));
}
```

The common upper bound is defined by `auto_damping_max`. The parameter `auto_damping_min_from_manual` selects either zero or the configured damping value as the lower bound.

The coordinate frame used for the inertia estimate follows the frame of the corresponding stiffness matrix:

- approach translation and rotation use the surface frame;
- set-up translation uses the selected base or surface frame, while set-up rotation uses the surface frame;
- Cartesian pose hold uses the robot base frame; and
- the phase gate uses base-frame translation and surface-frame rotation.

The damping matrices are cached by phase group. Entering tool orientation or surface approach calculates the approach damping. Entering set-up, grinding, or set-up-impedance hold calculates the set-up damping. Cartesian pose hold uses its hold damping, while an active operator gate uses the gate-hold damping. Leaving a phase group clears its cached state so that a later entry uses the current robot configuration.

For the phase-gate hold, separate damping ratios are derived from the configured translational and rotational gate-hold damping values. Cartesian pose hold can similarly derive one damping ratio from its configured translational damping and apply it to the translational and rotational inertia-scaled damping calculations.

The active configuration enables inertia-scaled damping for approach, set-up, and the phase gates. Standard Cartesian pose hold uses its configured damping matrices. The exact stiffness, damping-factor, and limiting values used during the experiments are listed in Chapter 4.

When the inertia calculation satisfies the numerical validity checks, the calculated damping matrices replace the configured matrices for the active phase group. The configured matrices remain available as the initial and fallback values.

3.7 Virtual Centre of Compliance

The theoretical transformation from a centre-defined Cartesian impedance to an equivalent TCP impedance is derived in Section 2.7. The implementation applies this transformation inside `computeCartesianImpedanceWrench`. The phase-dependent translational and rotational gain blocks are first expressed in the robot base frame and are then shifted from the selected virtual centre of compliance p_c to the TCP.

The coupled branch is activated during set-up and during the dedicated set-up-impedance hold mode:

Listing 3.34: Selection of the virtual-compliance-centre branch.

```
const bool coupled =
    params.use_virtual_compliance_center &&
    (phase == ControlPhase::kSetup ||
     (phase == ControlPhase::kPoseHold &&
      params.use_setup_impedance_hold));

if (!coupled) {
  Vec6 wrench;
  wrench.head<3>() =
      Kp * dx.head<3>()
      + Dp * dv.head<3>();

  wrench.tail<3>() =
      KR * dx.tail<3>()
      + DR * dv.tail<3>();

  return wrench;
```

```
}
```

Tool orientation, surface approach, grinding, and standard Cartesian pose hold therefore use the decoupled impedance branch. A pre-grinding confirmation gate keeps `ControlPhase::kSetup`; the coupled set-up impedance consequently remains active while the controller waits, and the penetration ramp continues to advance.

The controller supports two definitions of the virtual centre. Both follow the lever convention

$$r_c = p_{\mathrm{TCP}} - p_c$$

introduced in Equation (2.51). The first definition specifies the centre position relative to the TCP in end-effector coordinates. The second specifies the lever r_c directly in surface coordinates.

Listing 3.35: Resolution of the compliance-centre lever in the robot base frame.

```
Vec3 r_c = Vec3::Zero();

if (params.compliance_center_in_tool_frame) {
  r_c =
      -(R_EE
        * params.compliance_center_offset_ee);
} else {
  r_c =
      R_base_surface
      * params.r_tcp_from_compliance_center_surface;
}
```

For the tool-frame definition, `compliance_center_offset_ee` represents $p_c - p_{\mathrm{TCP}}$. The multiplication by R_{EE} expresses this offset in the robot base frame, while the negative sign converts it to the implemented lever convention. Since the measured orientation R_{EE} is used during every callback, this lever follows the current end-effector orientation.

For the surface-frame definition, `r_tcp_from_compliance_center_surface` contains the components of r_c along $[t_1, t_2, n_s]$. Multiplication by `R_base_surface` expresses the lever in the robot base frame. The surface frame and configured lever remain fixed during one controller session, so this definition produces a constant base-frame lever throughout set-up.

When virtual-compliance-centre control is enabled, the configuration loader requires exactly one of these definitions:

Listing 3.36: Validation of the compliance-centre configuration.

```
if (params.use_virtual_compliance_center &&
    params.compliance_center_in_tool_frame ==
        params.compliance_lever_in_surface_frame) {
  throw std::runtime_error(
      "Virtual compliance-center control requires "
      "exactly one definition.");
}
```

The function `complianceShiftMatrix` constructs the point-shift matrix derived in Equation (2.53). The skew-symmetric lever matrix occupies the upper-right block because the Cartesian vectors follow the ordering $[\text{translation}, \text{rotation}]$.

Listing 3.37: Implementation of the spatial point shift and gain congruence.

```
Mat6x6 complianceShiftMatrix(
    const Vec3& r_c) {

  Mat6x6 adjoint = Mat6x6::Zero();

  adjoint.block<3, 3>(0, 0) =
      Mat3::Identity();

  adjoint.block<3, 3>(0, 3) =
      skewMatrix(r_c);

  adjoint.block<3, 3>(3, 3) =
      Mat3::Identity();

  return adjoint;
}

Mat6x6 shiftGainToTcp(
    const Mat6x6& gain_at_center,
    const Vec3& r_c) {

  const Mat6x6 adjoint =
      complianceShiftMatrix(r_c);

  return adjoint.transpose()
         * gain_at_center
         * adjoint;
}
```

The phase-dependent 3×3 translational and rotational matrices are combined into centre-defined spatial gain matrices. The same congruence is then applied independently to stiffness and damping:

Listing 3.38: Construction and evaluation of the coupled TCP impedance.

```
const Mat6x6 K_tcp =
    shiftGainToTcp(
        blockDiagonal(Kp, KR),
        r_c);

const Mat6x6 D_tcp =
    shiftGainToTcp(
        blockDiagonal(Dp, DR),
        r_c);

return K_tcp * dx + D_tcp * dv;
```

The function `blockDiagonal` places K_p and D_p in the translational blocks and K_R and D_R in the rotational blocks. These matrices have already been transformed into the frame selected for the active phase. The subsequent congruence changes the impedance reference point while preserving the selected principal-axis orientation.

The off-diagonal blocks generated by the point shift couple translation and rotation. A translational displacement or velocity error can therefore contribute to the commanded TCP moment, while a rotational error can contribute to the commanded TCP force. The coupling disappears when the configured lever is zero, in which case the shifted matrices reduce to their block-diagonal form.

During set-up, the desired orientation remains the orientation captured at the clearance transition. This orientation acts as a soft rotational reference while the selected contact point is advanced into the surface. The resulting rotation is determined by the coupled impedance, the contact wrench, the directional stiffness, and the physical tool–surface interaction rather than by a prescribed rotational trajectory.

The set-up and grinding phases share the same 3×3 stiffness and damping blocks. At the transition to grinding, the state machine assigns `ControlPhase::kGrinding` before the Cartesian wrench F is evaluated later in the same callback. The first grinding sample therefore uses the decoupled branch. The implementation applies this change as a direct switch from the shifted 6×6 matrices to the decoupled gain blocks, while the achieved penetration and desired orientation remain continuous across the transition.

The set-up-impedance hold mode permits the translational stiffness, rotational stiffness, commanded tilt, and compliance-centre coordinates to be adjusted through the operator interface. Centre-position commands are entered in end-effector coordinates, while lever commands are entered in surface coordinates. An accepted update is converted to the selected internal representation, after which the phase gains and inertia-scaled-damping cache are rebuilt.

After set-up, the reporting module resolves the selected centre in both end-effector and surface coordinates. It also evaluates the effective TCP stiffness and damping matrices, the largest translation–rotation coupling entry, and the eigenvalue ranges of the shifted matrices. These quantities support verification of the configured lever, its sign, and the resulting coupled impedance.

3.8 Control Modes and Phase Sequence

The controller state is represented by the enumeration `ControlPhase`. It contains the four phases of the run sequence together with Cartesian pose hold and manual guidance:

Listing 3.39: Control phases and standalone operating modes.

```
enum class ControlPhase {
  kToolOrientation,
  kSurfaceApproach,
  kSetup,
  kGrinding,
  kPoseHold,
  kManualGuidance
};
```

The run sequence follows

$$\texttt{kToolOrientation} \longrightarrow \texttt{kSurfaceApproach} \longrightarrow \texttt{kSetup} \longrightarrow \texttt{kGrinding}.$$

The orientation phase can be omitted through the controller configuration. The initial phase is selected before the real-time callback begins:

Listing 3.40: Selection of the initial phase.

```
const ControlPhase sequence_first_phase =
    params.enable_orientation_phase
        ? ControlPhase::kToolOrientation
        : ControlPhase::kSurfaceApproach;

const ControlPhase initial_phase =
    params.run_phase_sequence
        ? sequence_first_phase
        : ControlPhase::kPoseHold;

ControlPhase phase = initial_phase;
ControlPhase restart_phase = initial_phase;
```

The variable `restart_phase` stores the phase resumed after manual guidance. The complete sequence, standard pose hold, and set-up-impedance hold can also be selected during an active controller session. The requested mode is received through an atomic variable and applied inside the callback.

Figure 3.2 shows the phases, the condition that ends each one, and the two operator gates. Grinding ends in a hold that keeps the set-up impedance, so the tool stays pressed against the surface, and the same key that started the run starts it again from its first phase. That return is what closes the sequence into a loop.

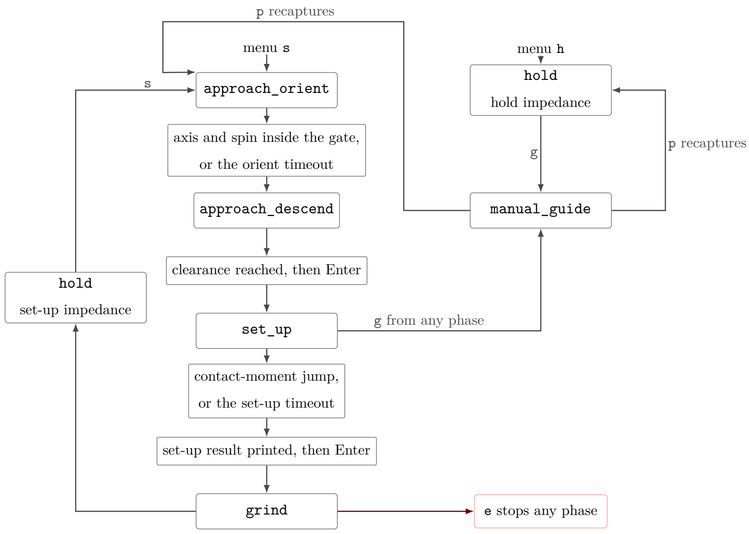


Figure 3.2: Run phase sequence and Cartesian pose hold.

3.8.1 Tool Orientation

The tool-orientation phase maintains the captured start position while the desired orientation is advanced toward the surface-relative target described in Section 3.5.2. The orientation command follows a rate-limited interpolation from the captured start orientation.

The transition condition is evaluated from two errors. The tool-axis error measures the angular separation between the current and desired tool axes. When the in-plane twist command is enabled, the remaining rotation about the target axis is evaluated separately as the spin error.

Listing 3.41: Tool-orientation convergence and phase transition.

```
const bool orientation_reached =
    tool_axis_error
        <= params.approach_orient_error_threshold
    &&
    (!params.command_tool_twist ||
     spin_error
        <= params.approach_orient_spin_error_threshold);

const bool orient_timed_out =
    params.approach_orient_timeout > 0.0
    &&
    phase_time >= params.approach_orient_timeout;

if (phase_time >= params.approach_orient_min_time &&
    (orientation_reached || orient_timed_out)) {
  phase = ControlPhase::kSurfaceApproach;
  phase_start_time = time;

  contact_force_bias = external_force;
  contact_moment_bias = external_moment;
}
```

The configured minimum duration allows the orientation command to begin settling before convergence is accepted. The timeout provides a second transition path and allows the sequence to continue with the orientation reached within the available interval. The external-wrench reference is updated when surface approach begins.

3.8.2 Surface Approach

During surface approach, the desired TCP position advances from the captured start position along the descent direction:

Listing 3.42: Surface-approach translation reference.

```
const double phase_time =
    time - phase_start_time;

const double distance =
    std::min(
        params.descend_speed
            * (phase_time - gate_paused_time),
        params.descend_max_distance);

desired.p_d =
    p_start + distance * descend_direction;

desired.pdot_d =
    params.descend_speed * descend_direction;
```

The variable `gate_paused_time` removes operator-confirmation time from the descent trajectory. The commanded distance is limited by `descend_max_distance`.

The transition to set-up is determined from the geometric clearance of the leading part of the rectangular tool face. The estimated external force is displayed as a diagnostic quantity during approach, while the phase transition uses

Listing 3.43: Geometric transition from surface approach to set-up.

```
const double height_above_surface =
    surface_normal.dot(
        p_EE - surface_point_runtime)
    - leading_contact_projection;

const bool clearance_reached =
    height_above_surface
    <= params.descend_surface_clearance;

if (clearance_reached && !gate_blocking) {
  active_tool_contact_offset_ee =
      tool_contact_offset_ee;

  first_contact_tcp = p_EE;
  first_contact_point =
      projected_surface_point;

  setup_push_start =
      -controlled_point_height;

  R_contact_start = R_EE;

  contact_force_bias = external_force;
```

```
  contact_moment_bias = external_moment;

  phase = ControlPhase::kSetup;
  phase_start_time = time;
}
```

The selected contact-point offset is locked at this transition. The controller also stores the current TCP pose, the projected point on the surface, the signed starting value of the set-up penetration, and the current force and moment estimates.

When the optional pre-set-up gate is enabled, reaching the clearance first captures the measured TCP position. The controller then holds this position with the dedicated phase-gate gains:

Listing 3.44: Operator gate between approach and set-up.

```
if (!gate_setup_armed) {
  gate_setup_armed = true;
  gate_setup_hold_pd = p_EE;
  signals.gate_continue.store(false);
}

if (signals.gate_continue.load()) {
  gate_setup_passed = true;
  signals.gate_continue.store(false);
} else {
  gate_blocking = true;
  pause_hold_active = true;

  desired.p_d = gate_setup_hold_pd;
  desired.pdot_d.setZero();

  gate_paused_time += period.toSec();
}
```

The geometric clearance condition remains satisfied while the gate is active. After confirmation, the contact-point and wrench references are captured from the current measured state and the controller enters set-up.

Reaching the maximum permitted descent distance before the clearance condition sets the stop request and records the approach as unsuccessful.

3.8.3 Set-Up

The set-up phase advances the selected contact point from the projected surface point into the plane. The scalar penetration begins at the signed contact-point coordinate captured during the approach transition and progresses toward `setup_push_end` at the configured speed.

Listing 3.45: Set-up penetration and TCP reference.

```
double setup_push_velocity = 0.0;
```

```

const double push =
    setupPush(
        params,
        phase_time,
        setup_push_start,
        setup_push_velocity);

const Vec3 edge_target =
    first_contact_point
    + push * descend_direction;

desired.p_d =
    edge_target
    - R_contact_start
      * tool_contact_offset_ee;

desired.pdot_d =
    setup_push_velocity
    * descend_direction;
```

The function `setupPush` integrates a constant signed speed and clamps the result at the configured final penetration. It is evaluated on the elapsed phase time, so the ramp runs whenever the set-up phase is active. Reaching the clamp is what ends the advance, not leaving the phase.

The desired orientation remains `R_contact_start`, which is the orientation captured at the clearance transition. The tool can consequently rotate relative to this soft reference under the coupled impedance and the physical contact wrench.

During set-up, the changes in external force and moment are calculated relative to the values captured at the approach transition:

Listing 3.46: Set-up termination condition.

```
const Vec3 df_ext =
    external_force
    - contact_force_bias;
const double df_ext_norm =
    df_ext.norm();

const Vec3 m_tcp =
    (external_moment
     - contact_moment_bias)
    - p_EE.cross(df_ext);
const double m_tcp_norm =
    m_tcp.norm();

const bool stopped_on_moment =
    phase_time >= params.setup_min_time
    &&
    m_tcp_norm
        >= params.setup_moment_threshold;

if (!gate_grind_armed &&
    !stopped_on_moment &&
    phase_time < params.setup_timeout) {
  break;
}
```

The force change $\Delta\hat{f}_{\text{ext}}$ is recorded and included in the set-up evaluation. The moment change governs the phase exit after the configured minimum duration, and the set-up timeout provides the alternative exit condition.

The moment is referenced to the TCP before it is compared with the threshold, which is the term `p_EE.cross(df_ext)` above. The estimated external moment is reported about the base origin, so the share the force produces through the TCP position is removed first; what remains is M_{TCP}. Section 3.10.2 carries the same moment on to the contact point.

Before continuing, the controller evaluates and prints the set-up result once. This report includes the achieved pose, contact-point position, alignment change $\Delta\theta_{\mathrm{align}}$, force and moment changes, moment transferred to the contact point, active impedance matrices, and compliance-centre diagnostics.

When the pre-grinding gate is enabled, the controller remains in the set-up phase after the exit condition has been reached:

Listing 3.47: Operator gate between set-up and grinding.

```
if (params.pause_before_grind &&
    !gate_grind_passed) {
  if (!gate_grind_armed) {
    gate_grind_armed = true;
    signals.gate_continue.store(false);
  }

  if (!signals.gate_continue.load()) {
    break;
  }

  gate_grind_passed = true;
  signals.gate_continue.store(false);
}

grind_push = push;
phase = ControlPhase::kGrinding;
phase_start_time = time;
```

The coupled set-up impedance stays active while the controller waits, and so does the penetration ramp. The commanded depth therefore keeps advancing until it reaches the configured final penetration, and the preload the tool holds at the gate depends on how long the wait lasts. After confirmation, the current penetration is stored in `grind_push` and becomes the normal preload of the grinding phase.

3.8.4 Grinding

Grinding maintains the final set-up penetration. The controller can either add a tangential sweep or maintain only the normal penetration.

For an active sweep, the selected tangent is obtained from the surface frame. The scalar sweep trajectory is then added to the pressed contact-point reference:

Listing 3.48: Grinding reference with an optional tangential sweep.

```
const Vec3 n = descend_direction;
Vec3 edge_target;

if (params.grind_sweep_enabled) {
  const Vec3 grind_tangent =
      (params.grind_tangent_axis == 2)
          ? Vec3(R_base_surface.col(1))
          : Vec3(R_base_surface.col(0));

  double sweep_s = 0.0;
  double sweep_s_dot = 0.0;

  grindSweep(
      time - phase_start_time,
      params.grind_amplitude_m,
      grindStrokeDuration(params),
      sweep_s,
      sweep_s_dot);

  edge_target =
      first_contact_point
      + grind_push * n
      + sweep_s * grind_tangent;

  desired.pdot_d =
      sweep_s_dot * grind_tangent;
} else {
  const double edge_penetration =
      n.dot(
          tool_contact_point
          - first_contact_point);

  edge_target =
      tool_contact_point
      + (grind_push - edge_penetration) * n;
}

desired.p_d =
    edge_target
    - R_contact_start
      * tool_contact_offset_ee;
```

The function `grindSweep` uses a fifth-order smooth-step trajectory. The first stroke moves from the centre position to the positive amplitude. Subsequent strokes alternate smoothly between the positive and negative amplitudes with zero velocity at each reversal.

When the sweep is disabled, the reference is corrected only along the surface normal n_s. Its tangential coordinates follow the current contact-point position, which preserves the set-up penetration without imposing a tangential trajectory.

Grinding uses the set-up stiffness and damping values. The Cartesian impedance routine selects the decoupled branch after the phase has changed to `kGrinding`, as described in Section 3.7.

3.8.5 Pose Hold and Manual Guidance

The pose-hold state supports two impedance configurations. Standard pose hold uses the dedicated hold gains, while set-up-impedance hold applies the set-up gains and can keep the virtual centre of compliance p_c. Both modes regulate a captured Cartesian pose.

When the set-up-impedance hold is selected during a run, the measured pose is first captured as the current controller state. The reference is then moved smoothly toward the pose at which the hold session began. Translation uses the configured approach speed, while orientation uses the configured maximum angular rate. After both references have reached their stored values, the controller continues as a stationary hold.

Manual guidance can be entered from any impedance-controlled phase. The controller then returns Coriolis compensation together with joint-space damping:

Listing 3.49: Manual-guidance torque and pose recapture.

```
const Vec7 tau_guide =
    coriolis
    - params.manual_guidance_damping * dq;

if (signals.proceed_requested.load()) {
  signals.proceed_requested.store(false);

  restartFromPoseReached(true);
  phase = restart_phase;

  hold_return_p = p_start;
  hold_return_R = R_d;
  hold_returning = false;
}
```

After operator confirmation, the measured position, orientation, joint configuration, and the selected surface and external-wrench references are recaptured. The controller then resumes the phase stored in `restart_phase`.

3.8.6 Run Termination

The callback finishes normally when a positive experiment duration has elapsed or the operator issues the stop command:

Listing 3.50: Normal termination of the real-time callback.

```
if ((params.experiment_duration > 0.0 &&
     time >= params.experiment_duration)
    ||
    signals.stop_requested.load()) {
  return MotionFinished(
      Torques(tau_array));
}
```

The same stop signal is assigned when the maximum approach distance is reached. Robot-side safety monitoring can also interrupt the torque-control session through a libfranka exception. The subsequent error handling and collision configuration are described in Section 3.9.

3.9 Robot Preparation and Safety Handling

Robot preparation is performed before the real-time impedance callback begins. It includes the initial FCI connection, error recovery, collision configuration, motion to the selected initial posture, and the optional gripper and tool-transfer actions.

3.9.1 Robot Connection and Collision Configuration

At program start, the robot address is read from the controller configuration and used to establish the FCI connection. Error recovery is requested before the collision behaviour and robot model are loaded:

Listing 3.51: Robot connection, error recovery, and model initialisation.

```
const ControllerConfig startup_config =
    readControllerConfig(parameter_files);

Robot robot(startup_config.robot_ip);

try {
  robot.automaticErrorRecovery();
} catch (const franka::Exception& e) {
  fprintf(stderr,
          "Automatic error recovery failed: %s\n",
          e.what());
  return -1;
}

configureCollisionBehavior(
    robot, startup_config);

Model model = robot.loadModel();
```

Failure of the recovery request terminates the program before a motion command is issued. Further recovery or unlocking is then performed through Franka Desk before restarting the controller.

The function `configureCollisionBehavior` first applies the standard robot behaviour provided by the libfranka example utilities. The custom thresholds defined in `safety.conf` are subsequently assigned when their selector is enabled:

Listing 3.52: Assignment of the configured collision thresholds.

```
void configureCollisionBehavior(
    Robot& robot,
    const ControllerConfig& params) {
```

```

  setDefaultBehavior(robot);

  if (!params.use_custom_collision_behavior) {
    return;
  }

  const Array7 torque_acc =
      filledArray7(
          params.collision_torque_acc);

  const Array7 torque_nom =
      filledArray7(
          params.collision_torque_nom);

  const Array6 force_acc =
      filledArray6(
          params.collision_force_acc);

  const Array6 force_nom =
      filledArray6(
          params.collision_force_nom);

  robot.setCollisionBehavior(
      torque_acc, torque_acc,
      torque_nom, torque_nom,
      force_acc, force_acc,
      force_nom, force_nom);
}
```

The two joint-torque parameters define the acceleration-phase and nominal thresholds in N m. Each scalar is assigned uniformly to all seven joints. The two Cartesian parameters are similarly assigned to all six force and moment components in the acceleration and nominal phases. The same array is passed for the lower and upper threshold of each group.

The controller returns the calculated joint torque τ_{cmd} directly to libfranka:

Listing 3.53: Transfer of the calculated torque command to libfranka.

```
const Array7 tau_array =
    vec7ToArray(tau_cmd);

return Torques(tau_array);
```

The application transfers the calculated Cartesian wrench F, null-space torque τ_{null}, and complete joint-torque command τ_{cmd} directly to libfranka. Robot-side monitoring and the configured collision behaviour provide command boundedness and motion interruption.

3.9.2 Initial Joint Posture

The parameter `q_init_case` selects one of the joint configurations stored in `initial_pose.conf`. The corresponding seven joint angles are assigned to `q_init` when the configuration is loaded.

The phase sequence and both Cartesian hold modes begin from this posture. The start-up interface moves the robot to it using the libfranka `MotionGenerator`:

Listing 3.54: Motion to the configured initial joint posture.

```
const auto move_to_q_init = [&]() {
  const Vec7 q_init =
      Map<const Vec7>(params.q_init.data());

  MotionGenerator motion_generator(
      0.4, params.q_init);

  robot.control(motion_generator);

  q_init_reached = true;
};
```

The speed factor is fixed at 0.4 for this preparation motion. A local flag records whether the posture has already been reached and prevents repeated motion when the operator performs several preparation actions before selecting a run mode.

Selection of the phase sequence, standard pose hold, or set-up-impedance hold calls `ensure_q_init`. Manual guidance forms the alternative start path and begins from the current robot pose. The guided pose is captured before the subsequent sequence or hold mode starts.

3.9.3 Gripper Verification

The start-up interface provides separate commands for opening, grasping, and recalibrating the Franka Hand. Each action reads the gripper state before and after the command and verifies the resulting finger position.

For opening, successful verification requires the libfranka motion command to finish, the calibrated maximum stroke to contain the requested width, and the measured final width to reach the target within a 2 mm tolerance. Grasp verification requires the grasp command to finish, the `is_grasped` flag to be active, and the measured width to lie within the configured inner and outer tolerances.

Before either action, the reported maximum finger width is compared with the largest width used by the controller. A smaller calibrated stroke indicates that gripper homing was performed while an object restricted the finger motion. The start-up interface then requests recalibration with empty fingers.

A configured gripper action can also be executed immediately before torque control. A previously verified manual action is kept and is therefore not repeated. When the requested action is grasping, an already verified tool grasp is also kept:

Listing 3.55: Selection of the gripper action before torque control.

```
if (params.perform_gripper_action_before_run &&
    !params.start_with_manual_guidance &&
    !params.startup_gripper_action_completed) {

  Gripper gripper(params.robot_ip);

  if (params.gripper_startup_grasp_tool) {
    const franka::GripperState state =
        gripper.readOnce();

    const bool already_holding =
        state.is_grasped &&
        state.width >=
            params.gripper_grasp_width
            - params.gripper_grasp_epsilon_inner &&
        state.width <=
            params.gripper_grasp_width
            + params.gripper_grasp_epsilon_outer;

    gripper_ok =
        already_holding
            ? true
            : graspTool(params, gripper);
  } else {
    gripper_ok =
        openGripper(params, gripper);
  }
}
```

The parameter `abort_on_gripper_action_failure` determines whether a failed gripper action terminates the program or allows the selected controller session to continue.

3.9.4 Tool Transfer

The controller also provides an optional sequence for collecting the tool from its holder or returning it. The holder configuration is represented by the joint posture `q_pickup`. A stand-off posture is calculated at the same end-effector orientation and at the configured distance along $-z_{\mathrm{EE}}$.

The stand-off configuration is obtained through damped least-squares inverse kinematics. Each iteration calculates the position error e_p and orientation error e_R, forms the current end-effector Jacobian $J(q)$, and limits the joint-update norm. The resulting posture is accepted only when all joints remain within their configured limits with a 0.05 rad margin.

After successful calculation, the tool-transfer path is executed as

$$q_{\text{init}} \longrightarrow q_{\text{above}} \longrightarrow q_{\text{pickup}} \longrightarrow q_{\text{above}}.$$

The corresponding implementation is:

Listing 3.56: Joint-space motion sequence for collecting or returning the tool.

```
Array7 q_above{};

if (!solveStandoffPosture(
        model,
        robot.readOnce(),
        params.q_pickup,
        params.pickup_standoff,
        q_above)) {
  return false;
}

int bad_joint = 0;
if (!withinJointLimits(
        q_above, bad_joint)) {
  return false;
}

ensure_q_init();

MotionGenerator to_above(
    0.4, q_above);
robot.control(to_above);

MotionGenerator descend(
    std::max(
        0.05,
        params.pickup_descend_speed_factor),
    params.q_pickup);
robot.control(descend);

const bool action_ok =
    grasp
        ? graspTool(params, gripper)
        : openGripper(params, gripper);

MotionGenerator lift(
    std::max(
        0.05,
        params.pickup_descend_speed_factor),
    q_above);
robot.control(lift);
```

When collecting the tool, the gripper is opened and verified before the arm moves toward the holder. At the pickup posture, the selected gripper action is executed and verified before the robot returns to the stand-off posture.

3.9.5 Exception Handling

Robot communication, motion-generation, and torque-control errors are reported through `franka::Exception`. Configuration, file, and numerical errors are reported through `std::exception`. Both exception types leave the active control operation and terminate the program with an error code.

A robot-side safety event can therefore end the torque callback independently of the controller phase or experiment timer. The corresponding exception is reported after control has stopped, and the robot is recovered through Franka Desk before a new program execution.

3.10 Data Recording and Run Evaluation

The controller records the quantities required for the subsequent run evaluation in a bounded in-memory ring buffer. Its memory is allocated before `robot.control` starts, so the real-time callback performs no dynamic buffer growth.

Listing 3.57: Allocation of the bounded logging buffer before torque control.

```
const int log_every_n_cycles =
    std::max(
        1,
        params.log_every_n_cycles);

const std::size_t max_log_rows =
    static_cast<std::size_t>(
        std::max(
            0,
            params.max_log_rows));

std::vector<LogData> log_data(
    max_log_rows);

std::size_t control_cycle_count = 0;
std::size_t log_write_index = 0;
std::size_t log_rows_written = 0;
bool log_buffer_wrapped = false;
```

The parameter `log_every_n_cycles` defines the sampling interval in control cycles, while `max_log_rows` defines the maximum number of samples stored in memory. The active configuration records every control cycle and allocates space for 120 000 rows.

The structure `LogData` stores one sample of the measured state, controller reference, calculated command, and active controller mode. The recorded quantities are summarised in table 3.3.

Table 3.3: Signal groups stored during the real-time control loop.

Signal group	Recorded quantities
Time and controller state	Run time, encoded control phase, and encoded null-space mode.
TCP state and reference	Measured and desired TCP positions, translational and rotational errors, measured linear and angular velocities, and desired linear velocity.
Tool and surface geometry	Current controlled contact-point position, contact-point offset in the end-effector frame, projected clearance point, TCP position at the clearance transition, desired contact target, and virtual surface penetration.
Alignment	Tool-axis alignment-error components along $[t_1, t_2, n_s]$ and their Euclidean norm.
Cartesian wrench	Commanded and model-estimated external wrenches, including the clearance-transition reference for the latter.
Null-space controller	Active null-space damping coefficient and the Euclidean norm of the calculated null-space torque.
Disturbance experiment	Disturbance scale, torque-limit scale, force-application point, commanded Cartesian force, and equivalent joint torque.
Joint-torque command	The seven final commanded joint torques returned to libfranka.

The internal variables and CSV columns named `first_contact_tcp` and `first_contact_point` refer to the geometric clearance transition described in Section 3.8.2. They represent the captured TCP position and projected contact point at this transition, rather than a force-detected contact event.

The selected values are copied into the current ring-buffer row at the configured sampling interval:

Listing 3.58: Assignment of one sampled controller state to the ring buffer.

```
++control_cycle_count;

if (max_log_rows > 0 &&
    control_cycle_count
```

```
        % static_cast<std::size_t>(
              log_every_n_cycles)
        == 0) {

  LogData& row =
      log_data[log_write_index];

  row.time = time;
  row.phase = static_cast<int>(phase);
  row.nullspace_mode =
      static_cast<int>(
          params.nullspace_mode);

  row.p_EE = p_EE;
  row.p_d = desired.p_d;
  row.tool_contact_point =
      tool_contact_point;
  row.first_contact_tcp =
      first_contact_tcp;
  row.first_contact_point =
      first_contact_point;
  row.edge_target = edge_target_log;
  row.tool_contact_offset_ee =
      tool_contact_offset_ee;

  row.e_p = e_p;
  row.e_R = e_R;
  row.angular_deviation_surface =
      R_base_surface.transpose()
      * toolSurfaceAlignmentErrorBase(
            params,
            R_EE,
            R_base_surface);
  row.angular_deviation =
      row.angular_deviation_surface.norm();

  row.pdot = pdot;
  row.pdot_d = desired.pdot_d;
  row.omega = omega;

  row.f = f;
  row.m = m;
  row.external_force = external_force;
  row.external_moment = external_moment;
  row.contact_force_bias =
      contact_force_bias;
  row.contact_moment_bias =
      contact_moment_bias;

  row.push = push_log;
  row.disturbance = disturbance;
  row.tau_nullspace_norm =
      tau_nullspace.norm();
  row.nullspace_damping =
      params.nullspace_damping;
  row.tau_cmd = tau_cmd;
}
```

The current implementation represents the null-space action through the active mode, damping coefficient, and resulting torque norm. These quantities permit the secondary torque to be distinguished from the Cartesian and disturbance contributions during post-processing.

After assigning one row, the write index is advanced cyclically:

Listing 3.59: Advancement of the bounded ring-buffer index.

```
log_write_index =
    (log_write_index + 1)
    % max_log_rows;

if (log_rows_written < max_log_rows) {
```

```
  ++log_rows_written;
} else {
  log_buffer_wrapped = true;
}
```

Once the allocated capacity is reached, new samples replace the oldest rows. The buffer therefore stores the most recent `max_log_rows` samples instead of increasing its memory requirement during control.

3.10.1 Chronological Reconstruction and CSV Output

After the torque callback has finished, the ring-buffer contents are converted to chronological order. When the buffer has wrapped, the oldest stored sample begins at `log_write_index`:

Listing 3.60: Reconstruction of the chronological sample order after control.

```
std::vector<LogData> ordered_log_data;
ordered_log_data.reserve(
    log_rows_written);

if (log_buffer_wrapped) {
  ordered_log_data.insert(
      ordered_log_data.end(),
      log_data.begin()
          + static_cast<std::ptrdiff_t>(
                log_write_index),
      log_data.end());

  ordered_log_data.insert(
      ordered_log_data.end(),
      log_data.begin(),
      log_data.begin()
          + static_cast<std::ptrdiff_t>(
                log_write_index));
} else {
  ordered_log_data.insert(
      ordered_log_data.end(),
      log_data.begin(),
      log_data.begin()
          + static_cast<std::ptrdiff_t>(
                log_rows_written));
}

result.log =
    std::move(ordered_log_data);
```

The ordered samples and final controller state are returned in a `RunResult`. File output is subsequently performed by `writeRunLogs`, outside the real-time callback:

Listing 3.61: Post-run CSV output and terminal summary.

```
void writeRunLogs(
    const ControllerConfig& params,
    const RunResult& result) {

  if (result.descend_failed) {
    printSection("descend stopped");
    printf(
        "  maximum distance reached "
        "before the clearance height.\n");
```

```
  }

  printJointStartEndTableDeg(
      result.q_start,
      result.final_q);

  writeLogToCsv(
      result.log,
      params.csv_file_name);

  printFinalSummary(
      result.final_p_d,
      result.final_p_EE,
      result.final_e_p,
      result.final_e_R,
      params.csv_file_name);
}
```

The CSV writer creates the parent output directory when required and uses a fixed column order. Cartesian positions, velocities, forces, moments, disturbances, and joint torques τ are written in their stored SI units. Angular deviations are converted from radians to degrees for the output file.

The CSV writer calculates each external-wrench change by subtracting the stored reference:

Listing 3.62: Writing the external-wrench change to the CSV file.

```
writeVec3(
    log_file,
    row.external_force
        - row.contact_force_bias);

writeVec3(
    log_file,
    row.external_moment
        - row.contact_moment_bias);
```

The commanded Cartesian wrench F and the model-estimated external wrench are therefore recorded as separate quantities. The former is generated by the impedance controller, while the latter is supplied by the robot state and used for set-up evaluation.

When several controller sessions are executed without restarting the program, the session number is inserted before the CSV filename extension:

Listing 3.63: Generation of separate CSV filenames for repeated sessions.

```
if (session > 1) {
  config.csv_file_name =
      sessionFileName(
          config.csv_file_name,
          session);
}
```

For example, a second session based on `controller_log.csv` is written as `controller_log_s2.csv`.

3.10.2 Set-Up Evaluation

A dedicated set-up report is generated once when the moment threshold or set-up time-out is reached. The report captures the final TCP pose, contact-point position, external wrench change, clearance-transition references, active gain matrices, and set-up termination condition.

A moment depends on the point it is referred to. The reported one is referred to the contact point, not to the TCP. M_{TCP} is the estimated external moment about the TCP, and r_{contact} the lever between the two points defined in eq. (2.50). The transfer is

$$M_{\mathrm{contact}} = M_{\mathrm{TCP}} + r_{\mathrm{contact}} \times \Delta \hat{f}_{\mathrm{ext}}. \tag{3.11}$$

$\Delta \hat{f}_{\mathrm{ext}}$ is the model-estimated external force, less the clearance-transition reference. M_{TCP} is the corresponding moment change. Force does not depend on the reference point and carries no point subscript. The cross product removes the share the force produces through the lever. What remains is the moment the contact itself applies.

The lever r_{contact} is not the compliance-centre lever r_c. It follows from the tool geometry and enters the evaluation only. r_c shapes the commanded wrench through $\mathrm{Ad}(r_c)$. It reaches the measurement through the contact, so the two never share an expression.

Listing 3.64: Calculation of the external moment at the contact point.

```
const Vec3 r_contact =
    p_EE - tool_contact_point;

const Vec3 m_contact =
    m_tcp
    + r_contact.cross(df_ext);
```

The terminal set-up report contains:

- the set-up termination condition and phase duration;
- the external-force and external-moment changes;
- the TCP and contact-point displacements from the clearance transition;
- the tool-alignment error before and after set-up;
- the alignment-error components along t_1, t_2, and n_s;
- the force, moment, displacement, and rotation resolved in the surface frame; and
- the active translational and rotational stiffness and damping matrices.

The displayed force–displacement and moment–angle ratios provide a pointwise diagnostic of the completed set-up state. They are calculated only when the corresponding displacement or rotation exceeds its numerical minimum.

When compliance-centre diagnostics are enabled, the report additionally resolves r_c in surface coordinates and p_c in end-effector coordinates. It reconstructs the shifted TCP stiffness and damping matrices and reports their diagonal entries, largest translation–rotation coupling terms, and eigenvalue ranges. The complete 6×6 matrices can also be printed for verification of the configured lever and gain transformation.

The final post-run summary displays the joint displacement between the start and end configurations, the desired and measured final TCP positions, the final translational and rotational errors, and the path of the generated CSV file.

The controller implementation described in this chapter forms the basis of all reported experiments. Chapter 4 presents the calibration procedure, the controller parameters used during the measurements, the varied experimental conditions, and the quantities derived from the recorded signals.

Chapter 4

Experimental Methodology

GREEN — revised and confirmed. The chapter was revised in full and every paragraph is settled. No further editing is to be applied to it, and any remaining concern is raised as a comment rather than as a change.

comment: "In the settled prose, use "commanded tool orientation offset" for the deliberate angular command and "commanded rotation axis" or "commanded rotation direction" for its components. Retain "tilt" only for a physical inclination or a calibrated surface angle."

This chapter presents the experimental configuration, calibration procedure, controller parameters, test sequence, and evaluation quantities used to characterise the contact-induced alignment of the grinding tool. The main test series consists of surface-contact measurements in which the commanded tilt, directional Cartesian stiffness, and virtual centre of compliance p_c were varied under otherwise matched conditions. Complementary Cartesian pose-hold trials quantified the two null-space torque τ_{null} contributions under a repeatable internally commanded disturbance.

4.1 Experimental Design

The cases are experimental stages within the three research questions. Cases A and C answer the stiffness question; Case C tests whether the geometric selection rule extends to combined tangent-plane directions; and Cases D to H establish the compliance centre from a reference condition carrying no lever at all, separating the frame the lever is defined in from its direction and its position. Table 4.1 gives this mapping before the complete parameter matrix.

Table 4.1: Research-question structure of the surface-contact investigation.

RQ	Experimental purpose	Principal evidence
RQ1	Effect of directional rotational and translational stiffness on alignment	Stiffness variations in B and C
RQ2	Effect of compliance-centre lever direction, sign, and magnitude on alignment	Reference without a lever in J; tool-fixed tangential lever in K; in-plane and tool-axis position in M and N
RQ3	Derivation and test of an orientation-dependent geometric selection rule	Four-direction generality and fixed-lever comparison in H; definition frame of the lever in L

Within that structure, the surface-contact measurements were divided into cases. Case A varied the rotational stiffness about the tilted tangent, while Case B varied the translational stiffness perpendicular to that tangent. Case C commanded the tilt in four directions in the tangent plane, compared a direction-selected lever with a fixed one, and varied the normal lever coordinate.

Cases D to H examine the compliance centre with its offset held fixed to the tool. Case D places the centre on the TCP, so the press makes no moment through an offset and what acts is the contact alone. It is measured on both surface tangents and at both signs of the commanded offset, because the grinding face is three times longer across one tangent than the other and the two need not behave alike.

Case E displaces the centre along the tangent perpendicular to the commanded rotation axis, seven positions from -40 to $+40\,\mathrm{mm}$, at every combination of tangent and sign. That grid is what settles whether the assisting direction follows the sign of the commanded offset. Case F names one of those offsets in surface coordinates instead of end-effector coordinates, so the definition frame is separated from the offset itself.

Cases G and H vary the remaining two quantities. Case G moves the centre along the tool axis, the direction a first-order coupling model leaves without effect. Case H halves the commanded offset at two of the Case-E positions, so what the centre does is told apart from what the size of the commanded offset does.

The commanded angular offset defines the orientation difference present at the beginning of set-up. The rotational stiffness determines the resistance to rotation about the tilted tangent, whereas the translational stiffness perpendicular to that tangent affects the translational response coupled to the alignment motion. Shifting the Cartesian impedance from a selected centre of compliance p_c to the TCP introduces translation–rotation coupling and therefore changes the moment produced by the commanded press.

The surface plane and the grinding-face direction were calibrated independently and used to define the surface and tool geometry throughout the measurements. The rotational stiffness about t_1 and t_2 was varied separately, while the orthogonal rotational-stiffness entry remained at 5 N m/rad. This arrangement allowed the response associated with each surface tangent t_1 or t_2 to be evaluated independently.

Unless stated otherwise for a particular case, the surface-frame translational stiffness K_p used during set-up and grinding was

$$K_p = \begin{bmatrix} 2000 & 0 & 0 \\ 0 & 2000 & 0 \\ 0 & 0 & 350 \end{bmatrix} \mathrm{N/m}. \tag{4.1}$$

The corresponding surface-frame rotational stiffness K_R was

$$K_R = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 50 \end{bmatrix} \mathrm{N\,m/rad}. \tag{4.2}$$

also in the surface frame. Both matrices use the coordinate order $[t_1, t_2, n_s]$. The fixed phase, damping, null-space, and trajectory parameters are specified in Section 4.4.

Neither matrix is a neutral starting point. The entries were chosen for the roles they play during set-up, and the choice is stated here because it bounds what varying them could show. The controller regulates pose and not force: the commanded wrench is $K\,\Delta x + D\,\Delta\dot{x}$ throughout, with no force target and no force feedback anywhere in the loop. Whatever load the tool carries is a consequence of a commanded displacement acting through a stiffness.

The normal translational entry is the compliant one, 350 N/m against 2000 N/m in the plane. The press reference is ramped past the surface and the surface stops the tool, so the load that arises is that entry times the penetration the contact refuses. A stiff normal entry would turn a small geometric error into a large load, which is what makes the compliant choice the workable one. The stiff in-plane entries hold the contact point where the geometry places it instead of letting it slide.

The rotational entries are chosen the other way round. Those about the two surface tangents are nearly free at 5 N m/rad, because turning onto the surface is the motion the set-up phase exists to permit and a stiff entry there would hold the tool at the attitude it arrived with. The entry about the surface normal is stiff at 50 N m/rad, because rotation about the tool axis contributes nothing to alignment.

Two consequences follow for the experiments. The stiffness variations of Cases A and B ask how much the chosen values matter rather than search for better ones, and their outcome supports the choice: raising the rotational entry about the commanded tangent reduced the alignment response. The entry about the surface normal was not varied, because the alignment error carries no component about that axis by construction and the measured rotation about it stayed below 0.71° in every reported run, leaving that entry little to act on.

The normal translational entry was not varied either. Raising it would raise the press and, through the compliance-centre offset, the moment the press makes. Whether that improves the alignment is not established here, and the reported measurements give reason for doubt: across Cases D to H the press lay between 76 and 85 N while the alignment change spanned −4.60 to +7.65°, and two conditions pressing alike at approximately 80 N differ by 6.3° in what they achieve. The load is therefore not what separates them over this range, and a larger one cannot be assumed to help.

Table 4.2: Parameter variations used in the surface-contact measurements.

Case	Varied quantity	Commanded tool orientation offset	Tested values	n	Comparison
A	Rotational stiffness	+10° about t_1 or t_2	15 and 50 N m/rad about the commanded rotation axis. The orthogonal tangent remained at 5 N m/rad. The matched 5 N m/rad, 2000 N/m reference setting is counted here and shared with Case B.	18	Tangent-specific influence of rotational stiffness on the alignment response.
B	Translational stiffness	+10° about t_1 or t_2	300 and 800 N/m perpendicular to the commanded rotation axis. The matched 2000 N/m reference is the setting counted in Case A. One additional condition for each commanded rotation direction combined 300 N/m with 50 N m/rad.	18	Influence of the cross-axis translational stiffness and its interaction with rotational stiffness.
C	Commanded rotation direction and normal lever coordinate	+10° in four tangent-plane directions	Lever of 60 mm selected for each commanded rotation direction, the same lever held fixed while the direction turned, and normal coordinates of −60, +20, +60 and +120 mm.	33	Whether one lever serves every commanded rotation direction, and the effect of the normal coordinate.

Table 4.2: Parameter variations used in the surface-contact measurements (continued).

Case	Varied quantity	Commanded tool orientation offset	Tested values	n	Comparison
D	Contact alone; the centre of compliance was placed at the TCP	None, and $\pm 10°$ about t_1 or t_2	No offset, and one pair carrying +40 mm at the zero offset.	18	Response of the contact alone on each tangent, and the residual the arrangement leaves with nothing commanded, which is the floor every other residual is read against.
E	Direction and sign of the compliance-centre offset	$\pm 10°$ about t_1 or t_2	−40, −20, −10, 0, +10, +20 and +40 mm along the tangent perpendicular to the commanded rotation axis, positive in the direction that assists a positive offset.	72	Whether the assisting direction follows the sign of the commanded offset, and what the offset adds on each tangent.
F	Frame in which the offset is defined	None and $+10°$ about t_1	The 40 mm offset of Case E named in surface coordinates.	6	Whether the definition frame changes the response once the tool turns.
G	Compliance-centre position along the tool axis	$+10°$ about t_1	−40, −20, −10, +10, +20 and +40 mm along the tool axis. The 0 mm entry is the corresponding trial of Case E.	18	Dependence on a direction that produces no first-order moment.

Table 4.2: Parameter variations used in the surface-contact measurements (continued).

Case	Varied quantity	Commanded tool orientation offset	Tested values	n	Comparison
H	Commanded magnitude	$+5°$ about t_1 or t_2	0 and +40 mm along the assisting tangent, read against the same positions at $+10°$ in Case E.	12	Whether the correction scales with the commanded offset.

Each parameter setting was repeated three times. Cases A and C used the decoupled Cartesian impedance. Case C used the point-shifted impedance with the listed levers expressed in the surface frame according to $r_c = p_{\mathrm{TCP}} - p_c$. The stated gains, calibrated geometry, phase sequence, press motion, and null-space settings were held constant except where a varied value is stated explicitly.

The eight cases contain 171 surface-contact runs distributed across 57 parameter settings. Within each controlled comparison, only the stated parameter was changed. The remaining gain entries, calibrated surface and tool geometry, phase sequence, and evaluation procedure were held constant. The principal evaluation quantity was the change in the angle between the calibrated surface normal n_s and the grinding-face direction calculated from the measured end-effector orientation. The rotation during set-up, the displacement of the point or edge of the grinding face used for the set-up motion, the TCP displacement Δp_{TCP}, and the model-estimated external load F_n were evaluated alongside this angle. Their definitions and treatment across repeated runs are presented in Section 4.6.

The surface-contact measurements and Cartesian pose-hold trials formed complementary data sets. Contact trials measured alignment with projected damping as the active null-space term. The pose-hold trials in Section 4.7 quantified projected damping and singular-value conditioning under a repeatable internally commanded disturbance.

4.2 Experimental System

4.2.1 Robot, Tool, and Surface

The experiments were performed with a seven-degree-of-freedom Franka Emika Panda operated through the Franka Control Interface. Joint-torque commands were transmitted through libfranka at the nominal sampling period T_s:

$$T_s = 1\,\mathrm{ms}, \tag{4.3}$$

corresponding to a nominal control rate of 1 kHz. The real-time controller and its phase sequence are described in Chapter 3.

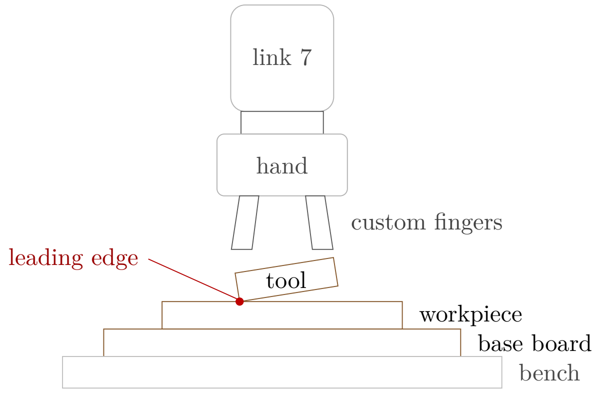
The rectangular grinding tool was held by the Franka Hand using custom gripper fingers. Its grinding face measured 40 mm along X_{EE} and 120 mm along Y_{EE}. The centre of the grinding face was located 20 mm along $+Z_{\mathrm{EE}}$ from the reported end-effector origin. The complete calibrated tool geometry is presented in Section 4.3.

The pads holding the tool were screwed to the gripper fingers. Clearance in that screwed joint permitted a small relative rotation of the tool about y_{EE}.

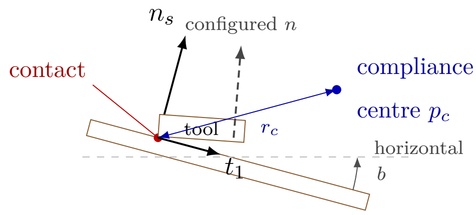
The grinding-face direction was reconstructed from the measured end-effector orientation and its calibrated direction in the end-effector frame. This frame-consistent metric quantifies the combined robot–gripper–tool response. The approximately $\pm 2°$ relative tool rotation about y_{EE} defines the mount-related contribution to its uncertainty.

The surface plate was mounted on a raised base board. All experiments used dry contact between the grinding face and the plate. The controller command followed the calibrated geometry and Cartesian impedance law. The surface plane and grinding-face direction were calibrated separately before the experimental campaign, as described in Section 4.3.

During approach, the controller determines which point or edge of the rectangular grinding face lies furthest in the descent direction. This point or edge is used for the geometric clearance calculation and the subsequent set-up motion. Its position was calculated from the measured end-effector pose and the calibrated tool geometry.



(a) side elevation



(b) contact geometry

Figure 4.1: Experimental system and surface-frame contact geometry.

4.2.2 Software and Real-Time Computer

The controller computer ran Ubuntu 20.04.3 with the `5.9.1-rt20` Linux kernel and the `PREEMPT_RT` patch. The controller was compiled using `g++ 9.3`, the C++14 language standard, and the `-O2` optimisation level. Eigen 3.3.7 and libfranka 0.7.1 were used for the matrix calculations and communication with the robot.

Before the contact measurements, the real-time computer and robot network were checked with the `communication_test` program supplied with libfranka 0.7.1 [8]. The program reports the fraction of control commands received by the robot and issues its lower communication-quality warning below an average success rate of 0.90. The experimental setup passed this criterion.

The controller operated through the libfranka joint-torque interface. At each control cycle, the measured robot state and the quantities provided by the robot model were used to calculate the commanded joint torque τ_{cmd}. The implementation of the real-time callback is documented in Section 3.2.

Each contact run followed the implemented sequence of tool orientation, surface approach, set-up, and grinding. Manual guidance supported preparation, while Cartesian pose hold provided the complementary null-space trials. Projected damping supplied the null-space contribution throughout the contact sequence.

4.2.3 Recorded Experimental Quantities

The quantities required for the subsequent evaluation were written to the controller's in-memory logging buffer during each run and exported to a CSV file after torque control had ended. The recorded quantities included:

- the measured end-effector position and orientation;
- the measured joint position q and velocity $\dot{q}$;
- the commanded Cartesian reference and Cartesian error e;
- the position of the point or edge of the grinding face used for the set-up motion;
- the active controller phase and null-space mode;
- the commanded Cartesian wrench F and final joint-torque command τ_{cmd}; and
- the model-estimated external force $\hat{f}_{\mathrm{ext}}$ and moment reported by libfranka.

The commanded wrench F and the model-estimated external wrench were recorded as separate quantities. After the clearance-transition reference was subtracted, the libfranka estimate provided a common supporting load signal across matched runs. The complete logging implementation and CSV structure are described in Section 3.10.

4.2.4 Reproducibility Record

The documented experimental system is summarised in Table 4.3. Controller parameters that differ between the individual experimental cases are reported separately in Section 4.4.

Table 4.3: Documented configuration of the experimental system.

Item	**Configuration used in the reported experiments**
Robot	Franka Emika Panda with seven actuated joints.
Robot interface	libfranka joint-torque interface with a nominal control period of 1 ms.
Gripper and tool mounting	Franka Hand with custom gripper fingers and an approximately $\pm 2°$ relative rotational play about y_{EE}.

Item	**Configuration used in the reported experiments**
Grinding tool	Rectangular grinding face measuring $40 \times 120\,\text{mm}$, with its centre located 20 mm along $+Z_{\text{EE}}$ from the reported end-effector origin.
Surface	Plane surface mounted on a raised base board and used under dry-contact conditions.
Operating system	Ubuntu 20.04.3.
Real-time kernel	Linux `5.9.1-rt20` with `PREEMPT_RT`.
Compiler	`g++ 9.3`, C++14, optimisation level `-O2`.
Software libraries	Eigen 3.3.7 and libfranka 0.7.1.
Recorded state quantities	Robot pose, joint position and velocity, Cartesian reference and error, controller phase, and position of the grinding-face point or edge used for the set-up motion.
Recorded command and load quantities	Commanded Cartesian wrench, commanded joint torque, and the model-estimated external force and moment reported by libfranka.
Contact condition	Dry contact; controller command determined by the calibrated geometry and Cartesian impedance law.

The reproducibility record contains the software versions, nominal control period, controller parameters, tool dimensions, and surface calibration. These quantities define the implemented control configuration and calibrated contact geometry used in the experiments.

4.3 Surface and Tool Calibration

The surface plane and the direction normal to the grinding face were calibrated independently before the contact experiments. The surface calibration provided the point, normal direction, and tangent directions used by the controller. The tool calibration provided the grinding-face direction expressed in the end-effector frame.

The calibrated quantities were written to the controller parameter files and were subsequently used for approach, clearance calculation, commanded tool orientation, gain orientation, compliance-centre placement, and evaluation of the experimental results.

The two calibration chains were independent during estimation and were combined only when the run-time tool direction was evaluated against the surface normal n_s. Their fit and validation stages are summarised in fig. 4.2.

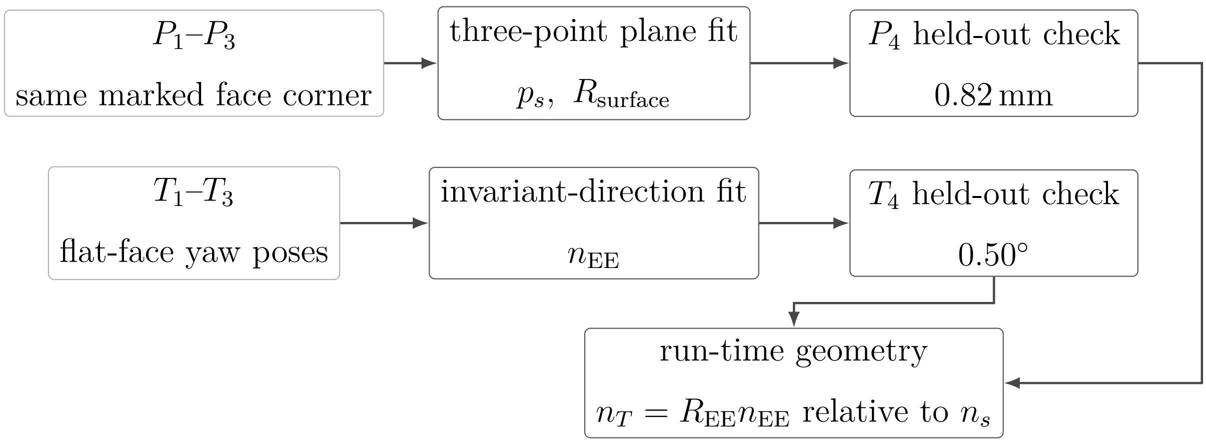


Figure 4.2: Surface and tool calibration flow.

4.3.1 Surface-Plane Calibration

Four positions distributed over the plate were probed with the same marked $+X_{\text{EE}}$, $+Y_{\text{EE}}$ corner of the grinding face. The first three corner positions, P_1–P_3, determined the plane. The fourth, P_4, was excluded from the fit and provided an independent geometric check.

At each position i, the read-only capture utility obtained the measured end-effector position $p_{\text{EE},i}$ and orientation $R_{\text{EE},i}$. The fixed end-effector-frame probe offset $r_{\text{probe,EE}}$ was the grinding-face centre plus the positive half-width and half-length vectors:

$$r_{\text{probe,EE}} = \begin{bmatrix} 0.020 \\ 0.060 \\ 0.020 \end{bmatrix} \text{m}. \tag{4.4}$$

The base-frame corner coordinate p_i stored for the plane fit was therefore

$$p_i = p_{\text{EE},i} + R_{\text{EE},i} r_{\text{probe,EE}}. \tag{4.5}$$

The plane used by the controller is represented by

$$n_s^\top (p - p_s) = 0, \tag{4.6}$$

where p_s is one point on the plane and n_s is its unit normal, expressed in the robot base frame.

The final calibrated plane point stored in the controller parameters was

$$p_s = \begin{bmatrix} 0.5153 \\ -0.1072 \\ 0.0031 \end{bmatrix} \text{m}. \tag{4.7}$$

This point is the centroid of the three fitted corner coordinates, not a captured grinding-face-centre position.

The controller stores the surface orientation through rotations a about the base x-axis and b about the base y-axis. With e_z denoting the base-frame z-axis unit vector, the corresponding normal is calculated from

$$n_s = R_y(b) R_x(a) e_z = \begin{bmatrix} \sin(b)\cos(a) \\ -\sin(a) \\ \cos(b)\cos(a) \end{bmatrix}. \tag{4.8}$$

The calibrated angles in the final controller configuration were

$$a = -1.585^\circ, \qquad b = +0.988^\circ. \tag{4.9}$$

These values give the base-frame surface normal n_s

$$n_s \approx \begin{bmatrix} 0.017245 \\ 0.027663 \\ 0.999469 \end{bmatrix}, \qquad \|n_s\| = 1. \tag{4.10}$$

This three-point physical-plane fit was copied into the controller configuration. The same normal n_s therefore defined the surface frame, the controller geometry, and the alignment values reported in Chapter 5. No separate evaluation normal was used.

The plane-validation residual $\varepsilon_{\text{plane}}$ is the perpendicular distance of the independently probed fourth point p_4 from the calibrated plane:

$$\varepsilon_{\text{plane}} = \left| n_s^\top (p_4 - p_s) \right|. \tag{4.11}$$

The resulting validation residual was

$$\varepsilon_{\text{plane}} = 0.82\,\text{mm}. \tag{4.12}$$

The first surface tangent t_1 is taken from the calibration itself. The direction a_s from the first to the second probed point is projected into the calibrated plane and normalised:

$$t_1 = \frac{(I - n_s n_s^\top) a_s}{\|(I - n_s n_s^\top) a_s\|}. \tag{4.13}$$

The tangent therefore follows the probed geometry rather than a base-frame axis, and the surface frame is rotated in the plane relative to the base x- and y-directions.

The second surface tangent t_2 completes the right-handed surface frame:

$$t_2 = n_s \times t_1. \tag{4.14}$$

For the calibrated normal, the resulting tangent directions are approximately

$$t_1 \approx \begin{bmatrix} -0.905786 \\ -0.422854 \\ 0.027332 \end{bmatrix}, \qquad t_2 \approx \begin{bmatrix} 0.423385 \\ -0.905776 \\ 0.017765 \end{bmatrix}. \tag{4.15}$$

The complete surface frame is therefore

$$R_{\text{surface}} = \begin{bmatrix} t_1 & t_2 & n_s \end{bmatrix} \approx \begin{bmatrix} -0.905786 & 0.423385 & 0.017245 \\ -0.422854 & -0.905776 & 0.027663 \\ 0.027332 & 0.017765 & 0.999469 \end{bmatrix}. \tag{4.16}$$

The surface normal n_s lies approximately 1.9° from the base z-direction, while the tangents are rotated substantially within the plane relative to the base x- and y-directions.

All surface-frame quantities in the contact campaign use the coordinate order

$$[\,t_1, t_2, n_s\,]. \tag{4.17}$$

This order applies to commanded angular offsets, rotational stiffness, surface-resolved displacement and rotation, and the compliance-centre levers r_c.

4.3.2 Grinding-Face Calibration

Four end-effector orientations, T_1–T_4, were captured with the complete grinding face seated flat on the validated plane. The face remained flat while its yaw about the plane normal was changed by approximately 20°–45° between samples. The first three rotations were used for estimation, and the fourth was held out for validation.

With only the yaw changed, the end-effector-frame grinding-face normal n_{EE} should map to one invariant base-frame direction $\bar{n}_B$. The matrices R_i are the rotational blocks of the captured poses T_i:

$$R_1 n_{\mathrm{EE}} \approx R_2 n_{\mathrm{EE}} \approx R_3 n_{\mathrm{EE}} = \bar{n}_B. \tag{4.18}$$

The estimator formed the mean rotation matrix $\bar{R} = (R_1 + R_2 + R_3)/3$ and applied the singular-value decomposition $\bar{R} = U\Sigma V^{\top}$. The matrix U contains the left singular vectors. The matrix Σ contains the singular values, and V contains the right singular vectors. The dominant right singular vector gave n_{EE}, while the corresponding left singular vector gave $\bar{n}_B$. The sign was selected toward nominal $+Z_{\mathrm{EE}}$. The final controller configuration was

$$n_{\mathrm{EE}} = \begin{bmatrix} -0.006011 \\ 0.000246 \\ 0.999982 \end{bmatrix}, \qquad \|n_{\mathrm{EE}}\| = 1, \tag{4.19}$$

expressed in the end-effector frame. The calibrated grinding-face normal differs from nominal $+Z_{\mathrm{EE}}$ by 0.34°. The calibration does not by itself identify the physical source of this offset.

The three fit residuals were 0.19°–0.41°. For the held-out orientation, the invariant-axis residual $\varepsilon_{\text{tool}}$ was

$$\varepsilon_{\text{tool}} = \angle(R_4 n_{\text{EE}}, \bar{n}_B) = 0.50°. \tag{4.20}$$

The fit and held-out limits were both 0.50°. The dominant-to-second singular-value gap was 0.210, above the required minimum of 0.005, which confirmed that the yaw samples made the axis observable. The fitted invariant base direction differed from the signed plane-normal target by 1.36°. This separate check satisfied its 2.0° limit; the plane normal did not participate in the tool-axis estimate.

During each controller cycle, the grinding-face direction in the robot base frame is calculated from the measured end-effector orientation:

$$n_T = R_{\text{EE}} n_{\text{EE}}. \tag{4.21}$$

The direction n_T is therefore calculated from the measured end-effector orientation and the calibrated grinding-face direction. This reconstruction provides the grinding-face direction used throughout the evaluation.

The commanded flat orientation turns the grinding face toward the calibrated surface according to

$$R_d n_{\text{EE}} = -n_s. \tag{4.22}$$

The negative sign follows from the axis-direction sign s_a:

$$s_a = -1, \tag{4.23}$$

which directs the positive grinding-face axis opposite the outward surface normal n_s. The remaining spin about the desired grinding-face axis, measured from t_1, was set to 115.05° and was held constant for all reported runs. For the flat orientation, the grinding-face axis is antiparallel to n_s. This spin places the long axis y_{EE} at 25.05° from the calibrated surface tangent t_2. In the following sections this angle is written as approximately 25°.

The parallel-jaw mounting permitted approximately ±2° of relative tool rotation about

y_{EE}. This relative motion contributes to the uncertainty of the alignment calculated from the end-effector orientation.

Figure 4.3 shows the calibrated face and the mounting axes in the surface plane.

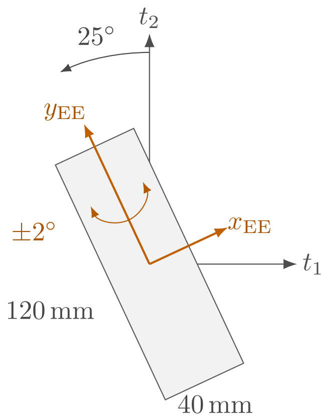


Figure 4.3: Calibrated grinding face and mounting axes in the surface plane.

4.3.3 Calibrated Grinding-Face Geometry

The grinding face was represented by a rectangle expressed in the end-effector frame. Its centre offset $r_{\mathrm{face,EE}}$ was configured as

$$r_{\mathrm{face,EE}} = \begin{bmatrix} 0 \\ 0 \\ 0.020 \end{bmatrix} \mathrm{m}. \tag{4.24}$$

The half-width vector $r_{\mathrm{width,EE}}$ and half-length vector $r_{\mathrm{length,EE}}$ were

$$r_{\mathrm{width,EE}} = \begin{bmatrix} 0.020 \\ 0 \\ 0 \end{bmatrix} \mathrm{m}, \qquad r_{\mathrm{length,EE}} = \begin{bmatrix} 0 \\ 0.060 \\ 0 \end{bmatrix} \mathrm{m}. \tag{4.25}$$

The complete grinding face therefore measured

$$40\,\mathrm{mm} \times 120\,\mathrm{mm}, \tag{4.26}$$

with the width along X_{EE}, the length along Y_{EE}, and its centre 20 mm along $+Z_{\mathrm{EE}}$ from the reported end-effector origin.

For a point on the grinding face with end-effector-frame offset $r_{g,\mathrm{EE}}$, its base-frame position p_g is calculated from

$$p_g = p_{\mathrm{EE}} + R_{\mathrm{EE}} r_{g,\mathrm{EE}}. \tag{4.27}$$

The controller transforms all four corners of the grinding face into the robot base frame and compares their positions along the descent direction. The point or edge of the grinding face that approaches the surface first is then used for the geometric clearance calculation and the subsequent set-up motion.

The grinding-face contact-point selection tolerance ε_g was

$$\varepsilon_g = 0.1\,\mathrm{mm} \tag{4.28}$$

are treated as belonging to the same edge or face. Their positions are averaged to form one geometric reference. This prevents a nominal edge configuration from switching between its two endpoints because of small numerical differences.

In the following sections, this geometry-based reference is called the point or edge of the grinding face used for the set-up motion. Its position is used for the clearance calculation, commanded set-up displacement, and reported grinding-face displacement.

The point or edge is obtained entirely from the measured end-effector pose, the calibrated grinding-face geometry, and the descent direction. The controller therefore uses a geometric surface reference rather than a separately measured physical contact position. The implementation of the corner transformation and geometric selection is described in Section 3.5.

4.4 Controller Parameters Used in the Experiments

The controller parameters were assigned according to the role of each phase. Approach used a comparatively soft isotropic Cartesian impedance in the surface frame. The operator gate applied a stiffer translational hold at the clearance pose. Set-up and grinding used lower translational stiffness along n_s than along t_1 and t_2, together with lower rotational stiffness about the surface tangents than about the surface normal n_s. The coordinate frame therefore forms part of every gain definition.

4.4.1 Fixed Phase and Motion Parameters

The phase-transition and trajectory parameters are listed in Table 4.4. The set-up push coordinate starts at the signed distance captured at the clearance transition, approximately $-0.020\,\mathrm{m}$, and moves toward $+0.060\,\mathrm{m}$. At the configured speed, the reference reaches this endpoint after approximately $1.6\,\mathrm{s}$ and is then held there until the phase ends. The endpoint is a Cartesian spring reference beyond the calibrated plane and is not interpreted as physical penetration of the same magnitude.

Table 4.4: Fixed phase-transition and trajectory parameters.

Quantity	**Value**	**Role**
Minimum orientation time	0.5 s	Prevents an immediate phase transition.
Tool-axis transition tolerance	0.035 rad	Maximum remaining tool-axis error before descent.
Descent speed	0.1 m/s	Reference speed along the descent direction.
Maximum descent distance	0.640 m	Terminates an unsuccessful approach.
Surface clearance	0.020 m	Geometric hand-off from approach to set-up.
Minimum set-up time	2.0 s	Earliest activation of the moment-based exit condition.
Set-up timeout	5.0 s	Time-based set-up termination.
Moment-change threshold	150 N m	Alternative set-up termination based on the model-estimated external moment.
Set-up push speed	0.080 m/s	Rate of the signed press coordinate.
Set-up push endpoint	+0.240 m	Final signed Cartesian spring reference.
Normal translational stiffness	350 N/m	Surface-frame entry along n_s.
Grinding reference	fixed tangential coordinates	Set-up press reference maintained.
Pre-set-up gate	enabled	Holds the clearance pose until operator confirmation.
Pre-grind gate	enabled	Holds the set-up pose until operator confirmation.

4.4.2 Phase-Dependent Cartesian Stiffness

During tool orientation and surface approach, the translational stiffness $K_{p,\mathrm{app}}$ and rotational stiffness $K_{R,\mathrm{app}}$ were isotropic in the calibrated surface frame:

$$K_{p,\mathrm{app}} = \begin{bmatrix} 150 & 0 & 0 \\ 0 & 150 & 0 \\ 0 & 0 & 150 \end{bmatrix} \mathrm{N/m}, \qquad K_{R,\mathrm{app}} = \begin{bmatrix} 90 & 0 & 0 \\ 0 & 90 & 0 \\ 0 & 0 & 90 \end{bmatrix} \mathrm{N\,m/rad}. \tag{4.29}$$

At the pre-set-up gate, the base-frame translational stiffness $K_{p,\mathrm{gate}}$ was increased. The gate rotational stiffness $K_{R,\mathrm{gate}}$ retained the approach value:

$$K_{p,\mathrm{gate}} = \begin{bmatrix} 5000 & 0 & 0 \\ 0 & 5000 & 0 \\ 0 & 0 & 5000 \end{bmatrix} \mathrm{N/m}, \qquad K_{R,\mathrm{gate}} = \begin{bmatrix} 90 & 0 & 0 \\ 0 & 90 & 0 \\ 0 & 0 & 90 \end{bmatrix} \mathrm{N\,m/rad}. \tag{4.30}$$

The surface-frame translational stiffness matrices used during set-up and grinding, $K_{p,\mathrm{set}}$ and $K_{p,\mathrm{grind}}$, were equal:

$$K_{p,\mathrm{set}} = K_{p,\mathrm{grind}} = \begin{bmatrix} 2000 & 0 & 0 \\ 0 & 2000 & 0 \\ 0 & 0 & 350 \end{bmatrix} \mathrm{N/m}. \tag{4.31}$$

The corresponding rotational stiffness matrices $K_{R,\mathrm{set}}$ and $K_{R,\mathrm{grind}}$ were also equal:

$$K_{R,\mathrm{set}} = K_{R,\mathrm{grind}} = \begin{bmatrix} 5 & 0 & 0 \\ 0 & 5 & 0 \\ 0 & 0 & 50 \end{bmatrix} \mathrm{N\,m/rad}. \tag{4.32}$$

The lower rotational tangent entries allow rotation about t_1 and t_2, while the larger normal entry resists twist about n_s. The effective normal translational stiffness $k_{p,n}$ is the normal entry because both matrices are diagonal in the surface frame:

$$k_{p,n} = n_s^\top K_{p,\mathrm{set}} n_s = 350\,\mathrm{N/m}, \tag{4.33}$$

and the tangential entries act along t_1 and t_2 without contributing to the normal direction.

The Cartesian pose-hold study used the isotropic base-frame translational stiffness $K_{p,\text{hold}}$ and rotational stiffness $K_{R,\text{hold}}$:

$$K_{p,\text{hold}} = \begin{bmatrix} 2000 & 0 & 0 \\ 0 & 2000 & 0 \\ 0 & 0 & 2000 \end{bmatrix} \text{N/m}, \qquad K_{R,\text{hold}} = \begin{bmatrix} 50 & 0 & 0 \\ 0 & 50 & 0 \\ 0 & 0 & 50 \end{bmatrix} \text{N m/rad}. \tag{4.34}$$

Manual guidance used joint damping of 0.5 N m s/rad and no Cartesian stiffness.

4.4.3 Inertia-Scaled Damping

The directional damping coefficients were calculated at entry to the corresponding phase group from the current Cartesian inertia, the active stiffness, and the configured damping ratio, as described in Section 3.6.2. Approach used $\zeta = 1.9$, while set-up and grinding used $\zeta = 1.0$. The calculated coefficients were used for the associated phase group.

The pre-set-up gate used separate damping ratios for translation and rotation. They were derived from the configured translational damping target $D^{\text{ref}}_{p,\text{gate}}$:

$$D^{\text{ref}}_{p,\text{gate}} = \begin{bmatrix} 200 & 0 & 0 \\ 0 & 200 & 0 \\ 0 & 0 & 200 \end{bmatrix} \text{N s/m}. \tag{4.35}$$

The rotational gate ratio was derived from the damping calculated for approach. For the pose-hold study, one damping ratio was derived from the translational damping target $D^{\text{ref}}_{p,\text{hold}}$:

$$D^{\text{ref}}_{p,\text{hold}} = \begin{bmatrix} 50 & 0 & 0 \\ 0 & 50 & 0 \\ 0 & 0 & 50 \end{bmatrix} \text{N s/m} \tag{4.36}$$

and applied to the translational and rotational inertia-scaled damping calculations.

When an inertia estimate could not be used, the configured approach fallbacks were the translational damping $D_{p,\text{app}}^{\text{ref}}$ and rotational damping $D_{R,\text{app}}^{\text{ref}}$:

$$D_{p,\text{app}}^{\text{ref}} = \begin{bmatrix} 20 & 0 & 0 \\ 0 & 20 & 0 \\ 0 & 0 & 20 \end{bmatrix} \mathrm{N\,s/m}, \qquad D_{R,\text{app}}^{\text{ref}} = \begin{bmatrix} 12 & 0 & 0 \\ 0 & 12 & 0 \\ 0 & 0 & 12 \end{bmatrix} \mathrm{N\,m\,s/rad}. \tag{4.37}$$

The corresponding set-up fallbacks were the translational damping $D_{p,\text{set}}^{\text{ref}}$ and rotational damping $D_{R,\text{set}}^{\text{ref}}$:

$$D_{p,\text{set}}^{\text{ref}} = \begin{bmatrix} 50 & 0 & 0 \\ 0 & 50 & 0 \\ 0 & 0 & 25 \end{bmatrix} \mathrm{N\,s/m}, \qquad D_{R,\text{set}}^{\text{ref}} = \begin{bmatrix} 10.01 & 0 & 0 \\ 0 & 10.01 & 0 \\ 0 & 0 & 10 \end{bmatrix} \mathrm{N\,m\,s/rad}. \tag{4.38}$$

Both set-up matrices are expressed in the surface frame with the coordinate order $[t_1, t_2, n_s]$.

The factor $\zeta = 1$ defines the coefficient for each decoupled direction in the calculation. The controller uses the resulting matrix as its inertia-scaled damping setting after the compliance-centre shift.

4.4.4 Compliance-Centre Settings

Cases A and C used the decoupled Cartesian impedance at the TCP. Case C used the point-shifted impedance during set-up. The centre of compliance p_c was specified through the lever r_c

$$r_c = p_{\text{TCP}} - p_c, \tag{4.39}$$

resolved in the surface frame $[t_1, t_2, n_s]$. The lever was captured with the clearance pose and remained constant during the set-up phase. The stiffness and damping matrices were shifted to the TCP according to the transformation derived in Section 2.7.

Cases E to H also used the point-shifted impedance during set-up.

For a commanded tilt about t_1, the varied component was r_{c,t_2}, while $r_{c,t_1} = 0$. For a commanded tilt about t_2, the varied component was r_{c,t_1}, while $r_{c,t_2} = 0$. The tested perpendicular components were -60, 0, and $+60$ mm, with $r_{c,n} = 0$.

The fourth setting placed the centre of compliance p_c at the centre of the grinding face. The surface-frame levers $r_c^{(t_1)}$ and $r_c^{(t_2)}$ correspond to the measurements about t_1 and t_2, respectively:

$$r_c^{(t_1)} = \begin{bmatrix} 0 \\ -3.5 \\ 19.7 \end{bmatrix} \text{mm}, \qquad r_c^{(t_2)} = \begin{bmatrix} 3.5 \\ 0 \\ 19.7 \end{bmatrix} \text{mm}. \tag{4.40}$$

The sign of the perpendicular component determines the direction of the moment generated by the translational response. Reversing this component reverses the corresponding coupling moment about the tilted surface tangent t_1 or t_2. The reported selections are the largest measured responses among the discrete lever values tested.

For the directional measurements the commanded rotation axis lies in the surface tangent plane and is separate from the offset angle. Written from the surface-resolved angular components $(\theta_{t_1}, \theta_{t_2})$, the selected tangential lever is

$$r_{c,t} = \frac{\rho_c}{\sqrt{\theta_{t_1}^2 + \theta_{t_2}^2}} \left(\theta_{t_2} t_1 - \theta_{t_1} t_2\right). \tag{4.41}$$

Here $\rho_c > 0$ is the selected tangential lever magnitude. The lever is perpendicular to the commanded rotation axis within the tangent plane, which is what the phrase *aligned lever* means in Case C, and it reverses when the commanded orientation offset does. The four commanded directions are listed as surface-frame vectors in table 4.5.

Case C contains three internal comparisons. H1 rotates the 60 mm tangential lever with each of the four commanded rotation directions. H2 holds the H1 y_{EE} lever fixed while the other three directions are commanded. H3 uses the H1 y_{EE} tangential components and changes only the normal coordinate.

Table 4.5: Case-C commanded tool-orientation-offset components and compliance-centre lever vectors in the surface frame.

Condition	Commanded rotation-axis description	$[\theta_{t_1}, \theta_{t_2}]$ (deg)	$[r_{c,t_1}, r_{c,t_2}, r_{c,n}]$ (mm)
H1 aligned	About y_{EE}	$[-4.23, +9.06]$	$[+54.4, +25.4, 0]$
H1 aligned	$-45°$ diagonal	$[+3.41, +9.40]$	$[+56.4, -20.5, 0]$
H1 aligned	About x_{EE}	$[+9.06, +4.23]$	$[+25.4, -54.4, 0]$
H1 aligned	$+45°$ diagonal	$[-9.40, +3.41]$	$[+20.5, +56.4, 0]$
H2 fixed	$-45°$ diagonal	$[+3.41, +9.40]$	$[+54.4, +25.4, 0]$
H2 fixed	About x_{EE}	$[+9.06, +4.23]$	$[+54.4, +25.4, 0]$
H2 fixed	$+45°$ diagonal	$[-9.40, +3.41]$	$[+54.4, +25.4, 0]$
H3 normal	About y_{EE}	$[-4.23, +9.06]$	$[+54.4, +25.4, -60]$
H3 normal	About y_{EE}	$[-4.23, +9.06]$	$[+54.4, +25.4, +20]$
H3 normal	About y_{EE}	$[-4.23, +9.06]$	$[+54.4, +25.4, +60]$
H3 normal	About y_{EE}	$[-4.23, +9.06]$	$[+54.4, +25.4, +120]$

Figure 4.4 shows the four commanded rotation directions with the lever selected for each.

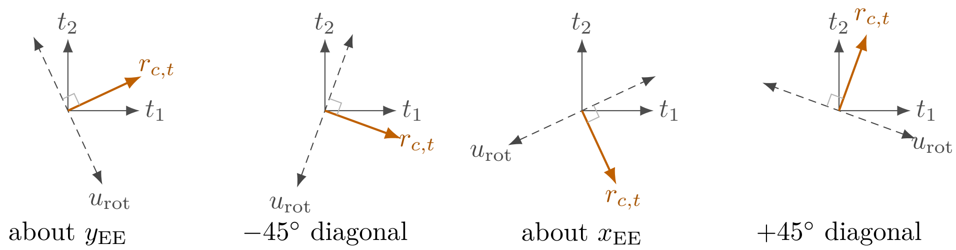


Figure 4.4: Selected tangential lever for each commanded rotation direction.

The H1 condition about y_{EE} with $r_{c,n} = 0$ also served as the zero-normal-offset reference for H3. Together, the four H1, three H2, and four additional H3 settings account for the eleven Case-C settings in table 4.2.

4.4.5 Settings for Cases D to H

Cases D to H used their own phase configuration, listed in Table 4.6. The press ran faster and deeper than in the earlier cases, and the pre-set-up gate was disabled so that approach handed over to set-up without an intermediate hold. The gate-hold impedance was therefore never active in these runs.

Table 4.6: Phase parameters for Cases D to H.

Parameter	Value	Role
Set-up push speed	0.080 m/s	Rate of the signed press coordinate.
Set-up push endpoint	+0.240 m	Final signed Cartesian spring reference.
Set-up timeout	5.0 s	Time-based set-up termination.
Minimum set-up time	2.0 s	Earliest activation of the moment-based exit.
Moment-change threshold	150 N m	Alternative set-up termination.
Pre-set-up gate	disabled	Approach hands over without stopping.
Pre-grind gate	enabled	Ends the trial once the set-up result has printed.
Translational stiffness frame	surface	K_p diagonal in $[t_1, t_2, n_s]$.

The reference reaches the endpoint after approximately 3.3 s and is held there for the remainder of the phase. The measured press settled near 81 N and lay between 76 and 85 N across the reported runs. The spread is not an independent variable: the press is generated by a spring against a commanded penetration, so a condition in which the tool rotates far presents a different contact geometry and settles a few newtons lower. The alignment comparisons therefore rest on a common commanded press rather than on an identical measured one.

The alignment criterion is evaluated while the phase runs but never ends it. A deviation below 2° held for 0.3 s marks the run as aligned and records the time; the crossing of half the deviation present at first contact is recorded separately, together with the smallest deviation reached and when. The last two are defined for every run, including those that never align, so a condition that fails is kept with its own numbers rather than left empty.

The commanded offsets were 0°, 5° and 10° about t_1, and 5° and 10° about t_2. The tangential lever was 60 mm for offsets about t_1 and 30 mm for offsets about t_2. Case G sampled the in-plane position at −50, −20, 0, +20 and +50 mm; Case H sampled the tool-axis position at −90, −60, −40, 0, +40, +60 and +90 mm. Every setting was repeated three times.

4.4.6 Null-Space Settings for the Contact Measurements

Projected damping formed the active null-space configuration throughout the surface-contact measurements. This held the redundant-motion policy constant across Cases A to H. The damping gain was

$$d_{\text{null}} = 2\,\mathrm{N\,m\,s/rad}. \tag{4.42}$$

The projected damping gain was held fixed throughout Cases A to H. Its role in the null-space torque τ_{null} is described in Section 2.8. The contact measurements establish alignment performance with this configuration. Matched pose-hold trials quantify projected damping and singular-value conditioning under a repeatable disturbance.

The recorded parameters confirmed one phase configuration throughout all reported contact runs: surface-frame translation, $K_n = 350\,\mathrm{N/m}$ toward $+0.240\,\mathrm{m}$, a 5.0 s set-up timeout, a moment threshold the contact never reached, a disabled pre-set-up gate, and null-space mode 1. The detailed values remain collected in tables 4.4, C.2 and C.4.

4.5 Contact Experiment Procedure

Before each run, the active parameter set was recorded and the robot was moved to the configured initial joint posture. Damped manual guidance could be used to adjust this posture, after which the selected start pose was captured. The tool was held by the Franka Hand using the configured grasp width of 66 mm and a grasping force of 60 N. The seating was repeated in the same manner for every measurement, while the approximately ±2° rotational play about y_{EE} remained part of the physical mounting condition.

The calibrated surface frame and grinding-face geometry were constructed before torque control began. The active Cartesian gains, damping settings, null-space parameters, and compliance-centre definition were then loaded, and the robot-side collision behaviour was configured.

During tool orientation, the TCP position was held while the calibrated grinding-face direction was rotated toward $-n_s$, including the commanded angular offsets of the active case. The controller remained in this phase for at least 0.5 s and continued to surface approach when the remaining tool-axis error was at most 0.035 rad.

Surface approach moved the Cartesian reference along the descent direction at 0.1 m/s, with a maximum commanded travel of 0.640 m. During this motion, the controller continuously calculated the point or edge of the rectangular grinding face that approached the surface first. The geometric transition was reached when the leading corner height was at most 20 mm above the calibrated plane. Corner heights agreeing within 0.1 mm were averaged according to the geometric selection described in Section 4.3.3.

At the clearance transition, the operator gate held the reached pose until the set-up motion was confirmed. The controller then stored the TCP pose, the orientation reference, the point or edge of the grinding face used for the set-up motion, its projection onto the plane, and the model-estimated external force $\hat{f}_{\mathrm{ext}}$ and moment stored as reference values. This transition was defined by the calibrated geometry; the model-estimated force $\hat{f}_{\mathrm{ext}}$ was recorded for evaluation but did not determine the transition.

During set-up, the signed press coordinate moved from its captured value, approximately −0.020 m, toward +0.240 m at 0.080 m/s along the descent direction. Cases A, B and D used the decoupled impedance, whereas Cases C and E to H applied the point-shifted 6×6 stiffness and damping matrices associated with the selected centre of compliance p_c. The orientation stored at the clearance transition remained the rotational reference, so the alignment motion arose from the contact response rather than from a time-varying tipping command.

After the minimum set-up time of 2 s, the phase could end when the norm of the model-estimated external-moment change from the clearance transition reached 150 N m. The independent 5 s timeout provided the second termination condition. The final signed coordinate is a virtual Cartesian spring reference and therefore describes the controller command rather than a measured penetration depth.

Every reported experimental set-up phase ended at the 5 s timeout; the 150 N m moment-change threshold was not reached. In this campaign, the threshold therefore functioned as an exceptional alternative bound rather than the routine set-up termination criterion.

The pre-grind gate then held the reached pose until the transition to grinding was confirmed. The controller then switched to the decoupled set-up gain matrices and maintained the final press reference. During the reported measurements, the tangential coordinates remained at their achieved set-up values.

The operator ended the run through the terminal interface. The buffered data were then written to the run-specific CSV file, and the stored parameter set was included with the measurement. Between the three repetitions of one setting, the calibrated geometry, fixed gain entries, phase timing, and procedure remained unchanged. Only the parameter assigned to the corresponding case was modified.

4.6 Evaluation Metrics and Repeated-Run Treatment

4.6.1 Alignment Change

The grinding-face direction used in the evaluation was calculated from the measured end-effector orientation and the calibrated direction expressed in the end-effector frame:

$$n_T = R_{\mathrm{EE}} n_{\mathrm{EE}}. \tag{4.43}$$

The alignment angle θ_{align} measures the angular difference between this direction and the calibrated surface:

$$\theta_{\mathrm{align}} = \cos^{-1}\left(-n_T^\top n_s\right). \tag{4.44}$$

The sign convention follows the orientation target $R_d n_{\mathrm{EE}} = -n_s$. The angle is calculated from the measured end-effector orientation and the calibrated tool geometry. Relative rotation of the tool in the gripper therefore contributes to its uncertainty.

The alignment improvement $\Delta\theta_{\mathrm{align}}$ is the initial alignment angle θ_{align} minus the final alignment angle θ_{align}:

$$\Delta\theta_{\text{align}} = \theta_{\text{align,before}} - \theta_{\text{align,after}}, \tag{4.45}$$

where the initial value is taken at the clearance transition and the final value at the end of set-up. Positive values represent a reduction of the angular difference between the grinding face and the calibrated surface.

4.6.2 Rotation, Displacement, and Model-Estimated Load

The finite set-up rotation vector $\Delta\theta_{\text{set}}$ was evaluated from

$$\Delta\theta_{\text{set}} = \phi_{\text{set}} u_{\text{set}}. \tag{4.46}$$

Here ϕ_{set} and u_{set} are the rotation angle and the unit rotation axis of the relative rotation $R_{\text{after}} R_{\text{before}}^{\top}$, obtained from its trace and its skew-symmetric part exactly as in Equation (2.16) and Equation (2.20). The vector is expressed in the robot base frame. Its magnitude is the angle turned during set-up, in radians, and its direction is the axis about which that rotation occurred.

The surface-resolved set-up rotation $\Delta\theta_{\text{set}}^{S}$ was obtained from

$$\Delta\theta_{\text{set}}^{S} = R_{\text{surface}}^{\top} \Delta\theta_{\text{set}}. \tag{4.47}$$

This representation separates the rotations about t_1, t_2, and n_s and permits direct comparison with the commanded tilt direction.

Let p_g denote the position of the point or edge of the grinding face used for the set-up motion. Its displacement vector Δp_g and displacement magnitude s_g were

$$\Delta p_g = p_{g,\text{after}} - p_{g,\text{before}}, \qquad s_g = \|\Delta p_g\|. \tag{4.48}$$

The TCP displacement vector Δp_{TCP} and its magnitude s_{TCP} were evaluated in the same form:

$$\Delta p_{\text{TCP}} = p_{\text{TCP,after}} - p_{\text{TCP,before}}, \qquad s_{\text{TCP}} = \|\Delta p_{\text{TCP}}\|. \tag{4.49}$$

The surface-frame components of both displacement vectors were recorded to show whether the angular change was accompanied by normal motion or tangential sliding.

The model-estimated external force $\hat{f}_{\text{ext}}$ was reported by libfranka. The clearance-transition reference $\hat{f}_{\text{ref}}$ was subtracted from it. The normal-force magnitude F_n was

$$F_n = \left| n_s^\top \left(\hat{f}_{\text{ext}} - \hat{f}_{\text{ref}} \right) \right| . \tag{4.50}$$

This change in the robot-model force estimate provided the supporting load quantity for matched-run comparisons. Its common calculation preserved a consistent load basis across all contact settings.

The controller orientation error e_R is referenced to the orientation stored at the clearance transition. It therefore describes deviation from the held controller reference, while θ_{align} describes the relationship between the grinding face and the calibrated surface. The latter was used as the principal alignment quantity.

4.6.3 Treatment of Repeated Runs

Each parameter setting was repeated three times. For a measured quantity y, the repetition value y_j carries the repetition index j. The arithmetic mean $\bar{y}$ was

$$\bar{y} = \frac{1}{3} \sum_{j=1}^{3} y_j . \tag{4.51}$$

The sample standard deviation s_y was

$$s_y = \sqrt{\frac{1}{2} \sum_{j=1}^{3} \left(y_j - \bar{y} \right)^2 } . \tag{4.52}$$

Cases A and C compare the alignment change $\Delta\theta_{\text{align}}$ with the corresponding stiffness coordinate, while using the rotation, displacement, and model-estimated load as supporting quantities. Case C compares direction-selected and fixed levers across four tilt directions and evaluates the normal compliance-centre coordinate.

Case D provides the reference response to commanded tilt without a lever. Cases E and F compare that response with a tangential lever named in the end-effector frame and in the surface frame. Cases G and H evaluate the alignment improvement $\Delta\theta_{\text{align}}$ and residual angle as functions of the compliance-centre position within the contact plane and along the tool axis.

The comparison was restricted to settings with matching calibrated geometry, phase timing, fixed gains, and null-space parameters. Differences were interpreted together with the within-setting variation and the zero-tilt reference reported in Section 5.1. The largest response among the sampled levels is reported as the best tested magnitude.

4.7 Secondary Null-Space Pose-Hold Study

Matched Cartesian pose-hold trials quantified the two null-space torque τ_{null} contributions. The pose-hold controller maintained the initial TCP pose, while an internally commanded joint torque τ_{dist} represented the effect of a point force applied at a specified robot-link location. This repeatable software command provided a common disturbance input for all settings.

The force point was assigned to link 3 at the link-frame offset r_p:

$$r_p = \begin{bmatrix} 0 \\ 0 \\ 0.200 \end{bmatrix} \text{m}. \tag{4.53}$$

At the initial posture q_0, the structural null direction $v_7(q_0)$ and the translational point Jacobian $J_p(q_0)$ defined the base-frame unit force direction u_f:

$$u_f = \frac{J_p(q_0)v_7(q_0)}{\|J_p(q_0)v_7(q_0)\|_2}. \tag{4.54}$$

The direction was held constant throughout the trial. The commanded disturbance-force magnitude was $F_d = 20\,\text{N}$. The disturbance force $f_d(t)$ was mapped by the current point Jacobian to the disturbance torque $\tau_{\text{dist}}(t)$:

$$\tau_{\text{dist}}(t) = J_p(q(t))^\top f_d(t), \qquad f_d(t) = F_d s(t) u_f. \tag{4.55}$$

The scale $s(t)$ increased from zero to one through a half-cosine between 5 s and 7 s, remained at one until 8 s, and returned to zero during the following second. The joint-torque-equivalent disturbance was limited to

$$\|\tau_{\text{dist}}\|_2 \leq 3\,\text{N m}. \tag{4.56}$$

Four null-space settings were tested with three repetitions each. The first setting disabled the null-space torque τ_{null}. The second used projected damping with $d_{\text{null}} = 2\,\text{N m s/rad}$. The remaining settings used singular-value conditioning with $k_\sigma = 1.5\,\text{N m}$ and $k_\sigma = 2\,\text{N m}$, respectively. The SVD probe step, sign deadband, and relative singular-value threshold remained at 0.08 rad, 10^{-6}, and 10^{-4}.

All twelve runs used the fixed pose-hold damping matrices $D_p = 50I_3\,\text{N s/m}$ and $D_R = 8I_3\,\text{N m s/rad}$. The archived configuration used these fixed matrices for pose hold.

The predefined task-retention limit was a peak Cartesian position error e_p of 2 mm. The integrated projected joint motion E_N measured the null-space motion during the disturbance:

$$E_N = \int_{5\,\text{s}}^{9\,\text{s}} \|N_q(q)\dot{q}\|_2 \,\mathrm{d}t. \tag{4.57}$$

The integral was converted to degrees for comparison with the angular quantities reported elsewhere. Singular-value conditioning was evaluated from the change in the smallest singular value σ_{min} between the initial and final states. Cartesian task retention was evaluated from the peak position-error norm during the pose hold. The acceptance outcome and measured differences between the four settings are reported in the results chapter.

The disturbance is the torque equivalent of a commanded point force. It provides a repeatable controller input for comparing the four null-space settings.

4.8 Robot-Side Monitoring and Procedural Limits

Before torque control started, the robot-side collision behaviour was configured with equal acceleration and nominal joint-torque thresholds of 140 N m. The six Cartesian force and moment components used equal thresholds of 240 N or 240 N m, according to the component type. These values define the conditions for the Franka reflex response.

The joint-torque callback returned the calculated controller command directly to libfranka. Robot-side monitoring supplied the reflex mechanism through the configured joint and Cartesian collision thresholds. These values document the operating configuration used throughout the measurements.

The motion sequence added several procedural limits. Surface approach was bounded to 0.640 m, and the transition to set-up occurred at a geometrically defined clearance of 20 mm. The operator gate held the reached pose with the translational stiffness of Equation (4.30) until the set-up motion was confirmed. Set-up was bounded by a 5 s timeout and by the alternative 150 N m change in the norm of the model-estimated external moment after the minimum phase time.

The set-up stiffness and signed press reference also define a command scale. For a completely blocked 0.060 m displacement along the calibrated normal, the blocked-reference normal-force scale $F_{n,\mathrm{block}}$ follows from Equation (4.31):

$$F_{n,\mathrm{block}} = 0.060\, n_s^{\top} K_{p,\mathrm{set}} n_s = 48.0\,\mathrm{N}, \tag{4.58}$$

and, because $K_{p,\mathrm{set}}$ is diagonal in the surface frame, a total magnitude of the same value,

$$\left\| K_{p,\mathrm{set}} \left(0.060 n_s\right) \right\|_2 = 48.0\,\mathrm{N}. \tag{4.59}$$

This value describes the virtual elastic command under a fully blocked reference. The actual robot motion reduces the tracking error, and the value is therefore a command-scale bound rather than a measured contact load.

The set-up moment threshold was applied to the change in libfranka's model-estimated external moment after subtraction of the value stored at the clearance transition. That estimate is referenced to the base frame origin, so its magnitude is dominated by the reach of the arm rather than by the load at the tool: at a press near 80 N it reaches 42 N m, where the moment referenced to the TCP is about 1 N m. The threshold therefore terminated no run, and set-up ended on its timeout throughout. It served as a controller termination condition rather than a calibrated contact-force or contact-moment boundary. The timeout, collision configuration, operator gates, logged torque command, displacement, and model-estimated wrench were considered together when assessing each run.

Chapter 5

Results and Discussion

GREEN — revised and confirmed. The chapter was revised in full and every paragraph is settled. No further editing is to be applied to it, and any remaining concern is raised as a comment rather than as a change.

comment: "In the settled prose, use "commanded tool orientation offset" for the deliberate angular command and "commanded rotation axis" or "commanded rotation direction" for its components. Retain "tilt" only for a physical inclination."

This chapter presents the measured response of the surface-contact tests and complementary Cartesian pose-hold trials. The eight contact cases introduced in table 4.2 are evaluated using the alignment change $\Delta\theta_{\text{align}}$ relative to the calibrated surface, together with the measured rotation, displacement, and model-estimated external load. The results are interpreted as answers to the three linked research questions rather than as separate case-specific findings: Cases A and B establish the stiffness response; Case C tests the tilt-dependent selection rule; and Cases D to H establish the compliance-centre direction, frame, and position behaviour.

5.1 Data Quality and Evaluation Basis

All contact runs completed the set-up phase and were included in the analysis. Each parameter setting was repeated three times, and the tables report the mean and standard deviation of the corresponding repetitions. One repetition in the 10° t_2 condition differed substantially from the other two. It was included so that the reported mean and dispersion represent the complete measured response. Its possible relation to the tool mounting is discussed in section 5.7.

The primary result is the alignment improvement $\Delta\theta_{\text{align}}$, calculated from the change during set-up in the angle between the calibrated surface normal n_s and the grinding-face direction calculated from the measured end-effector orientation. A positive value denotes a reduction of this angle and therefore an improvement in alignment. Because the tool could rotate by approximately $\pm 2°$ relative to the gripper about y_{EE}, the quantity represents the combined response of the robot, gripper, and tool. Measurement uncertainty is treated in section 5.7.

Repeatability was assessed from the three repetitions of each contact setting. The median within-setting standard deviation of the alignment change $\Delta\theta_{\text{align}}$ was 0.04° about t_1 and 0.14° about t_2. Differences smaller than approximately 0.1° are therefore treated as being of the same order as the observed experimental scatter. This value is used as a descriptive comparison scale rather than as a statistical significance limit.

The scatter is not uniform across the conditions. It is largest where the contact is near the transition between correcting and not correcting, and reaches 0.42° about t_1 and 2.35° about t_2 at single positions of Case E. Those settings are named where they are reported, and the comparison scale does not apply to them.

The reported values were calculated from settled intervals. During the final second of set-up, the median change in the model-estimated normal load was 0.10 N, and the median change in the alignment angle θ_{align} was 0.00°. The largest corresponding changes were 1.10 N and 0.60°. For settings that kept the zero compliance-centre lever r_c, the steady model-estimated normal load F_n remained between 76 and 85 N. The contact comparisons were therefore made under similar estimated preload levels, except where the lever position itself altered the contact response.

The alignment angle θ_{align} at the beginning of set-up differed from the commanded angular offset. The orientation phase leaves a steady-state error, because joint friction acts against a finite rotational stiffness and the phase exits once the error falls inside its tolerance. Raising the rotational stiffness about the surface tangents from 90 to 180 N m/rad reduced that error from 1.3 to 0.5°, at the cost of doubling the high-frequency content of the commanded joint torque during the phase, and the larger value was not carried further for that reason.

The difference is neither uniform nor always a deficit. A commanded +10° about t_1 arrived as $9.34 \pm 0.02°$ and the reversed one as $9.37 \pm 0.07°$, both short of the command, whereas a commanded +10° about t_2 arrived as $10.04 \pm 0.03°$ and overshot it. At half the command the same pattern holds, 4.29° about t_1 against 5.19° about t_2. What the two tangents share is the scatter, at most 0.07° within a group, so every condition of a comparison starts from the same attitude even where that attitude is not the commanded one.

The tables therefore report the measured angle rather than the commanded one. The commanded offset names the condition, because it is the setting that was varied and its sign is what the direction rule is read against, but every quantity computed from an angle uses what the tool actually held when contact began.

Figure 5.1 shows the angular deviation over one set-up phase for two compliance-centre positions. The deviation holds near its initial value while the press builds, falls as the tool rolls onto the surface, and settles on the residual the condition reaches. Each curve is one trial.

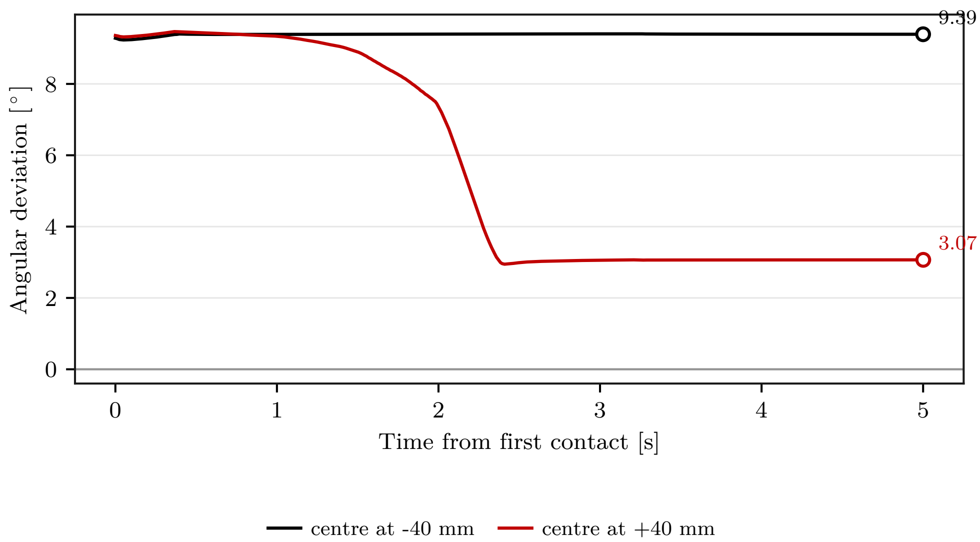


Figure 5.1: Angular deviation during set-up for two compliance-centre positions.

5.1.1 The Three Reported Quantities

Three quantities describe each run, and they answer different questions. Reporting only the first, as an earlier evaluation did, mistakes a gripper that stayed tilted for a tool that stayed tilted.

The *angular deviation* θ_{align} is the angle between the tool normal n_T and the calibrated surface normal n_s. It has an absolute zero, so a value can be read as how flat the tool is, but n_T is not measured: it is reconstructed from the end-effector orientation and a calibrated direction in that frame, $n_T = R_{\text{EE}} n_{\text{EE}}$. The pads that clamp the tool to the gripper carry play, so the tool can turn relative to the frame the reconstruction rests on. What this quantity therefore reports is the attitude of the gripper.

The *set-up rotation* $\Delta\theta_{\text{set}}$ is the rotation since first contact, resolved on the surface axes. It comes from the joint angles and the forward kinematics alone: neither the tool calibration nor the plane zero enters it, and the play cannot reach it. It has no absolute zero, so it says how far the robot turned and not where the tool ended up. This is the quantity that describes what the controller did.

The *TCP height above the plane* settles the physical outcome. A tool lying flat puts its grinding face on the surface, so the TCP stands one face offset above it; a tool resting on one edge stands higher, by the half dimension it is tilted over times the sine of the tilt. The face is 60 mm half-length across t_1 and 20 mm half-width across t_2, so a 10° tilt lifts the TCP by 10.4 and 3.5 mm respectively. Only the TCP position and the plane enter it, so like the rotation it is beyond the reach of the play, and unlike the rotation it carries an absolute zero.

Measured that way, the tool ended flat in 48 of the 54 reported conditions. The height sat within 2 mm of the flat-tool value in every one of them and 3.5 to 9.3 mm above it in the six that did not, with nothing in between, so the classification does not rest on where a threshold is placed.

Table 5.1: The conditions whose tool did not end flat. Every one commands its offset about t_1.

Condition	Excess height	$\Delta\theta_{\text{align}}$	$\Delta\theta_{\text{set}}$
Case A, 50 N m/rad about t_1	3.6 mm	+2.38	+2.73
Case B, combined about t_1	3.5 mm	+2.46	+2.80
Case E, −20 mm at +10° about t_1	5.3 mm	+0.70	+0.99
Case E, −40 mm at +10° about t_1	6.0 mm	−0.12	+0.14
Case E, +20 mm at −10° about t_1	5.2 mm	+4.43	−5.01
Case E, +40 mm at −10° about t_1	9.3 mm	−0.13	−0.09
Case F, surface frame at +10° about t_1	5.9 mm	−0.13	+0.13

The pattern is the whole of the difference between the two tangents. About t_1 the tool ends tilted whenever the rotational stiffness is raised, the compliance centre is placed on the opposing side, or the offset is named in the surface frame. About t_2 it never ends tilted, at any setting tested, including those whose deviation grew by several degrees.

The reason is the play. A commanded offset about t_2 turns the tool about the axis the pads are free in, so the contact can flatten the tool while the gripper keeps most of the commanded attitude, and the deviation records that attitude. The arithmetic closes: for every condition the deviation after set-up equals the deviation at contact plus the set-up rotation, to within 0.1°. Nothing in it describes the tool.

Taking the tool as flat, the play follows as the deviation that remains. About t_1 it is 1.1 to 1.8° in the conditions that end flat, which matches the clearance the mount was expected to carry. About t_2 it reaches 8.5 to 11.9° in the conditions where the robot does not turn. A clearance alone does not explain that, and the pads were not characterised independently, so the figure is reported as what the measurement implies rather than as a property of the mount.

What follows therefore reports the set-up rotation as the response of the controller, the flatness as the physical outcome, and the deviation as the attitude of the gripper. Where the three agree, which is every condition about t_1, any of them carries the result. Where they do not, which is every condition about t_2, only the first two mean anything.

5.2 Baseline Contact Response (Case D)

Case D places the centre of compliance on the TCP, so the commanded press makes no moment through a lever and what acts is the contact and the finite rotational stiffness. The commanded offset is 10°, applied about each surface tangent and in both signs, and nothing else differs between those four conditions. A fifth commands no offset at all.

That fifth condition says what the deviation reads when nothing is asked of the tool. The gripper arrives at 0.74° and the press leaves it at 2.30°, so pressing turns the gripper by $1.56 \pm 0.01°$ while the tool stays on the surface. The deviation therefore does not fall to zero even in the condition that commands nothing, which is the clearest single statement of what it measures.

Table 5.2: Case-D response with the centre of compliance on the TCP (mean ± SD, n=3). The rotation is signed about the commanded tangent; flatness is read from the TCP height above the plane.

Commanded offset	Initial deviation	$\Delta\theta_{\text{set}}$	$\Delta\theta_{\text{align}}$	Tool flat
None	0.74	−0.97	-1.56 ± 0.01	yes
+10° about t_1	9.32	+7.57	$+5.94 \pm 0.03$	yes
−10° about t_1	9.43	−10.83	$+7.19 \pm 0.24$	yes
+10° about t_2	10.07	−1.63	-1.74 ± 0.17	yes
−10° about t_2	9.59	−1.25	$+1.20 \pm 0.19$	yes

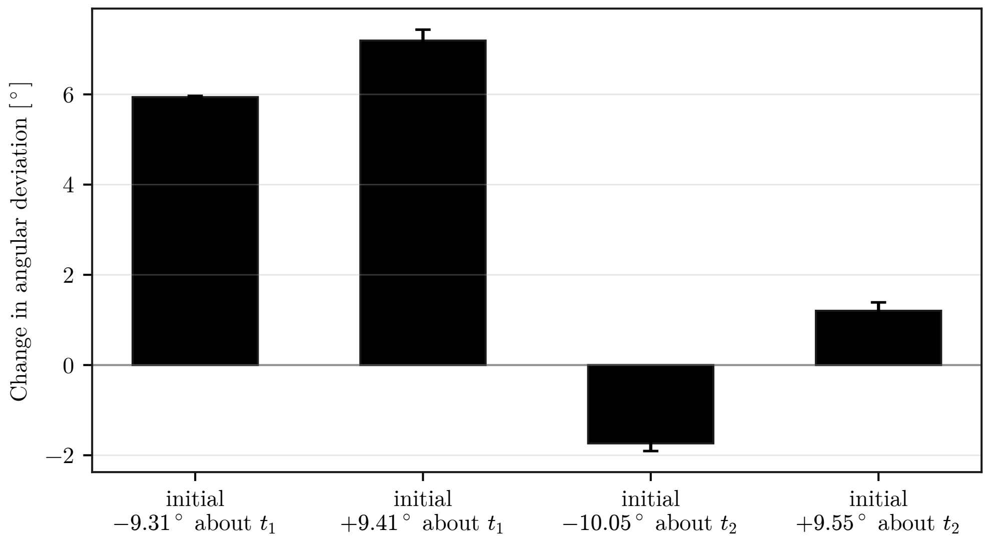


Figure 5.2: Alignment change with the centre of compliance on the TCP, by commanded offset.

The tool ends flat in all five conditions. The contact by itself brings the grinding face onto the surface whatever the commanded offset, which is the outcome the set-up phase exists to produce, and no setting of this case prevents it.

What differs between the tangents is who provides the rotation. About t_1 the robot turns 7.57 and $-10.83°$ at the two signs, close to the offset it was given, so the impedance carries the motion. About t_2 it turns -1.63 and $-1.25°$, a small fraction of the commanded offset, and the remainder is taken up between the tool and the gripper. The deviation records the gripper and therefore reads as a failure about t_2, from 10.07 to 11.81°, where the tool itself lies flat.

A commanded offset about t_2 turns the tool about the axis the mount is free in, which is what allows the contact to flatten it without the robot moving. The asymmetry between the tangents is therefore a property of the mount and not of the grinding face, and it sets what the compliance centre has to achieve on each axis: about t_1 the robot already does the work, whereas about t_2 it has to be made to do it.

5.3 Compliance-Centre Lever Variations

5.3.1 Case E: Tangential Lever Direction and Sign

Case E moves the centre of compliance along the tangent perpendicular to the commanded rotation axis, which is the direction eq. (2.59) selects. Seven positions are sampled from -40 to $+40$ mm, and the sign of the commanded offset is reversed at every one of them. The positive direction is the one that assists a positive commanded offset, so a rule that follows the sign of the offset appears as a mirror between the two.

Table 5.3: Case-E alignment change by centre position along the assisting tangent (mean ± SD, n=3).

Commanded offset	Initial	-40	-20	-10	0	$+10$	$+20$	$+40$ mm
$+10°$ about t_1	9.34	-0.12	$+0.70$	$+5.69$	$+5.94$	$+5.84$	$+6.08$	$+6.19$
$-10°$ about t_1	9.37	$+6.22$	$+6.08$	$+6.08$	$+7.19$	$+6.30$	$+4.43$	-0.13
$+10°$ about t_2	10.04	-4.60	-2.06	-1.95	-1.74	-1.43	-1.24	$+4.18$
$-10°$ about t_2	9.59	$+7.65$	$+1.46$	$+1.25$	$+1.20$	$+1.24$	$+1.23$	-4.71

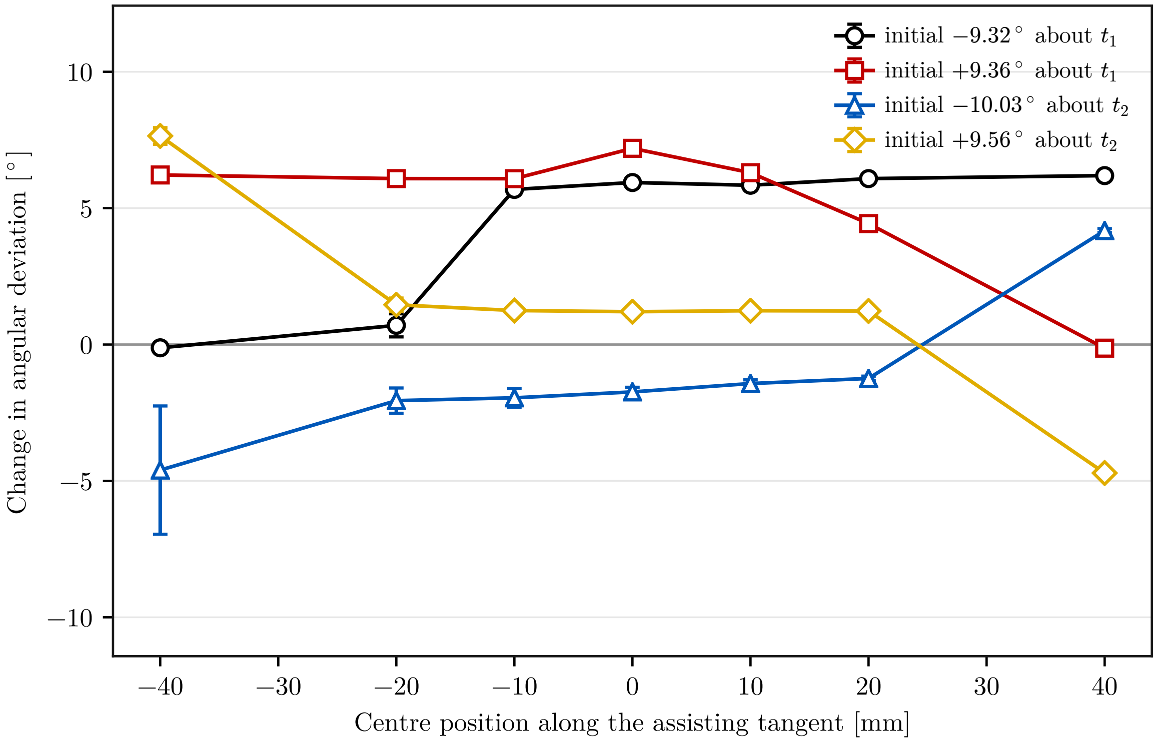


Figure 5.3: Alignment change by centre position along the assisting tangent, for both surface tangents and both signs of the commanded offset.

The response reverses with the sign of the commanded offset on both tangents. About t_1 the positive offset improves by +6.19° at +40 mm and by −0.12° at −40 mm, while the negative offset gives −0.13 and +6.22° at the same two positions. About t_2 the same exchange holds, between +4.18 and −4.60° for the positive offset and −4.71 and +7.65° for the negative one. The direction that assists is therefore selected by the signed commanded rotation direction.

The two tangents differ in what the wrong direction costs, and the flatness separates them. About t_1 the opposing direction leaves the tool tilted: at −20 and −40 mm the TCP stands 5.3 and 6.0 mm above the flat-tool height, and at +20 and +40 mm of the reversed offset 5.2 and 9.3 mm. These are the only conditions of the case in which the tool does not reach the surface. About t_2 it ends flat at every position, so what the opposing direction costs there is not the flatness but the share of the rotation the robot performs.

Read as the rotation the robot made, the sweep about t_2 runs from −4.51 at −40 mm through −1.63 at the TCP to +4.43° at +40 mm. The offset moves the flattening from the mount to the robot, and at the opposing end it drives the robot against the motion the contact is producing. **comment:** “Clarify that the tested lever about t_2 did not rotate the robot far enough to remove the full initial deviation. A tangential lever is still required to align the EE-inferred orientation from a t_2 initial deviation; the grinding face nevertheless ended flat because the remaining rotation was absorbed within the tool mount. Carry the same distinction into the conclusion."

What the offset achieves also differs between the tangents. About t_1 the assisting side is level: from -10 to $+40$ mm the rotation stays between $+7.45$ and $+7.87°$, against $+7.57°$ with the centre on the TCP. Displacing the centre adds nothing there that the contact was not already delivering through the robot. About t_2 it is what makes the robot move at all, carrying the rotation from -1.63 to $+4.43°$ at the positive offset and from -1.25 to $-10.51°$ at the negative one.

The transition between the two regimes is not gradual. About t_1 at a positive offset the change collapses from $+5.69$ to $+0.70°$ between -10 and -20 mm, and the three repetitions at -20 mm scatter by $0.42°$ against a median within-setting scatter of $0.05°$ elsewhere. The -40 mm position of the positive t_2 condition scatters by $2.35°$ for the same reason. Near the transition the contact resolves one way or the other from nominally identical starts.

Figure 5.4 reads one comparison in the order the load acts: the deviation first, then the press along the surface normal, then M_{t_1}, the moment about the commanded tangent, each as the controller commanded it and as the robot model estimated it. The press is negative because n_s points out of the plate, and it reaches -77 to -82 N in all three conditions, so they are compared under the same normal load.

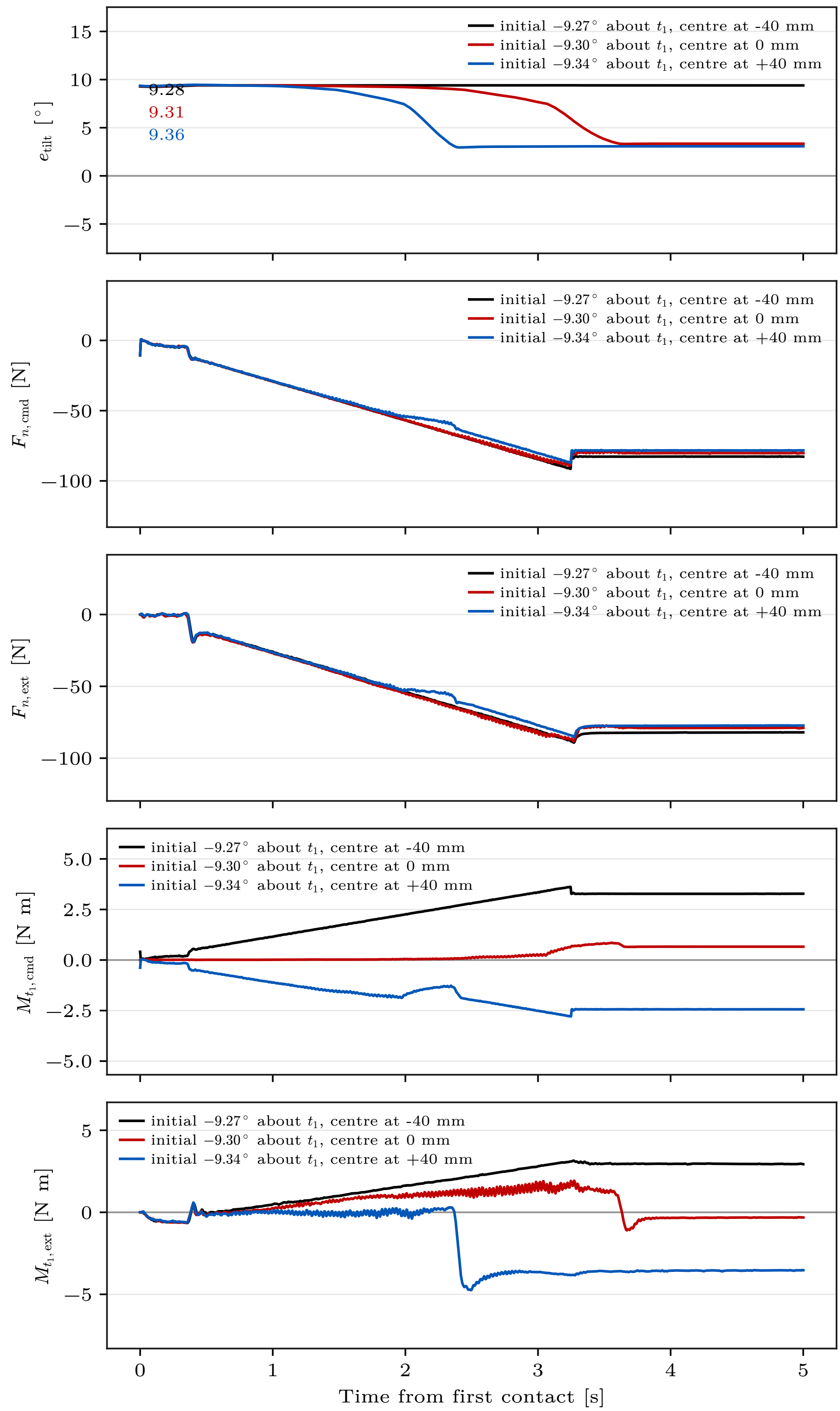


Figure 5.4: Deviation, normal press F_n and moment M_{t_1} for three centre positions, each commanded and model-estimated. All three carry a commanded offset of $+10°$ about t_1; the deviation each arrived with is printed beside the start of its curve.

Both moments are referred to the TCP, which is the only point at which the commanded and the estimated one compare without a contact model between them. They track each other closely. At -40, 0 and $+40$ mm the commanded moment settles at $+3.28$, $+0.66$ and -2.44 N m and the estimated one at $+2.94$, -0.32 and -3.54 N m, so the environment returns within about 1 N m of what the controller asks for.

The two agree in sign at the outer positions and disagree only at the centred one, where the commanded moment is $+0.66$ and the estimated -0.32 N m. Neither is the negative of the other, because the estimate reports the external wrench in the same sense as the command: a press directed into the plate is negative in both, so agreement means the contact is returning what the controller applies. What separates them is the moment the estimate does not attribute to the contact, 0.34 to 1.10 N m across the three conditions, which is where joint friction and the error of the model itself appear. At the centred position that gap is larger than either value, so the sign of the smaller pair is not resolved by the measurement; at the outer positions the moments exceed it several times over and the sign is.

What the centre position changes is the sign of that moment, not its size. The magnitudes at the two outer positions are alike, and the conditions end at 9.39 and 3.07°. The larger moment is the one that leaves the tool tilted. Under a common press this rules out both the load and the size of the moment as what separates the conditions, and leaves the direction in which it acts.

Referring the same two wrenches to the centre of compliance instead gives a nearly constant pair: $+0.66$, $+0.69$ and $+0.01$ N m commanded against -0.32, -0.46 and -0.29 N m estimated. That is the point shift working as constructed: the centre sees a block-diagonal impedance, so the moment there is the rotational spring acting on the orientation error and little else. The displacement does not change what the centre carries; it changes where that point sits relative to the contact, and with it the moment at the TCP. Figure 5.5 resolves both referenced moments and the rotation on the surface axes for the two extremes of the same comparison.

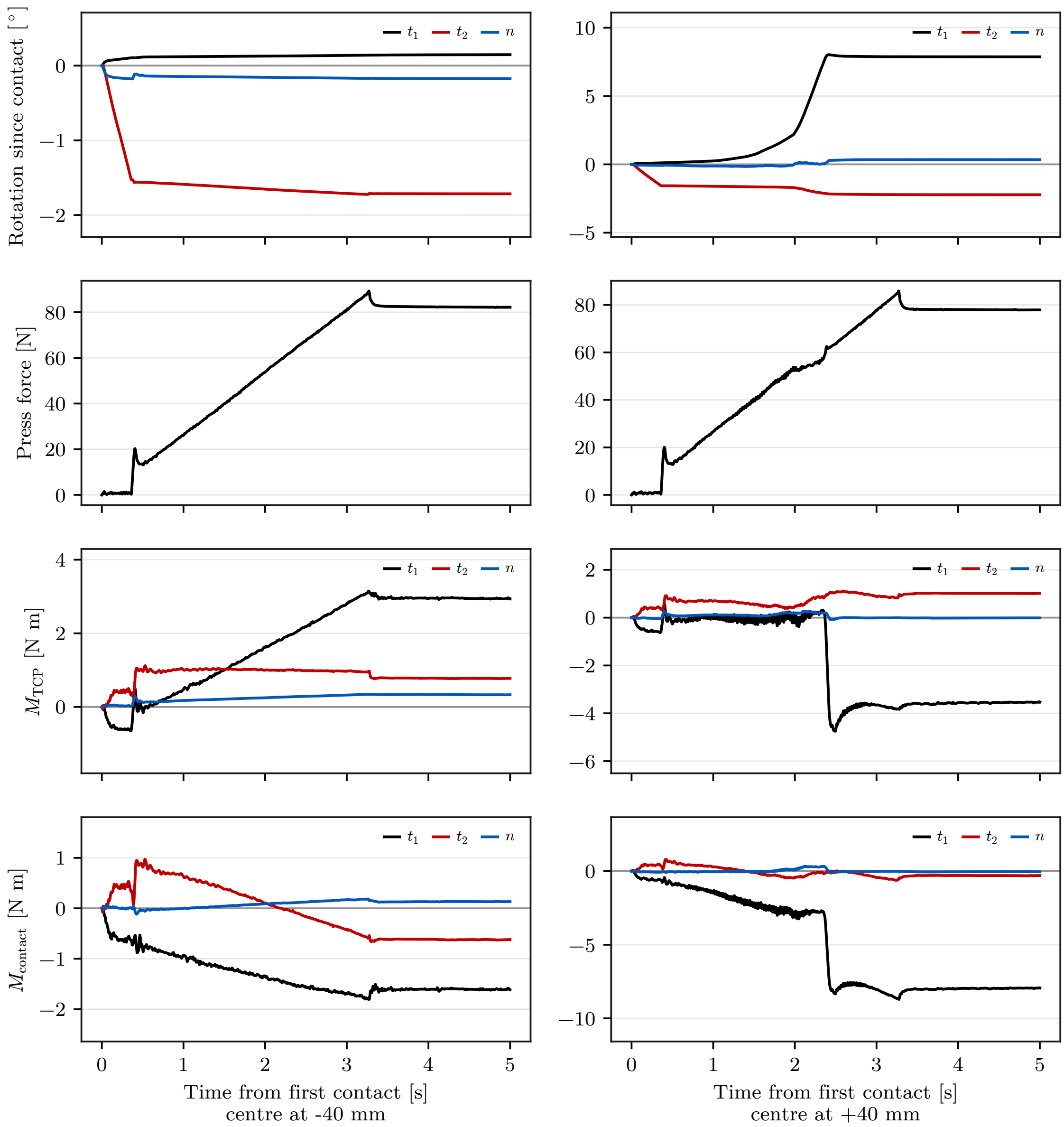


Figure 5.5: Rotation and model-estimated external force and moments during set-up.

5.3.2 Case G: Position Along the Tool Axis

Case G moves the centre of compliance along the tool axis at a fixed commanded offset of +10° about t_1, which the tool arrived with as $9.31 \pm 0.03°$. A press normal to the surface makes no moment about a lever parallel to it, so a first-order coupling model leaves this coordinate without effect and any dependence on it is not produced by the normal component of the press alone.

Table 5.4: Case-G alignment change by centre position along the tool axis (mean ± SD, n=3).

Position	−40	−20	−10	0	+10	+20	+40 mm
$\Delta\theta_{\text{align}}$	+5.37	+5.77	+5.84	+5.94	+5.94	+6.09	+6.25
Standard deviation	0.04	0.03	0.02	0.03	0.03	0.06	0.09
Residual	3.97	3.55	3.48	3.38	3.36	3.17	3.07

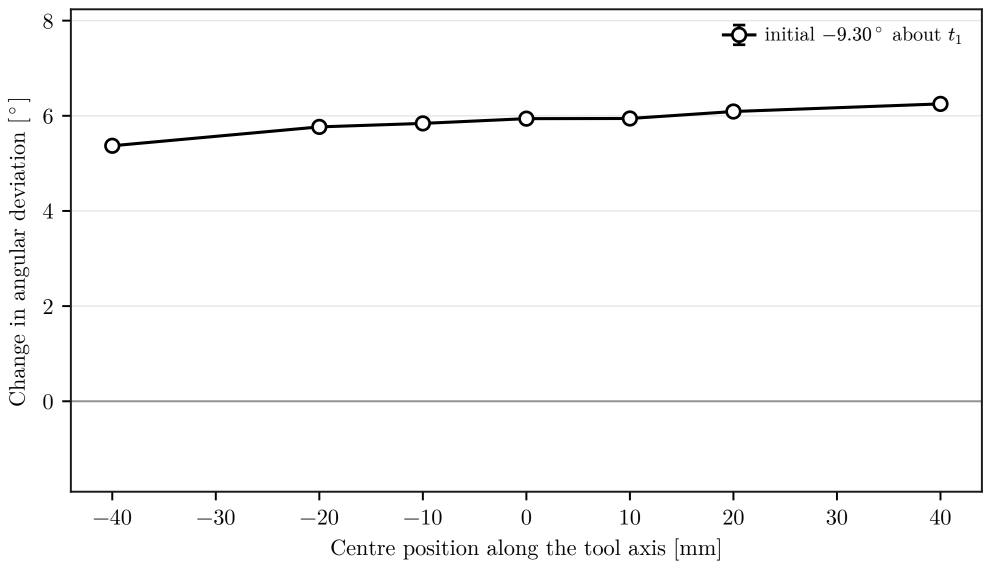


Figure 5.6: Alignment change by centre position along the tool axis.

The coordinate does act, and it acts monotonically: the change rises from +5.37 to +6.25° across the sampled range and the rotation the robot makes from +7.26 to +7.60°. The tool ends flat at every position, so the coordinate changes how the flattening is shared rather than whether it happens. The whole span is 0.88°, against the 6.3° the tangential direction of Case E carries at the same commanded offset. The zero-moment statement holds to first order and is not exact: a purely normal press is an idealisation, and tangential force components, the finite rotation, the evolving contact geometry and compliance in the tool mounting each leave the coordinate something to act through.

5.3.3 Case F: Frame in Which the Offset Is Defined

Case F names the same physical offset in surface coordinates instead of end-effector coordinates. At zero commanded offset the two definitions describe one point; as the tool turns they separate, because a surface-fixed point keeps its direction while a tool-fixed one follows the tool.

Table 5.5: Case-F alignment change by the frame the offset is named in (mean ± SD, n=3).

Commanded offset	Initial	Tool frame (Case E)	Surface frame
None	0.76	-1.09 ± 0.04	-1.71 ± 0.17
$+10°$ about t_1	9.33	$+6.19 \pm 0.09$	-0.13 ± 0.01

At 10° about t_1 the two definitions separate by 6.32°: the surface-frame definition leaves $-0.13 \pm 0.01°$ where the tool-frame definition gives $+6.19 \pm 0.09°$ at the same distance. Naming the offset in surface coordinates therefore does not weaken the response, it removes it, and removes with it the part the contact produces on its own. Resolved along the surface axes the rotation about the commanded tangent is +0.13°, so the tool did not turn toward the surface at all.

At zero commanded offset the two definitions describe one point, so they are expected to agree, and they do not: the tool-frame trial gave $-1.09 \pm 0.04°$ against $-1.71 \pm 0.17°$ in surface coordinates, a difference of 0.62°. That is six times the descriptive comparison scale and larger than either scatter. Both trials carry the same commanded spin, the same distance and the same assisting direction, so the geometry does not account for it; what remains is the tool-mount uncertainty of section 5.7, which acts on the residual and not on a difference between conditions. The zero-offset pair therefore does not deliver the consistency check it was measured for, and the separation at 10° should be read as a comparison between two conditions rather than as a deviation from an established agreement.

5.3.4 Case H: Dependence on the Commanded Magnitude

Case H repeats the zero and the assisting position at half the commanded offset, +5° instead of +10°, on both tangents. It separates what the compliance centre does from what the size of the commanded offset does.

Table 5.6: Alignment change at both commanded magnitudes (mean ± SD, n=3).

	About t_1		About t_2	
Centre position	+5°	+10°	+5°	+10°
Initial deviation	4.29	9.34	5.19	10.04
0 mm	+1.53	+5.94	−1.58	−1.74
+40 mm	+1.63	+6.19	+3.39	+4.18

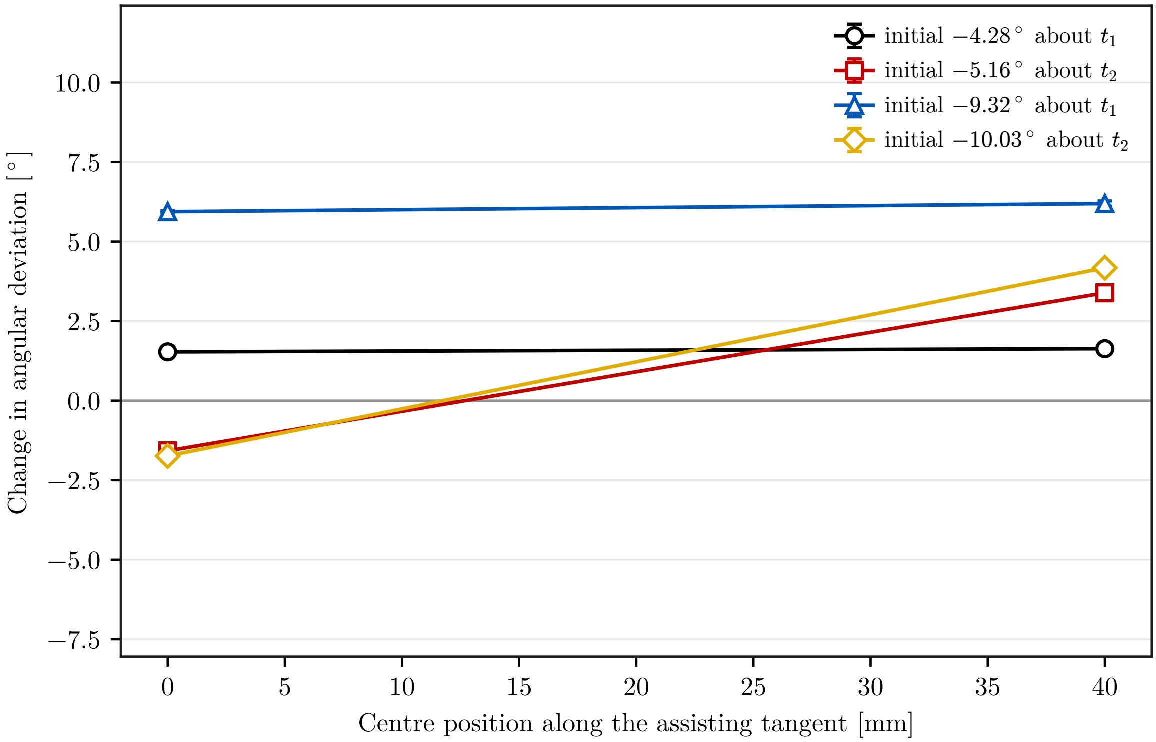


Figure 5.7: Alignment change at both commanded magnitudes, by centre position along the assisting tangent.

The pattern of Case E survives the halving. About t_1 the centre position changes little at either magnitude, 0.10° at 5° and 0.25° at 10°. About t_2 it decides the sign of the response at both, carrying −1.58 to +3.39° at the smaller offset and −1.74 to +4.18° at the larger one. Read as the rotation the robot made, the same pair runs from −1.40 to +5.91° and from −1.63 to +4.43°. The tool ends flat in all eight conditions.

The rotation is not proportional to the commanded offset. About t_1 the robot turns +7.57° for a commanded 10° and +2.55° for a commanded 5°, so halving the offset removes two thirds of the motion rather than half. The gripper is left 3.38° from flat in the first case and

2.76° in the second, and the mount carries the difference in both. What the commanded offset changes is therefore mostly how much of the flattening the robot performs.

5.3.5 Case C: Generalisation Across Rotation Directions

Case C commands the offset in directions between the two surface tangents and asks whether the compliance-centre offset has to turn with it. Each direction is run twice: once with the offset selected for that direction by the rule of eq. (2.59), and once with the offset held at the one that assists a command about t_1. The two principal directions are the corresponding trials of Case E.

Table 5.7: Case-C response with the offset selected for the commanded direction against one held fixed (mean, n=3).

Commanded direction	$\Delta\theta_{\text{set}}$		$\Delta\theta_{\text{align}}$	
	selected	fixed	selected	fixed
$-45°$ from t_1	+4.46	+4.39	+5.30	+4.18
$+45°$ from t_1	+4.85	+4.80	+1.51	−0.12
$+90°$, about t_2	+4.43	−1.61	+4.18	−1.62

The selected offset gives the larger response at every direction. The margin is 1.12° at −45°, 1.63° at +45° and 5.80° about t_2, and the tool ends flat in all six conditions.

The two diagonals separate the rule from the axis it was derived on. At the diagonals the set-up rotation is almost the same whether the offset is selected or fixed, 4.46 against 4.39° and 4.85 against 4.80°: the fixed offset still has a component along the direction the rule would choose, because the two are 45° apart. About t_2 they are perpendicular, the fixed offset has no such component, and the response collapses from +4.43 to −1.61°. The cost of not turning the offset with the command therefore grows with the angle between them, which is what a vector rule predicts and what a rule tied to one axis would not.

5.4 Stiffness Variations

5.4.1 Case B: Effect of Translational Stiffness

Case B varies the translational stiffness across the commanded tangent, the entry that resists the translation accompanying the alignment motion, and adds one condition combining the lowest translational entry with the highest rotational one.

Table 5.8: Case-B response by translational stiffness across the commanded tangent, and the combined condition (mean, n=3).

Commanded offset	Quantity	300	800	2000 N/m	Combined
+10° about t_1	$\Delta\theta_{\mathrm{set}}$	+7.59	+7.56	+7.57	+2.80
	$\Delta\theta_{\mathrm{align}}$	+6.26	+6.05	+5.94	+2.46
	tool flat	yes	yes	yes	**no**
+10° about t_2	$\Delta\theta_{\mathrm{set}}$	−2.24	−1.78	−1.63	−1.03
	$\Delta\theta_{\mathrm{align}}$	−2.33	−1.89	−1.74	−1.15
	tool flat	yes	yes	yes	yes

The translational entry does almost nothing. Across a factor of nearly seven, from 300 to 2000 N/m, the set-up rotation about t_1 moves by 0.02° and the tool ends flat at every setting. About t_2 the rotation moves by 0.61°, and in the direction that makes the deviation worse rather than better.

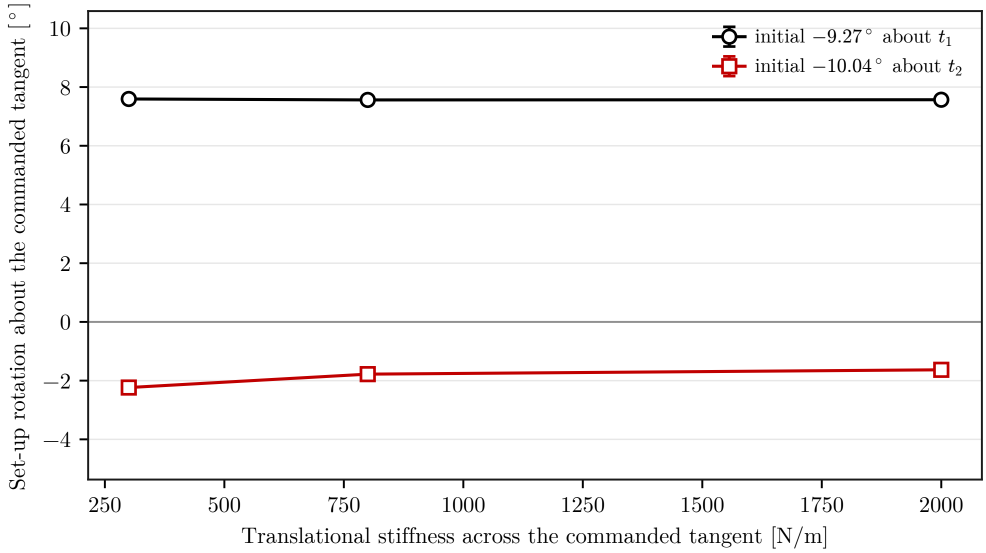


Figure 5.8: Set-up rotation about the commanded tangent by translational stiffness across it.

The combined condition is not a combination of two effects. Its response, +2.80° about t_1, matches the +2.73° of the rotational entry alone at the same value, and the tool ends tilted in both. The rotational entry accounts for the result and the translational one adds nothing measurable to it, which is the reading the separate variations already support.

5.4.2 Case A: Effect of Rotational Stiffness

Case A varies the rotational stiffness about the commanded tangent and leaves every other entry where the design places it. The 5 N m/rad column is the corresponding condition of Case D and is not a separate measurement.

Table 5.9: Case-A response by rotational stiffness about the commanded tangent (mean, n=3).

Commanded offset	Quantity	5	15	50 N m/rad
+10° about t_1	$\Delta\theta_{\text{set}}$	+7.57	+7.29	+2.73
	$\Delta\theta_{\text{align}}$	+5.94	+6.08	+2.38
	tool flat	yes	yes	**no**
+10° about t_2	$\Delta\theta_{\text{set}}$	−1.63	−1.52	−0.48
	$\Delta\theta_{\text{align}}$	−1.74	−1.63	−0.60
	tool flat	yes	yes	yes

About t_1 the rotational entry decides whether the tool ends flat. At 5 and 15 N m/rad the robot turns 7.6 and 7.3° and the tool lies flat. At 50 N m/rad it turns only 2.7° and the tool ends tilted, with the TCP standing 3.6 mm above the flat-tool height. The spring that holds the attitude captured at first contact is then strong enough to keep the tool off the surface.

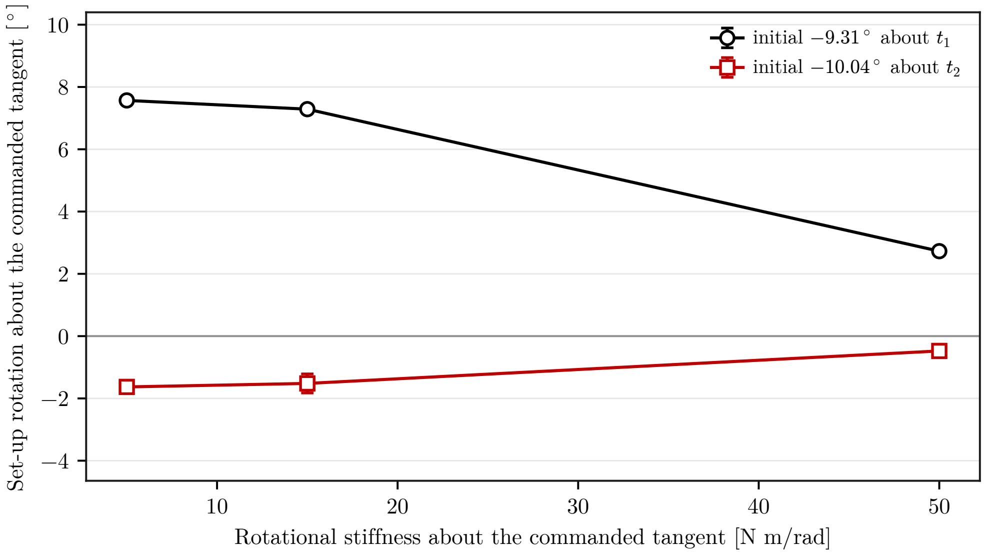


Figure 5.9: Set-up rotation about the commanded tangent by rotational stiffness.

About t_2 the entry changes little, because the robot barely turns at any of the three settings. The tool ends flat throughout, and what the deviation records is how much of the commanded attitude the gripper kept.

5.5 Null-Space Pose-Hold Results

Matched Cartesian pose-hold trials evaluated the null-space terms under a repeatable internally commanded disturbance. The point-force equivalent reached 20.00 N in all twelve trials. Its peak equivalent joint-torque norm lay between 2.17 and 2.45 N m. The torque-limit scale remained equal to one, so the requested waveform was applied without clipping. All trials completed and were included in the analysis.

Projected null-space damping reduced the cumulative redundant joint excursion from $(7.596 \pm 0.916)°$ without null-space torque τ_{null} to $(5.687 \pm 0.228)°$ at 2 N m s/rad. This corresponds to a reduction of approximately 25 % relative to the zero-torque condition.

The final change in the minimum singular value $\sigma_{\min}$ also became less negative, from $(-2.13 \pm 0.47) \times 10^{-3}$ to $(-1.26 \pm 0.10) \times 10^{-3}$.

In both sigma-only settings, the minimum singular value $\sigma_{\min}$ was approximately at its initial level by the end of the trial. Its final change was $(1.95 \pm 0.07) \times 10^{-5}$ at $k_\sigma = 1.5\,\mathrm{N\,m}$ and $(1.93 \pm 0.03) \times 10^{-5}$ at $2\,\mathrm{N\,m}$. Both commands therefore achieved the same final singular-value recovery. The $1.5\,\mathrm{N\,m}$ setting did so with lower cumulative excursion, $(0.283 \pm 0.007)°$ rather than $(1.619 \pm 0.130)°$, and a lower mean peak Cartesian position error e_p, $0.889 \pm 0.024\,\mathrm{mm}$ rather than $0.983 \pm 0.053\,\mathrm{mm}$. The largest position error e_p measured in the complete pose-hold study was $1.304\,\mathrm{mm}$, below the predefined $2\,\mathrm{mm}$ acceptance limit.

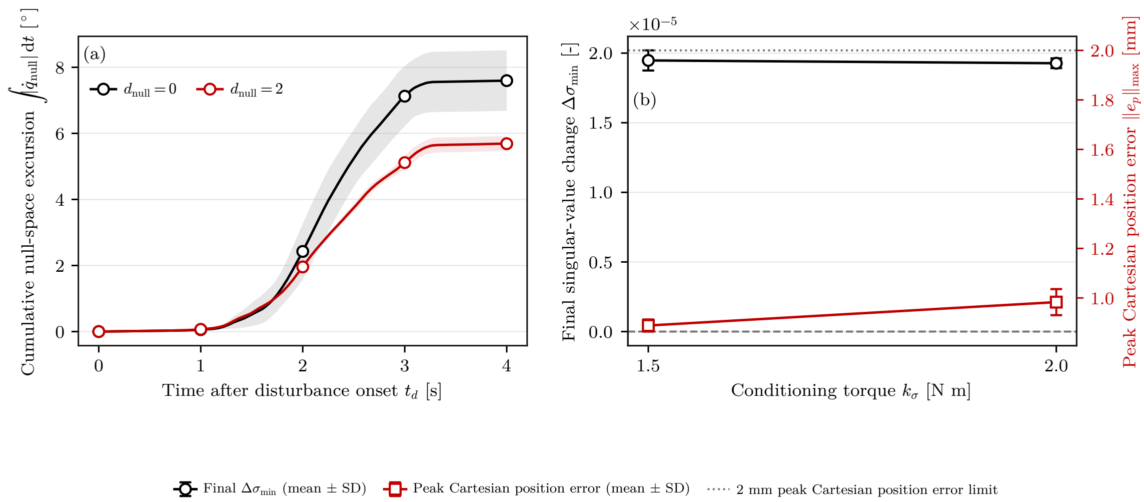


Figure 5.10: Null-space response during the Cartesian pose-hold disturbance test. In (a), the solid curves show the mean cumulative null-space excursion and the shaded bands indicate ± one standard deviation across the three repetitions.

Matched pose-hold trials isolate the projected damping and sigma-torque terms under a repeatable internally commanded disturbance. The contact measurements establish alignment performance with projected damping as the active null-space configuration.

5.6 Discussion

5.6.1 Relative Influence of the Varied Parameters

The measured effect sizes differed substantially across the varied parameter families. For t_1, increasing the tilted-axis rotational stiffness from 5 to $50\,\mathrm{N\,m/rad}$ reduced the alignment improvement $\Delta\theta_{\text{align}}$ by 0.15°. For t_2, the highest tested rotational stiffness reduced the response by 0.44° relative to the 15 N m/rad condition. Reducing the perpendicular translational stiffness from 2000 to 300 N/m changed the response by −0.06° about t_1 and by +0.73° about t_2.

The compliance-centre position p_c produced the largest measured changes. Across Case E the alignment change ranges from −4.60 to +7.65°, a span of 12.25°, against at most 0.44° from either stiffness family over its tested range.

Both spans were measured under the same press and at the same commanded offset, so the comparison is controlled. Within the tested configuration and parameter range the compliance-centre position therefore had the stronger influence.

The measurements support considering the compliance-centre lever r_c before refining the stiffness entries. The rotational stiffness controls the angular resistance of the impedance, and the perpendicular translational stiffness can modify the response for particular directions. The lever, however, directly determines whether the normal press produces a moment that assists or opposes the required rotation.

5.6.2 Role of the Compliance-Centre Lever

The sign dependence observed in Case E follows from the point-shifted impedance. With the press directed approximately along the negative surface normal n_s, $f_{\text{press}} \approx -F_n n_s$, the coupling moment is

$$m_c = -r_c \times f_{\text{press}}. \tag{5.1}$$

The tangential lever $r_{c,t}$ follows the rule derived in eq. (2.59), perpendicular to the commanded rotation axis within the tangent plane:

$$r_{c,t} = -\rho_c t_2 \quad \text{about } t_1, \qquad r_{c,t} = +\rho_c t_1 \quad \text{about } t_2, \qquad \rho_c > 0. \tag{5.2}$$

With $f_{\text{press}} \approx -F_n n_s$, this selection gives

$$m_c \approx -F_n \rho_c t_1 \quad \text{about } t_1, \qquad m_c \approx -F_n \rho_c t_2 \quad \text{about } t_2. \tag{5.3}$$

The induced moment is therefore opposite to the commanded rotation in both cases. The lever lies along the tangent perpendicular to the commanded rotation axis. Reversing the orientation offset reverses the commanded rotation axis and the selected lever with it. Case E tested this vector rule on both surface tangents and at both signs of the commanded offset, and Case C four directions in the tangent plane.

A lever component along the tangent perpendicular to the commanded rotation axis therefore creates a moment about that axis. Reversing the lever reverses the moment. One sign assists the rotation that brings the grinding face towards the surface, whereas the opposite sign opposes it. Exchanging the roles of t_1 and t_2 changes the cross-product direction, which is consistent with the assisting lever signs being opposite for the two tilted axes.

The component of r_c parallel to the press direction contributes no moment because its cross product with f_{press} is zero. A centre displaced along the tool axis is close to that case, because the tool stands nearly normal to the surface. At a 10° tilt, a 90 mm axial displacement has a tangential projection of approximately 15.6 mm, about one quarter of the 40 mm tangential offset. To first order the coordinate should therefore do little.

The normal coordinate nevertheless changes the response. Case C measured a 0.9° range along n_s, and Case G a 0.88° range along the tool axis, rising monotonically from +5.37 to +6.25° across the sampled positions. The zero-moment statement assumes a purely normal force. Tangential force components, finite translation and rotation, the evolving contact geometry, and compliance within the tool mounting could each make the coordinate act. The point-shifted transformation also changes the effective rotational resistance when a normal offset is introduced.

For an axial displacement d and a tilt angle θ, the effective tangential component is $d \sin \theta$. At 10°, an axial displacement of approximately 345 mm would be required to produce a tangential component of 60 mm. The tested axial placement therefore generated substantially less coupling moment than the direct tangential placement.

The lower model-estimated normal loads F_n at the assisting lever positions are consistent with the same mechanism. A larger alignment motion allows part of the commanded displacement to be accommodated through rotation, reducing the remaining translational compression. This explanation remains model-based because the external load was estimated by the robot model. Relative tool–gripper rotation remains part of the alignment uncertainty.

The controller regulated pose through Cartesian impedance rather than controlling the contact force directly. Changing the compliance-centre lever r_c therefore also changed the resulting contact load, so the measured improvement characterises the complete coupled contact response rather than an isolated moment effect.

The lever is a commanded setting and does not change sign on its own during a run. Case E measured the consequence of its sign directly, and at both signs of the commanded offset. About t_1 the response moved between $+6.19$ and $-0.12°$ as the centre was carried from one side of the TCP to the other, and reversing the commanded offset exchanged the two ends. About t_2 the same exchange spans $+4.18$ to $-4.60°$. The signed commanded rotation direction therefore defines the assisting offset direction. For every reported run, that direction was selected before set-up from the commanded rotation direction.

5.6.3 What the Contact Contributes Without a Lever

Case D separates the contact from the coupling. With the centre of compliance at the TCP the commanded press produces no moment, and what remains is the response of the contact and the finite rotational stiffness. That response is large about t_1: 5.94° of the measured 9.32° was recovered with the centre on the TCP, and 7.19° of 9.43° at the reversed offset.

About t_2 the same contact recovers almost nothing at one sign, $+1.20°$, and increases the misalignment at the other, from 10.07 to 11.81°. The contact face is 20 mm half-width across t_2 and 60 mm across t_1, so the two directions present different support to a pressing tool. The measured asymmetry follows the face geometry rather than the commanded offset.

This changes what the compliance centre has to achieve on each axis. About t_1 a correction is already present and the centre adds almost nothing to it: across the whole assisting side of Case E the response stays between $+5.69$ and $+6.19°$, against $+5.94°$ with the centre on the TCP. About t_2 the centre is what produces the correction at all, carrying -1.74 to $+4.18°$ at the positive offset and $+1.20$ to $+7.65°$ at the negative one. The same displacement is therefore decisive on one tangent and irrelevant on the other.

5.6.4 Frame and Magnitude of the Lever

Case F separates the frame the offset is named in from the offset itself. At 10° about t_1 the two definitions differ by 6.32°: the surface-frame definition leaves −0.13° where the tool-frame one gives +6.19° at the same distance, and where the contact alone gives +5.94°. Naming the point in surface coordinates removes the correction rather than weakening it.

The usable magnitude is set by the face dimension the tool rocks over. About t_2 an offset of 60 mm carried a 5° commanded offset out to a 20.7° excursion, and 80 mm aborted the motion on a wrist-torque reflex, which is why the reported sweeps stop at 40 mm. About t_1, where the face is three times longer, the same distances stayed in range throughout. The magnitude that suits one tangent therefore does not suit the other.

5.6.5 Implications for Controller Parameter Selection

The selection below is quasi-static. It rests on the balance between the moment the press produces through the lever and the restoring stiffness that opposes the rotation, both evaluated at the settled end of set-up, and it says nothing about the transient between contact and that state. Two things justify treating the problem this way here. The reported values were read from settled intervals, over which the load and the angle no longer changed. And the range over which a setting may be varied is itself bounded statically: at a tool-axis displacement of 120 mm the commanded torque of the sixth joint reached 21.4 N m against its limit of 12 N m and the run aborted, while across the reported sweep that joint stayed near 6 N m and rose to 15.1 N m only at the last position. A selection method for this arrangement therefore has to respect a wrist torque before it optimises a response.

The measured ordering suggests a practical sequence for selecting controller parameters within a similar contact geometry. The compliance-centre lever r_c is considered first because it determines the direction and approximate magnitude of the coupling moment. Its tangential direction should be perpendicular to the commanded rotation axis, and its sign should be selected so that the induced moment assists alignment.

After the lever has been selected, the rotational stiffness about the commanded rotation axis can be refined. Within the tested range, increasing this stiffness reduced the available alignment motion. The perpendicular translational stiffness can then be adjusted where the measurements show a directional benefit. In the present tests, reducing it increased the response about t_2, while the t_1 variation remained within the descriptive comparison scale.

This measured sequence provides an initial tuning procedure for the investigated contact geometry and parameter range. Cases C and E establish the direction and sign rule at the tested conditions, and Cases E and G establish where the centre should sit along each coordinate. On the tangent the best position depends on the sign of the commanded offset; along the tool axis it was the largest sampled displacement, where the response was still rising.

Table 5.10: Case-C evaluation of the direction rule at a common 60 mm magnitude.

Commanded rotation axis	Selected $[r_{c,t_1}, r_{c,t_2}]$	Alignment improvement	Fraction removed
About y_{EE}	$[+54.4, +25.4]$ mm	$+2.97 \pm 0.12$	42 %
$-45°$ diagonal	$[+56.4, -20.5]$ mm	$+7.10 \pm 0.07$	77 %
About x_{EE}	$[+25.4, -54.4]$ mm	$+2.27 \pm 0.03$	33 %
$+45°$ diagonal	$[+20.5, +56.4]$ mm	$+3.47 \pm 0.07$	47 %

The selected lever assisted in all four Case-C directions. The common magnitude isolates the vector direction rule in eq. (5.2). The measured fractions describe the direction-selection response at this shared magnitude.

Where the sign of the initial tilt is unknown, a fixed tangential lever can assist one direction and oppose the other, and Case E measured the cost of the wrong sign at both: about t_1 the response fell from $+6.19$ to $-0.12°$, and about t_2 from $+4.18$ to $-4.60°$. A placement along the tool axis carries no such choice, because it produces no first-order moment about either tangent, and Case G measured its largest response at the furthest sampled position. Matching the tangential direction to the signed offset is therefore required only where that displacement is used. The reported controller applied this matching during experimental set-up.

5.7 Limitations

5.7.1 Experimental Range

The parameter matrix includes commanded offsets of 5° and 10° about both surface tangents. The rotational and translational stiffnesses were varied at three levels. The combined comparison used one low-translational-stiffness and high-rotational-stiffness pair for each axis. The compliance-centre comparison sampled seven positions along the assisting tangent at both signs of the commanded offset, and six along the tool axis. Case C additionally sampled four combined in-plane tilt and lever directions at one common magnitude. These settings establish the direction and sign of the assisting displacement and the dependence of the response on the centre position. The combined stiffness result applies to the selected pair, and the direction result covers the four sampled in-plane directions.

The measured centre positions are geometry-specific design values defined by the tool-face dimensions, commanded press, and calibrated surface. The transferable controller results are the rule relating lever direction and sign to the signed tilt direction, and the finding that displacing the centre along the tool axis improves the response without a sign to choose. The distances are then selected for the contact geometry.

A fixed Panda configuration, rectangular tool, plane surface, and commanded press trajectory provided a common contact geometry across all comparisons. This controlled basis isolates the effects of the tested controller parameters and defines the reference response of the complete robot–gripper–tool system.

5.7.2 Measurement and Tool-Mount Uncertainty

The alignment metric combines the measured end-effector orientation with the calibrated grinding-face direction and surface normal n_s. This measurement chain quantifies the combined robot–gripper–tool response. The approximately ±2° rotational freedom about y_{EE} defines the mount-related contribution to its uncertainty.

For the tested mounting orientation, y_{EE} lay approximately 25° from t_2. Relative tool motion may therefore have contributed more strongly to the t_2 conditions than to the

t_1 conditions. The 10° t_2 repetition gave an alignment change $\Delta\theta_{\text{align}}$ near zero. It was included in the reported mean and standard deviation, preserving the observed variability.

Within-setting scatter grew with the size of the correction on both tangents. The largest spread about t_1 was measured in the 6.05° compliance-centre condition. These repeatability measurements provide the quantitative scale for the combined alignment metric.

Within-setting scatter is not the only scale that bounds a comparison. The direction of the grinding face in end-effector coordinates was measured by seating the face on the surface at several orientations about the surface normal and solving for the direction those seatings share. Within one such measurement the seatings agreed to between 0.12 and 0.28°. Repeated measurements of the same arrangement, however, differed by 1.70 and 2.17°, and the difference appeared as the tool settled in the mount under repeated pressing. The absolute zero of the alignment metric is therefore known to one to two degrees, and no better.

This bounds the absolute residuals and not the comparisons. A constant error in the measured face direction shifts every condition by the same amount and cancels in a difference between conditions. The rotation since first contact, resolved along the surface axes, is measured from joint angles alone and carries neither the face direction nor the surface normal; it agreed with the change in the alignment metric to 0.05° on every trial compared. The reported changes are therefore not limited by this uncertainty, whereas a statement that the tool ended flat within a degree would be. Figure 5.11 places the two quantities side by side: the left panel overlays them for one trial, and the right panel puts the magnitude of the reported change against the rotation measured from joint angles for every archived trial.

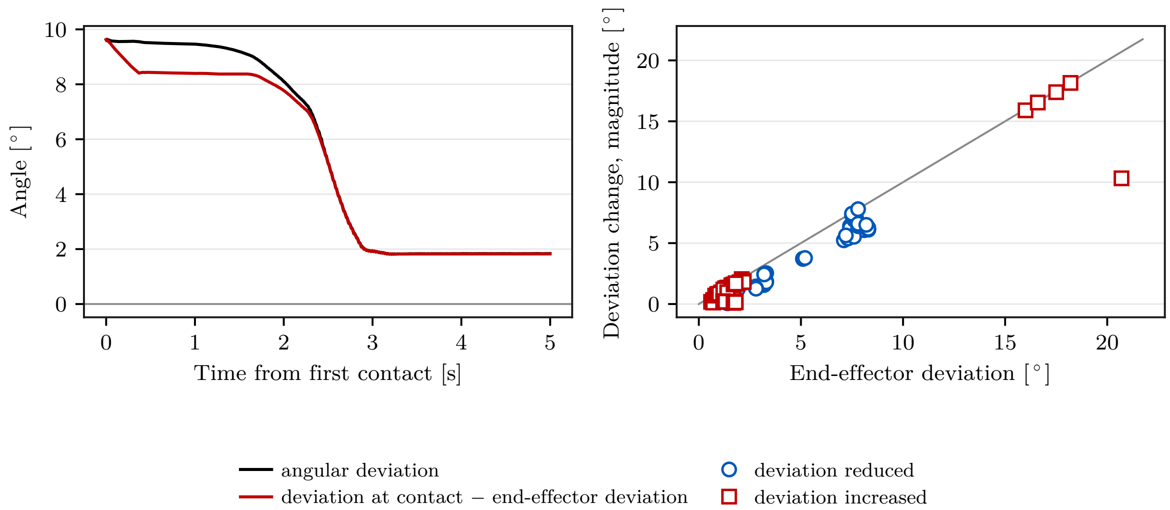


Figure 5.11: Agreement between the reported angular deviation and the rotation measured from joint angles alone.

Many conditions about t_1 ended near a residual of 3.3°, which invites reading that value as a limit of the arrangement. Other conditions passed it: the reversed offset about t_1 with the centre on the TCP left $2.23 \pm 0.24°$, and the assisting position about t_2 left $1.95 \pm 0.31°$ at the full commanded offset and $1.80 \pm 0.06°$ at half of it. The residual is a property of the settings rather than of the mount.

The model-estimated force change from the clearance transition provided the supporting normal-load quantity. It was evaluated with the alignment metric across matched runs. Both quantities follow a common robot-state and model basis, preserving a consistent comparison across the contact campaign.

5.7.3 Null-Space Configuration and Real-Time Operation

The surface-contact results establish alignment performance with projected damping as the active null-space configuration. Matched pose-hold trials quantify projected damping and singular-value conditioning under an internally commanded point-force equivalent.

The communication check described in section 4.2.2 passed the 0.90 average command-success criterion before the experiments. The controller then completed every reported contact and pose-hold trial through the nominal 1 kHz Franka Control Interface. The reported campaign therefore ran to completion at the nominal control rate.

Chapter 6

Conclusion and Future Work

GREEN — revised and confirmed. The chapter was revised in full and every paragraph is settled. No further editing is to be applied to it, and any remaining concern is raised as a comment rather than as a change.

comment: "In the settled prose, use "commanded tool orientation offset" for the deliberate angular command and "commanded rotation axis" or "commanded rotation direction" for its components. Retain "tilt" only for a physical inclination."

6.1 Conclusion

A real-time Cartesian impedance controller was implemented for the contact-induced alignment of a grinding tool with a plane surface. During the set-up phase, the Cartesian stiffness and damping were spatially shifted from a selected centre of compliance p_c to the TCP. This shift introduced translation–rotation coupling. The normal interaction load could therefore generate a moment about the surface tangents.

The point-shift adjoint and congruence transformation are established theory. The implementation contribution is their phase-dependent integration into the real-time contact controller.

The controller was evaluated in eight surface-contact cases covering commanded tool orientation offsets, directional stiffness, and the position and magnitude of the compliance-centre lever r_c. Complementary Cartesian pose-hold trials quantified the projected null-space damping and singular-value-conditioning terms. The compliance-centre cases were measured at both signs of the commanded offset and on both surface tangents, so the direction that assists is established against its own reversal rather than against a single reference.

The stiffness variations produced smaller changes than moving the virtual centre of compliance p_c. Raising the rotational stiffness about the commanded rotation axis reduced the measured alignment improvement $\Delta\theta_{\text{align}}$. Reducing the perpendicular translational stiffness increased the response about t_2 only. The combined stiffness result was non-additive for the selected settings.

Within the investigated range, the centre of compliance p_c had the largest measured influence. For each principal surface tangent t_1 and t_2, one tangential direction increased the alignment improvement $\Delta\theta_{\text{align}}$ while the opposite direction removed it about t_1 and reversed it about t_2, in both cases leaving the tool worse than with the centre on the TCP. Reversing the commanded tool orientation offset exchanged the two directions. The displacement must therefore be selected from both the commanded rotation axis and the required direction of the induced moment.

With $r_c = p_{\text{TCP}} - p_c$ and $m_c = -r_c \times f_{\text{press}}$, the experimentally supported selection rule is a lever perpendicular to the commanded rotation axis in the tangent plane, with $\rho_c > 0$. The assisting tangential lever is therefore perpendicular to the commanded rotation axis and reverses with the commanded rotation direction.

The direction rule was supported for four commanded rotation directions in the tangent plane. A lever selected perpendicular to the commanded rotation axis assisted alignment at every tested direction, whereas holding one lever fixed reduced the response in the matched comparisons. The usable magnitude is bounded by the face dimension the tool rocks over rather than by the response: about t_2, where the face is 20 mm half-width, a 60 mm displacement carried a 5° commanded offset out to a 20.7° excursion and 80 mm aborted the motion on a wrist-torque reflex. The correction also does not scale with the commanded offset: halving the offset about t_1 reduced the improvement from 5.94 to 1.53° while the residual moved only from 3.38 to 2.76°.

Displacing the centre of compliance p_c along the tool axis raised the alignment improvement across the whole sampled range, and the response continued to increase up to the largest tested magnitude. That direction produces no first-order moment against a normal press, so the measurement indicates a tangential component in it. Orientation offsets commanded about the grinding-face axes produced responses between the two principal surface-tangent responses, which is consistent with the surface-frame interpretation of the directional asymmetry. The result remains bounded by the relative motion between the tool and its mount.

The contact brings the grinding face onto the surface in every tested condition but those that place the compliance centre on the opposing side about t_1, raise the rotational stiffness about it, or name the offset in the surface frame. What the compliance centre changes is therefore which of the robot and the tool mount provides the rotation that flattens the tool, not whether the tool ends flat. About t_1 the robot turns 7.6° of a commanded 10° without any displacement, and displacing the centre adds nothing to it. About t_2, where the mount is free, it turns 1.6° without a displacement and 4.4° with one.

Naming the lever in surface coordinates rather than in the tool frame changed the response once the tool turned. At a commanded offset of 10° the two definitions differed by 6.32°, although they describe one physical point at zero offset. The usable magnitude also differs between the tangents, being bounded by the face dimension the tool rocks over, so neither the frame nor one distance transfers between them.

In the matched Cartesian pose-hold trials, projected null-space damping reduced the cumulative redundant joint excursion by approximately 25 %. Both singular-value-conditioning settings returned the minimum singular value $\sigma_{\min}$ to approximately its initial level after the commanded disturbance. These findings apply to the isolated free-space hold conditions tested.

The experiments therefore support the use of a point-shifted Cartesian impedance to convert the normal interaction into an alignment moment without an explicit rotational trajectory during set-up. The measured results provide a geometric rule for selecting the lever direction. Its magnitude remains a condition-dependent design parameter rather than a universal setting.

6.2 Limitations

6.2.1 Experimental Range

The contact experiments used one Panda configuration, tool mount, plane surface, press trajectory, and bounded parameter range. The direction-and-sign rule was reproduced on both principal surface tangents and supported at four in-plane directions. The measured lever magnitudes and stiffness effects remain specific to the investigated geometry and tested settings.

6.2.2 Measurement and Tool-Mount Uncertainty

The alignment evaluation reconstructs the grinding-face direction from the end-effector pose and a calibrated tool direction. The resulting alignment angle θ_{align} therefore describes the combined robot–gripper–tool response rather than a separately measured grinding-face orientation. Relative motion within the tool mount bounds the reported alignment improvements $\Delta\theta_{\text{align}}$, particularly the t_2 results.

The supporting external wrench was obtained from the change in the libfranka model estimate relative to its clearance-transition reference. It was not measured by an independent contact-load sensor. The reported force and moment comparisons are consequently limited to this common model-based measurement chain.

6.2.3 Null-Space Configuration and Real-Time Operation

Projected damping was the active null-space configuration during the surface-contact experiments. The damping and singular-value terms were isolated only in free-space pose hold under an internally commanded point-force equivalent. Their combined mode, response to a physical disturbance, and effect during contact were not evaluated.

6.3 Future Work

Three extensions to the compliance-centre study are considered most important. Finer lever-magnitude sampling would extend the measured response around and beyond the best tested regions. A finer angular sweep, for example in 15° or 30° increments, should then be combined with magnitude variation at each direction. Case C already samples four directions, including two diagonals between the surface-axis cases; the next step is therefore a denser joint characterization of direction and magnitude, not merely the addition of intermediate directions.

An angular measurement of the grinding face taken independently of the end-effector pose would separate tool rotation from motion within the mount. A more rigid tool mount would further reduce this relative motion. This bears particularly on the reversed t_1 result and on the magnitude of the reported alignment changes $\Delta\theta_{\mathrm{align}}$. An independent force/torque sensor would additionally allow the model-estimated external wrench to be compared with a direct contact-load measurement. Further contact geometries, tool faces, and press trajectories would establish how far the selection rule extends beyond the investigated arrangement.

The null-space study can be extended with a physical disturbance applied to an intermediate robot link and with the combined damping and singular-value-conditioning mode. A corresponding surface-interaction comparison would show whether the pose-hold ordering is preserved during contact.

Finally, the compliance-centre selection can be incorporated into the controller as a geometry-based scheduling step. After the initial tilt is estimated at the clearance transition, the lever direction could be selected so that the induced moment opposes the measured angular error. Such a strategy would replace a lever fixed in advance with a run-specific placement while continuing to use the point-shifted Cartesian impedance implemented in this work.

Bibliography

[1] Alin Albu-Schäffer, Christian Ott, Udo Frese, and Gerd Hirzinger. Cartesian impedance control of redundant robots: Recent results with the DLR-light-weight-arms. In *Proceedings of the IEEE International Conference on Robotics and Automation (ICRA)*, volume 3, pages 3704–3709, 2003. doi: 10.1109/ROBOT.2003.1242165.

[2] Benjamin Alt, Florian Stöckl, Silvan Müller, Christopher Braun, Julian Raible, Saad Alhasan, Oliver Rettig, Lukas Ringle, Darko Katic, Rainer Jäkel, Michael Beetz, Marcus Strand, and Marco F. Huber. RoboGrind: Intuitive and interactive surface treatment with industrial robots. In *2024 IEEE International Conference on Robotics and Automation (ICRA)*, pages 2140–2146, 2024. doi: 10.1109/ICRA57147.2024.10611143.

[3] Stefano Chiaverini, Bruno Siciliano, and Olav Egeland. Review of the damped least-squares inverse kinematics with experiments on an industrial robot manipulator. *IEEE Transactions on Control Systems Technology*, 2(2):123–134, 1994. doi: 10.1109/87.294335.

[4] John J. Craig. *Introduction to Robotics: Mechanics and Control.* Pearson Prentice Hall, Upper Saddle River, NJ, 3 edition, 2005.

[5] Jacques Denavit and Richard S. Hartenberg. A kinematic notation for lower-pair mechanisms based on matrices. *Journal of Applied Mechanics*, 22(2):215–221, 1955.

[6] Alexander Dietrich, Christian Ott, and Alin Albu-Schäffer. An overview of null space projections for redundant, torque-controlled robots. *The International Journal of Robotics Research*, 34(11):1385–1400, 2015. doi: 10.1177/0278364914566516.

[7] Ernest D. Fasse and Jan F. Broenink. A spatial impedance controller for robotic

manipulation. *IEEE Transactions on Robotics and Automation*, 13(4):546–556, 1997. doi: 10.1109/70.611320.

[8] Franka Emika GmbH. libfranka: Communication Test Example. https://github.com/frankarobotics/libfranka/blob/0.7.1/examples/communication_test.cpp, 2019. Official libfranka 0.7.1 example, file `examples/communication_test.cpp`.

[9] Franka Emika GmbH. Franka control interface documentation. https://frankaemika.github.io/docs/, 2023. Accessed for robot interface, libfranka model access, and torque control documentation.

[10] Franka Emika GmbH. libfranka: Cartesian Impedance Control Example. https://github.com/frankaemika/libfranka/blob/master/examples/cartesian_impedance_control.cpp, 2023. Official libfranka example, file `examples/cartesian_impedance_control.cpp`.

[11] Christian Friedrich, Patrick Frank, Marco Santin, and Matthias Haag. Highly dynamic physical interaction for robotics: Design and control of an active remote center of compliance. In *2025 IEEE International Conference on Robotics and Automation (ICRA)*, 2025. doi: 10.1109/ICRA55743.2025.11128451.

[12] Richard Hartisch and Kevin Haninger. Flexure-based environmental compliance for high-speed robotic contact tasks. In *2022 IEEE/ASME International Conference on Advanced Intelligent Mechatronics (AIM)*, pages 1608–1613, 2022. doi: 10.1109/AIM52237.2022.9863334.

[13] Neville Hogan. Impedance control: An approach to manipulation—parts i–iii. *Journal of Dynamic Systems, Measurement, and Control*, 107(1):1–24, 1985.

[14] International Federation of Robotics. World robotics 2023 report. Technical report, IFR Statistical Department, Frankfurt am Main, 2023.

[15] Oussama Khatib. A unified approach for motion and force control of robot manipulators. *IEEE Journal on Robotics and Automation*, 3(1):43–53, 1987.

[16] Kevin M. Lynch and Frank C. Park. *Modern Robotics: Mechanics, Planning, and Control*. Cambridge University Press, 2017. URL https://hades.mech.northwestern.edu/images/7/7f/MR.pdf. Free preprint.

[17] Matthias Mayr and Julian M. Salt-Ducaju. A C++ implementation of a cartesian impedance controller for robotic manipulators. *Journal of Open Source Software*, 9 (93):5194, 2024. doi: 10.21105/joss.05194.

[18] Kang Min, Fenglei Ni, Zhaoyang Chen, and Hong Liu. A force control method integrating human skills for complex surface finishing. *Machines*, 12(11):756, 2024. doi: 10.3390/machines12110756.

[19] Richard M. Murray, Zexiang Li, and S. Shankar Sastry. *A Mathematical Introduction to Robotic Manipulation*. CRC Press, 1994.

[20] James L. Nevins and Daniel E. Whitney. Computer-controlled assembly. *Scientific American*, 1989. Background source on robotic assembly and compliance.

[21] Christian Ott, Alin Albu-Schäffer, Andreas Kugi, and Gerd Hirzinger. On the passivity-based impedance control of flexible joint robots. *IEEE Transactions on Robotics*, 24(2):416–429, 2008.

[22] Mathew Jose Pollayil, Franco Angelini, Guiyang Xin, Michael Mistry, Sethu Vijayakumar, Antonio Bicchi, and Manolo Garabini. Choosing stiffness and damping for optimal impedance planning. *IEEE Transactions on Robotics*, 39(2):1281–1300, 2023. doi: 10.1109/TRO.2022.3216078.

[23] Bruno Siciliano, Lorenzo Sciavicco, Luigi Villani, and Giuseppe Oriolo. *Robotics: Modelling, Planning and Control*. Springer, London, 2009.

[24] Stefano Stramigioli, Claudio Melchiorri, and Stefano Andreotti. Intrinsically passive grasping and manipulation. *IEEE Transactions on Robotics and Automation*, 2000. URL http://robby.caltech.edu/~jwb/courses/CDS270/papers/passive.grasp.pdf. Open-access preprint.

[25] Markku Suomalainen, Sylvain Calinon, Emmanuel Pignat, and Ville Kyrki. Improving dual-arm assembly by master-slave compliance. In *2019 International Conference on Robotics and Automation (ICRA)*, pages 8676–8682, 2019. doi: 10.1109/ICRA.2019.8793977.

[26] Valeria Villani, Francesco Pini, Francesco Leali, and Cristian Secchi. Survey on human–robot collaboration in industrial settings: Safety, intuitive interfaces and

applications. *Mechatronics*, 55:248–266, 2018. doi: 10.1016/j.mechatronics.2018.02.009.

[27] Sen Wang, Guodong Chen, Hui Xu, and Zheng Wang. A robotic peg-in-hole assembly strategy based on variable compliance center. *IEEE Access*, 2019. doi: 10.1109/ACCESS.2019.2954459.

[28] Paul C. Watson. Remote center compliance system. U.S. Patent 4,098,001, 1978. Assigned to The Charles Stark Draper Laboratory.

[29] Chong Wu, Kai Guo, and Jie Sun. Dual PID adaptive variable impedance constant force control for grinding robot. *Applied Sciences*, 13(21):11635, 2023. doi: 10.3390/app132111635.

Appendix A

Controller Source Listings

ORANGE — sufficient. The listings are abridged to the mathematical paths used by the thesis and connect each important equation to an implementation excerpt.

This appendix contains abridged source excerpts from the controller implementation described in Chapter 3. Only the paths needed to connect the equations to the implementation are shown. The excerpts use fixed-size Eigen aliases such as `Vec3`, `Vec7`, `Mat3`, `Mat6x6`, and `Mat6x7`. The identifier `R_base_surface` denotes the surface-frame orientation used in the chapters.

A.1 Surface Task Frame and Spatial Gains

The frame order is $[t_1, t_2, n_s]$. Because $t_2 = n_s \times t_1$, this column order is right-handed. A configured tangent that is close to the surface normal n_s is replaced by a fallback direction, so the frame stays defined for any configured plane.

Listing A.1: Surface frame and gain transformation used by the controller.

```
Mat3 makeSurfaceFrameFromNormalTangent(
    const Vec3& normal_input, const Vec3& tangent1_input) {
  const Vec3 normal =
      normalizedOrFallback(normal_input, Vec3(1.0, 0.0, 0.0));
  Vec3 tangent1 =
      tangent1_input - normal * normal.dot(tangent1_input);

  if (tangent1.norm() <= 1e-9) {
    Vec3 fallback(0.0, 1.0, 0.0);
    if (std::abs(normal.dot(fallback)) > 0.95) {
      fallback = Vec3(0.0, 0.0, 1.0);
    }
    tangent1 = fallback - normal * normal.dot(fallback);
  }

  tangent1.normalize();
  Vec3 tangent2 = normal.cross(tangent1);
```

```
  tangent2.normalize();

  Mat3 R_base_surface;
  R_base_surface.col(0) = tangent1;
  R_base_surface.col(1) = tangent2;
  R_base_surface.col(2) = normal;
  return R_base_surface;
}

Mat3 makeSpatialGainMatrix(
    const Vec3& diagonal_in_task_frame, const Mat3& R_task) {
  return R_task * diagonal_in_task_frame.asDiagonal()
       * R_task.transpose();
}
```

A.2 Tool Orientation and Rotation Error

The physical tool normal n_{EE} is configured in the EE frame. The desired orientation is the minimum rotation of the captured start orientation that maps this axis onto the signed normal; it is not obtained by assigning the complete surface frame to the EE frame. The commanded angular offsets of a case are applied to that axis before the rotation is formed. A commanded tool orientation offset about t_1 or t_2 therefore enters as a rotation of the orientation target rather than as a trajectory.

Listing A.2: Tool-axis target, commanded tool orientation offsets, and rotation-vector error.

```
Vec3 surfaceToolAxisInBase(
    const ControllerConfig& params, const Mat3& R_base_surface) {
  const double sign =
      (params.tool_axis_target_sign >= 0.0) ? 1.0 : -1.0;
  return sign * R_base_surface.col(2);  // normal = 3rd column
}

Vec3 desiredToolAxisInBase(
    const ControllerConfig& params, const Mat3& R_base_surface) {
  const Vec3 flat_axis =
      surfaceToolAxisInBase(params, R_base_surface);
  const double deg_to_rad = M_PI / 180.0;
  const Vec3 rotation_vector =
      deg_to_rad *
      (params.tool_target_offset_tangent1_deg
           * R_base_surface.col(0) +
       params.tool_target_offset_tangent2_deg
           * R_base_surface.col(1));
  const double angle = rotation_vector.norm();
  if (angle <= 1e-12) {
    return flat_axis;
  }
  return Eigen::AngleAxisd(angle, rotation_vector / angle)
       * flat_axis;
}

Vec3 orientationError(
    const Mat3& R_current, const Mat3& R_desired) {
  Mat3 R_error = R_current.transpose() * R_desired;
  Eigen::AngleAxisd angle_axis(R_error);
  if (std::abs(angle_axis.angle()) < 1e-9) {
    return Vec3::Zero();
  }
  return R_current * angle_axis.axis() * angle_axis.angle();
}
```

The shortest arc that carries the current tool axis onto the target turns about an axis perpendicular to the tool axis, so it leaves the spin about that axis at whatever the start pose held. That spin decides which edge of a rectangular face leads into the surface, so it is commanded separately through `tool_target_offset_normal_deg` rather than inherited from q_{init}.

Listing A.3: Commanded spin about the tool axis (abridged).

```
const Mat3 R_axis =
    rotationBetweenUnitVectors(tool_axis_start, tool_axis_target)
    * R_start;
if (!params.command_tool_twist) {
  return R_axis;
}

// Both directions are measured in the plane perpendicular to the
// tool axis.
const Vec3 current = perpendicular(R_axis * face_long_axis_ee);
const Vec3 zero_reference =
    perpendicular(R_base_surface.col(0));
if (current.norm() <= 1e-9 || zero_reference.norm() <= 1e-9) {
  return R_axis;  // no twist is defined
}

const Vec3 target =
    Eigen::AngleAxisd(
        (M_PI / 180.0) * params.tool_target_offset_normal_deg,
        tool_axis_target)
    * zero_reference.normalized();
const Vec3 current_unit = current.normalized();
double twist = std::atan2(
    tool_axis_target.dot(current_unit.cross(target)),
    current_unit.dot(target));

// A face centred on the tool axis maps onto itself every half turn,
// so take the representative nearest the start pose.
if (face_center_off_axis.norm() <= 0.05 * face_scale) {
  twist -= M_PI * std::round(twist / M_PI);
}
return Eigen::AngleAxisd(twist, tool_axis_target) * R_axis;
```

A.3 Geometric Clearance and Phase Targets

The leading point, edge, or face of the rectangular tool is selected along the descent direction. The transition into set-up is based on the exact most-leading corner height; the average of corners tied within 0.1 mm supplies the controlled contact point and projected reference. The transition depends on this height; the external-force estimate does not enter the decision. The following excerpt is abridged from the phase switch.

Listing A.4: Phase-dependent tool-feature targets (abridged).

```
const Vec3 descend_direction = -R_base_surface.col(2);
double leading_contact_projection = 0.0;
const Vec3 tool_contact_offset_ee =
    selectLeadingToolContact(R_EE, leading_contact_projection);
const Vec3 tool_contact_point =
    p_EE + R_EE * tool_contact_offset_ee;

case ControlPhase::kSurfaceApproach: {
  // The gate freezes this clock, so waiting does not go on ramping
  // the commanded distance.
```

```
  const double distance = std::min(
      params.descend_speed * (phase_time - gate_paused_time),
      params.descend_max_distance);
  desired.p_d = p_start + distance * descend_direction;
  desired.pdot_d = params.descend_speed * descend_direction;

  const Vec3 surface_normal = (-descend_direction).normalized();
  const double height_above_surface =
      surface_normal.dot(p_EE - surface_point_runtime)
      - leading_contact_projection;
  const double controlled_point_height =
      surface_normal.dot(tool_contact_point - surface_point_runtime);
  const Vec3 projected_surface_point =
      tool_contact_point - controlled_point_height * surface_normal;
  const bool clearance_reached =
      height_above_surface <= params.descend_surface_clearance;

  if (clearance_reached && !gate_blocking) {
    active_tool_contact_offset_ee = tool_contact_offset_ee;
    first_contact_tcp = p_EE;  // captured at clearance
    first_contact_point = projected_surface_point;
    setup_push_start = -controlled_point_height;
    R_contact_start = R_EE;
    contact_force_bias = external_force;
    contact_moment_bias = external_moment;
    phase = ControlPhase::kSetup;
  }
  break;
}

case ControlPhase::kSetup: {
  const double push =
      setUpPush(params, phase_time - gate_grind_paused_time,
                setup_push_start, setup_push_velocity);
  const Vec3 edge_target =
      first_contact_point + push * descend_direction;
  desired.p_d =
      edge_target - R_contact_start * tool_contact_offset_ee;
  desired.pdot_d = setup_push_velocity * descend_direction;
  // Exit after minimum time on moment-delta norm, or on timeout.
  break;
}

case ControlPhase::kGrinding: {
  const Vec3 n = descend_direction;  // unit, into the surface
  Vec3 edge_target;
  if (params.grind_sweep_enabled) {
    edge_target = first_contact_point + grind_push * n
                + sweep_s * grind_tangent;
    desired.pdot_d = sweep_s_dot * grind_tangent;
  } else {
    const double edge_penetration =
        n.dot(tool_contact_point - first_contact_point);
    edge_target =
        tool_contact_point + (grind_push - edge_penetration) * n;
  }
  desired.p_d =
      edge_target - R_contact_start * tool_contact_offset_ee;
  break;
}
```

The identifiers `first_contact_*` are retained source names. They are captured at the configured clearance, not at a force-detected first contact. The model-estimated wrench recorded at the same instant is subtracted from every later sample.

Both operator gates latch once armed. The gate compares a held pose against the clearance height, and re-testing that comparison every cycle released the gate for single cycles on measurement noise; on each release the descent clock advanced and commanded the tool further down, so a longer wait produced a harder press. Latching removes that dependence on waiting time.

A.4 Centre of Compliance and the Offset Adjoint

The lever r_c passed to the implementation points from the selected centre of compliance p_c to the TCP:

$$r_c = p_{\mathrm{TCP}} - p_c.$$

The same congruence is applied to stiffness and damping.

Listing A.5: Pole-to-TCP adjoint and congruence transform.

```
Mat3 skewMatrix(const Vec3& v) {
  Mat3 s;
  s << 0.0, -v(2),  v(1),
       v(2),  0.0, -v(0),
      -v(1),  v(0),  0.0;
  return s;
}

Mat6x6 complianceShiftMatrix(const Vec3& r_c) {
  Mat6x6 adjoint = Mat6x6::Zero();
  adjoint.block<3, 3>(0, 0) = Mat3::Identity();
  adjoint.block<3, 3>(0, 3) = skewMatrix(r_c);
  adjoint.block<3, 3>(3, 3) = Mat3::Identity();
  return adjoint;
}

Mat6x6 shiftGainToTcp(
    const Mat6x6& gain_at_center, const Vec3& r_c) {
  const Mat6x6 adjoint = complianceShiftMatrix(r_c);
  return adjoint.transpose() * gain_at_center * adjoint;
}
```

The wrench routine selects between the decoupled and the coupled law and constructs the lever. The centre can be defined relative to the TCP in end-effector coordinates, or the lever can be given directly in surface coordinates. Both definitions use the convention $r_c = p_{\mathrm{TCP}} - p_c$.

Listing A.6: Selection of the decoupled or coupled spring wrench (abridged).

```
const bool coupled =
    params.use_virtual_compliance_center &&
    (phase == ControlPhase::kSetup ||
     (phase == ControlPhase::kPoseHold &&
      params.use_setup_impedance_hold));
if (!coupled) {
  // Two independent springs. dv.tail is -omega, so the damping
  // signs match.
  Vec6 wrench;
  wrench.head<3>() = Kp * dx.head<3>() + Dp * dv.head<3>();
  wrench.tail<3>() = KR * dx.tail<3>() + DR * dv.tail<3>();
  return wrench;
}

Vec3 r_c = Vec3::Zero();
if (params.compliance_center_in_tool_frame) {
  r_c = -(R_EE * params.compliance_center_offset_ee);
} else {
  r_c = R_base_surface
```

```
      * params.r_tcp_from_compliance_center_surface;
}
const Mat6x6 K_tcp =
    shiftGainToTcp(blockDiagonal(Kp, KR), r_c);
const Mat6x6 D_tcp =
    shiftGainToTcp(blockDiagonal(Dp, DR), r_c);
return K_tcp * dx + D_tcp * dv;
```

A.5 Task-Inertia Damping

The joint-space mass matrix M is solved rather than explicitly inverted. A 10^{-9} diagonal regularisation is used only for the task-inertia matrix Λ calculation; it is not the null-space pseudoinverse cutoff. Every intermediate result is checked before use, and a single non-finite or non-positive entry marks the estimate invalid.

Listing A.7: Task-inertia estimate and per-axis damping (abridged).

```
Eigen::LDLT<Mat7x7> mass_ldlt(joint_mass);
const Mat7x6 Minv_Jt = mass_ldlt.solve(J.transpose());
Mat6x6 lambda_inv = J * Minv_Jt;
lambda_inv = 0.5 * (lambda_inv + lambda_inv.transpose());

Mat6x6 lambda_inv_damped = lambda_inv;
lambda_inv_damped.diagonal().array() += 1e-9;
Eigen::LDLT<Mat6x6> lambda_ldlt(lambda_inv_damped);
Mat6x6 Lambda_base = lambda_ldlt.solve(Mat6x6::Identity());
Lambda_base = 0.5 * (Lambda_base + Lambda_base.transpose());

const Mat6x6 T_task = blockDiagonalRotation(R_task);
Mat6x6 Lambda_task = T_task.transpose() * Lambda_base * T_task;
Lambda_task = 0.5 * (Lambda_task + Lambda_task.transpose());

for (int i = 0; i < 3; ++i) {
  const double m = Lambda_task(i, i);
  const double I = Lambda_task(i + 3, i + 3);
  if (m <= 0.0 || I <= 0.0) {
    return result;  // estimate stays invalid
  }
  result.translational(i) = m;
  result.rotational(i) = I;
}

Vec3 criticalDampingFromStiffness(const Vec3& inertia,
                                  const Vec3& stiffness,
                                  double damping_ratio,
                                  const Vec3& min_damping,
                                  double max_damping) {
  Vec3 damping = Vec3::Zero();
  const double zeta = std::max(0.0, damping_ratio);
  for (int i = 0; i < 3; ++i) {
    const double m = std::max(0.0, inertia(i));
    const double k = std::max(0.0, stiffness(i));
    const double critical = 2.0 * zeta * std::sqrt(m * k);
    damping(i) = std::min(
        max_damping,
        std::max(std::max(0.0, min_damping(i)), critical));
  }
  return damping;
}
```

Only finite, positive directional inertia values λ_i are used. Otherwise, the phase-specific fixed damping matrix D is applied. Each inertia-scaled coefficient is bounded above by `auto_damping_max`; the value configured for the reported experiments is listed in

Appendix C. The lower bound `min_damping` is the configured manual matrix only when `auto_damping_min_from_manual` is set, and is zero otherwise.

A.6 Null-Space Torque

For $J \in \mathbb{R}^{6\times 7}$, Eigen returns six singular values and a full 7×7 right-singular matrix. Column six in zero-based C++ indexing is therefore v_7, the structural null vector; it is distinct from v_6, the right-singular vector paired with $\sigma_{\min} = \sigma_6$. The two sampled configurations are evaluated by re-querying the robot model for the Jacobian $J(q_\pm)$ at $q \pm \alpha n$, so the comparison uses the same kinematics as the control cycle.

Listing A.8: Thresholded-SVD joint-torque projector and null-space law (abridged).

```
Eigen::JacobiSVD<Mat6x7> svd_current(
    J, Eigen::ComputeFullU | Eigen::ComputeFullV);
const auto singular_values = svd_current.singularValues();
const double svd_cutoff =
    std::max(0.0, params.nullspace_svd_relative_tolerance)
    * singular_values.maxCoeff();

Mat7x6 J_pinv = Mat7x6::Zero();
for (int i = 0; i < 6; ++i) {
  if (singular_values(i) > svd_cutoff) {
    J_pinv += (1.0 / singular_values(i))
            * svd_current.matrixV().col(i)
            * svd_current.matrixU().col(i).transpose();
  }
}

Mat7x7 N_tau =
    Mat7x7::Identity() - J.transpose() * J_pinv.transpose();
N_tau = 0.5 * (N_tau + N_tau.transpose());
const Vec7 dq_nullspace = N_tau * dq;

// Mode 0 commands nothing, but the projector above still observes
// the redundant axis for the diagnostics.
if (params.nullspace_mode == NullspaceMode::kOff) {
  return Vec7::Zero();
}

const bool damping_enabled =
    params.nullspace_mode == NullspaceMode::kDampingOnly ||
    params.nullspace_mode == NullspaceMode::kDampingAndSigma;
Vec7 tau_damping = Vec7::Zero();
if (damping_enabled) {
  tau_damping.noalias() =
      -params.nullspace_damping * dq_nullspace;
}
if (params.nullspace_mode == NullspaceMode::kDampingOnly) {
  return tau_damping;
}
const bool sigma_only =
    params.nullspace_mode == NullspaceMode::kSigmaOnly;
const Vec7 sigma_fallback = sigma_only ? Vec7::Zero() : tau_damping;

Vec7 n = svd_current.matrixV().col(6);      // v_7
if (n.norm() <= 1e-9) { return sigma_fallback; }
n.normalize();
const double alpha =
    std::abs(params.nullspace_probe_step_rad);
if (alpha <= 1e-12) { return sigma_fallback; }

// The Jacobian is re-evaluated at the two sampled configurations.
Map<const Mat6x7> J_plus(model.zeroJacobian(
    Frame::kEndEffector, vec7ToArray(Vec7(q_current + alpha * n)),
```

```
    state.F_T_EE, state.EE_T_K).data());
Map<const Mat6x7> J_minus(model.zeroJacobian(
    Frame::kEndEffector, vec7ToArray(Vec7(q_current - alpha * n)),
    state.F_T_EE, state.EE_T_K).data());

const double sigma_difference =
    smallestSingularValue(J_plus) - smallestSingularValue(J_minus);
const double deadband =
    std::max(0.0, params.nullspace_sigma_deadband);
if (std::abs(sigma_difference) <= deadband) {
  return sigma_fallback;
}

const double sigma_direction = (sigma_difference > 0.0) ? 1.0 : -1.0;
const Vec7 best_direction = sigma_direction * n;
// alpha determines only where sigma is sampled. k_sigma is the
// commanded torque magnitude along the better direction.
const Vec7 tau_sigma =
    params.nullspace_sigma_gain * N_tau * best_direction;

return sigma_only ? tau_sigma : tau_damping + tau_sigma;
```

Here the code variable `alpha` represents α_{probe} and changes only the two sampled configurations. It does not multiply the active torque. Singular values at or below the cutoff are omitted from the inverse. The routine also fills a diagnostics structure that carries the sampled singular values, the selected direction, and the projected joint velocity $\dot{q}_{\text{null}}$ into the recorded data.

A.7 Core Control Callback

Listing A.9: Core impedance, coupled set-up, and torque composition.

```
const Vec3 e_p = desired.p_d - p_EE;
const Vec3 e_R = applyRotationalAxisMask(
    params, orientationError(R_EE, R_d_used), R_base_surface);

Vec6 dx;
dx.head<3>() = e_p;
dx.tail<3>() = e_R;
Vec6 dv;
dv.head<3>() = desired.pdot_d - pdot;
dv.tail<3>() = -omega;

const Vec6 wrench =
    computeCartesianImpedanceWrench(
        params, phase, Kp_used, Dp_used,
        KR_used, DR_used, R_base_surface,
        dx, dv, R_EE);

const Vec7 tau_nullspace = computeNullspaceTorque(
    params, model, state, D, dq);

AutomaticDisturbance disturbance;
if (phase == ControlPhase::kPoseHold &&
    params.disturbance_auto_enabled) {
  disturbance = computeAutomaticDisturbance(
      params, model, state, disturbance_force_direction_base,
      time - phase_start_time);
}

Array7 coriolis_array = model.coriolis(state);
Map<const Vec7> coriolis(coriolis_array.data());
const Vec7 tau_task = J.transpose() * wrench;
const Vec7 tau_cmd =
    tau_task + tau_nullspace + disturbance.tau + coriolis;
```

The null-space function is called in every non-manual phase. Manual guidance uses its separate Coriolis-plus-joint-damping branch. The disturbance term is non-zero only in the Cartesian pose-hold study of Section 4.7; it is zero in every surface-contact run.

Appendix B

Recorded Experimental Data

ORANGE — sufficient. The appendix records the signals actually used in the evaluation and identifies the common calibrated plane used by the controller and analysis.

The controller stores experimental signals in a preallocated ring buffer. This avoids file access and dynamic buffer growth inside the 1 kHz control callback. After the controller stops, the retained rows are written in chronological order to a CSV file.

B.1 Control Phases

ORANGE — sufficient. The numeric phase values are mapped directly to their recorded state names.

The integer `phase` identifies the active part of the experiment:

`phase`	0	1	2	3	4	5
state	`approach_orient`	`approach_descend`	`set_up`	`grind`	`hold`	`manual_guide`.

B.2 Signals Used in the Evaluation

ORANGE — sufficient. The table is a useful reproducibility record and keeps units, frames, and model-estimated quantities explicit.

Table B.1 lists the groups used to calculate the reported quantities. A suffix `_x,_y,_z` denotes base-frame components, and `_1.._7` denotes joint components.

Table B.1: Recorded signal groups used in the evaluation.

Column group	Meaning
`time`, `phase`	Run time and active control state.
`p_EE`, `p_d`	Measured and desired TCP positions [m].
`tool_contact`, `edge_target`	Selected contact point and its reference [m].
`tool_contact_offset_ee`	Offset of the selected contact point from the TCP, in EE coordinates [m].
`first_contact_tcp`, `first_contact`	TCP position and projected surface reference captured at the geometric clearance transition; the transition is not force triggered.
`e_p`, `e_R`, `pdot`, `pdot_d`, `omega`	Cartesian errors and velocities. During set-up, `e_R` is measured relative to the clearance orientation selected at the transition and held constant during set-up.
`angular_deviation_deg`	Angle between the EE-inferred tool axis and the calibrated physical-plane normal n_s stored in the controller configuration [°]. This recorded quantity produced the deviation values reported in the result tables.
`angular_deviation_t1_deg`, `angular_deviation_t2_deg`, `angular_deviation_normal_deg`	The same angular deviation resolved on t_1, t_2, and n_s [°]. These supply the per-tangent decomposition reported in the results.
`f`, `m`	Commanded Cartesian force [N] and moment [N m].
`external_force`, `external_moment`	Model-estimated external wrench from libfranka, expressed in the base orientation and referenced to stiffness frame K.
`contact_force_bias`, `contact_moment_bias`	External-wrench values captured at the clearance transition.
`force_after_contact`, `moment_after_contact`	The model-estimated wrench with those reference values already subtracted. The reported normal load is taken from this group.

Table B.1 continued

Column group	Meaning
push	Set-up penetration reference or grind preload retained after set-up [m].
nullspace_mode, sigma_current	Selected null-space law and current smallest singular value $\sigma_{\min}$.
sigma_plus, sigma_minus, sigma_difference, sigma_direction	Smallest singular value $\sigma_{\min}$ at the two sampled configurations $q \pm \alpha_{\text{probe}} v_7$, their difference, and the selected sign. Recorded in every mode, including when no torque is commanded.
nullspace_dq_1..7	Projected joint velocity $\dot{q}_{\text{null}}$ [rad/s].
tau_sigma_1..7, tau_sigma_norm, tau_nullspace_norm	Singular-value torque τ_σ and total null-space torque τ_{null} [N m].
nullspace_damping	Active projected damping gain [N m s/rad].
disturbance_scale, disturbance_force_ base_x..z, disturbance_ point_base_x..z	Commanded hold-disturbance waveform, point force [N], and the base-frame position of the link point it acts at [m].
tau_disturbance_1..7	Equivalent joint torque $J_p^\top f_d$ in the internally commanded hold study [N m].
tau_cmd_1..7	Final commanded joint torque [N m].

B.3 Evaluation Quantities

ORANGE — sufficient. The short summary routes each reported metric to its recorded source without reintroducing unused fields.

The contact-alignment angle θ_{align} was calculated from the measured end-effector orientation and n_s, the three-point physical-plane normal copied into the controller configuration. The before value was taken at the start of set-up and the after value from the settled set-up interval. The analysis did not substitute another plane normal. Selected contact-point displacement was calculated from `tool_contact`, while the reported normal load F_n was the model-estimated force change from the clearance transition.

For the internally commanded hold study, redundant motion is calculated by integrating the norm of the recorded projected joint velocity $\dot{q}_{\text{null}}$ over the disturbance interval. The conditioning response is the final change in $\sigma_{\min}$, and Cartesian task retention is represented by the peak position-error norm $\|e_p\|$.

Appendix C

Controller Parameters

ORANGE — sufficient. The active geometry, phase gains, damping policy, timings, null-space values, and collision settings are collected in one reproducibility record.

This appendix records the active controller configuration used for the reported experiments. It is a reproducibility record for the implementation described in chapters 3 and 4. The controller loads fifteen disjoint files in a fixed order: `Run_Settings.txt`, `safety.txt`, `Gripper_Action.txt`, `Auto_Damping.txt`, `Plane_Definition.txt`, `Tool_Orientation.txt`, `Tool_Geometry.txt`, `Q_Init.txt`, `Approach_Phase.txt`, `Clearance_Gate.txt`, `SetUp_Phase.txt`, `Grind_Phase.txt`, `Nullspace.txt`, `hold.txt`, and `guidance.txt`. The values below are those recorded with each run rather than the defaults compiled into the controller.

C.1 Surface and Tool Geometry

ORANGE — sufficient. The configured geometry and initial joint vector are stated with their exact keys and units.

The plane satisfies

$$n_s^\top (p - p_s) = 0, \qquad n_s = R_y(b) R_x(a) e_z,$$

and the surface-frame column order is $[t_1, t_2, n_s]$.

Table C.1: Surface and tool parameters.

Quantity	Configuration key	Value
Surface point	`surface_point`	$[0.515267, -0.107172, 0.003108]\,\mathrm{m}$
Surface tilts	a, b	$-1.585°,\ +0.988°$
Entered first tangent	`alignment_target_tangent1`	$[-0.905786, -0.422854, 0.027332]$
Tool axis in EE	`tool_axis_ee`	$[-0.006011, 0.000246, 0.999982]$
Signed alignment target	`tool_axis_target_sign`	$-n_s$
Spin about the tool axis, from t_1	`tool_target_offset_normal_deg`	115.05°
Grinding-face centre in EE	`tool_contact_face_center_ee`	$[0, 0, 0.020]\,\mathrm{m}$
Grinding-face size	width × length	$0.040 \times 0.120\,\mathrm{m}$
Feature tie tolerance	`tool_contact_feature_tie_tolerance`	$10^{-4}\,\mathrm{m}$
Initial-joint case	`q_init_case`	`horizontal_table_search`

This case selects the `q_init_table` entries, so the corresponding initial joint vector is

$$q_{\mathrm{init}} = [0.014777,\ 0.210911,\ -0.009320,\ -2.284835,\\ -0.002725,\ 2.493410,\ 0.780694]^\top\ \mathrm{rad}.$$

C.2 Phase Sequence and Impedance

ORANGE — sufficient. The phase-specific gain frames and transition parameters reconcile the values stated across Chapters 3 and 4.

Table C.2: Phase and mode gains.

Phase / quantity	Active stiffness and frame	Configured damping
Approach translation	$[150, 150, 150]$ N/m, surface $[t_1, t_2, n_s]$	$[20, 20, 20]$ N s/m
Approach rotation	$[90, 90, 90]$ N m/rad, surface $[t_1, t_2, n_s]$	$[12, 12, 12]$ N m s/rad
Pre-set-up gate translation	5000 N/m, isotropic base	200 N s/m
Pre-set-up gate rotation	90 N m/rad, inherited approach rotation	inherited cached approach D_R
Set-up translation (compliance-centre source)	$[2000, 2000, 350]$ N/m in Cases D to H, $[2000, 2000, 800]$ N/m in Cases A to C, surface $[t_1, t_2, n_s]$	$[50, 50, 25]$ N s/m
Set-up rotation (compliance-centre source)	$[5, 5, 50]$ N m/rad, surface $[t_1, t_2, n_s]$	$[10.01, 10.01, 10]$ N m s/rad
Grind translation (decoupled)	as the set-up entry above, surface $[t_1, t_2, n_s]$	$[50, 50, 25]$ N s/m
Grind rotation (decoupled)	$[5, 5, 50]$ N m/rad, surface $[t_1, t_2, n_s]$	$[10.01, 10.01, 10]$ N m s/rad
Hold translation	2000 N/m, isotropic base	50 N s/m
Hold rotation	50 N m/rad, isotropic base	8 N m s/rad
Manual guidance	no Cartesian stiffness; joint coordinates	0.5 N m s/rad joint damping

The enabled pre-set-up gate overrides only translational stiffness. The pre-grind gate was also enabled; it remains in set-up and keeps the coupled pressed law rather than using the pre-set-up gate gains.

With `setup_translation_surface_frame` set, the set-up and grind translational blocks are read from the surface-frame keys listed above. The base-frame keys `setup_Kp` and `setup_Dp` remain in the file at $[2000, 2000, 350]$ N/m and $[10, 10, 175]$ N s/m but take no part in the reported runs.

Table C.3: Phase-transition and trajectory parameters.

Quantity	Configuration key	Value
Orient minimum time	approach_orient_min_time	0.5 s
Tool-axis switch tolerance	approach_orient_error_threshold	0.035 rad
Descent speed / maximum distance	descend_speed, descend_max_distance	0.1 m/s, 0.640 m
Surface clearance	descend_surface_clearance	0.020 m
Set-up minimum time / timeout	setup_min_time, setup_timeout	2.0 s, 5.0 s
Set-up moment-change threshold	setup_moment_threshold	150 N m
Set-up push speed / endpoint	setup_push_speed, setup_push_end	0.050 m/s, 0.060 m in Cases A to C; 0.080 m/s, 0.240 m in Cases D to H
Grinding reference mode	grind_sweep_enabled	fixed tangential coordinates (configured t_2, 0.030 m, 0.2 Hz)
Pre-set-up / pre-grind gates	pause_before_set_up, pause_before_grind	enabled, enabled

C.3 Inertia-Scaled Damping and Centre of Compliance

Inertia-scaled damping follows eqs. (3.9) and (3.10). It is enabled for approach, set-up, grind, and the pause gate. The approach factor is $\zeta = 1.9$, while set-up and grind use $\zeta = 1.0$. The pause gate uses separate translational and rotational factors because the two gain blocks are expressed in different frames. The configured maximum coefficient `auto_damping_max` is 400. The configured values in table C.2 provide fallback damping for the phases using inertia-scaled damping. With `auto_damping_min_from_manual` left at zero, those manual values act only as that fallback and impose no lower bound on a calculated coefficient.

Standard Cartesian pose hold instead used its configured matrices directly. All twelve reported null-space runs recorded `hold_auto_damping=0`, with $D_p = 50 I_3\,\mathrm{N\,s/m}$ and $D_R = 8 I_3\,\mathrm{N\,m\,s/rad}$. The hold-damping values in table C.2 are therefore active values rather than fallbacks.

For set-up, the source damping blocks are calculated from TCP task inertia Λ and then shifted to the selected centre of compliance p_c. This preserves symmetry and positive semidefiniteness. The factor $\zeta = 1$ is the directional coefficient used by the online damping calculation.

The coupled 6×6 law is enabled only during set-up, and in Cases C and E to H. The lever was entered directly in surface coordinates through `coupled_use_direct_rc_surface`, so

$$r_c = R_{\text{surface}}\, r_c^S, \qquad r_c^S = [\, r_{c,t_1},\ r_{c,t_2},\ r_{c,n}\,]^\top,$$

with the tested components listed in table 4.2. The alternative offset-from-edge convention, in which the centre of compliance p_c is placed relative to the contact point projected onto the surface at the 20 mm clearance hand-off, is preserved in the implementation for archived configuration files and was not used for the reported runs. With `coupled_pole_freeze_at_contact=1`, the references are those captured at the hand-off, so the active r_c is constant throughout set-up.

C.4 Null-Space and Operating Parameters

ORANGE — sufficient. The selected mode, gains, probe settings, logging capacity, gripper state, and robot-side thresholds are stated explicitly.

The null-space settings are explained in section 4.4.6. The final two rows of Table C.4 are Franka collision/reflex thresholds configured through the libfranka collision-behaviour interface. Robot-side monitoring applied these thresholds throughout the reported runs.

Table C.4: Null-space and operating parameters.

Parameter	Symbol / key	Value
Null-space mode	`nullspace_mode`	1 (projected damping only)
Null-space damping	d_{null}	2.0 N m s/rad
Sigma torque magnitude	k_σ	1.0 N m (inactive in mode 1)
Probe distance	α_{probe}	0.08 rad
Sigma difference deadband	δ_σ	10^{-6}
Relative SVD cutoff	ρ	10^{-4}
Recorded-data sampling / capacity	`log_every_n_cycles`, rows	every callback (1), 120000
Run duration / stop	`experiment_duration`	0 s: indefinite until `e+Enter`
Gripper grasp	width, speed, force	0.066 m, 0.050 m/s, 60 N
Tool-mount rotational play	observed pad-mount clearance	approximately $\pm 2°$ about y_{EE}
Manual-guidance damping	`manual_guidance_damping`	0.5 N m s/rad
Joint-side collision threshold	acceleration/nominal	140 N m on each axis
Cartesian collision threshold	acceleration/nominal	240 N or N m on each channel